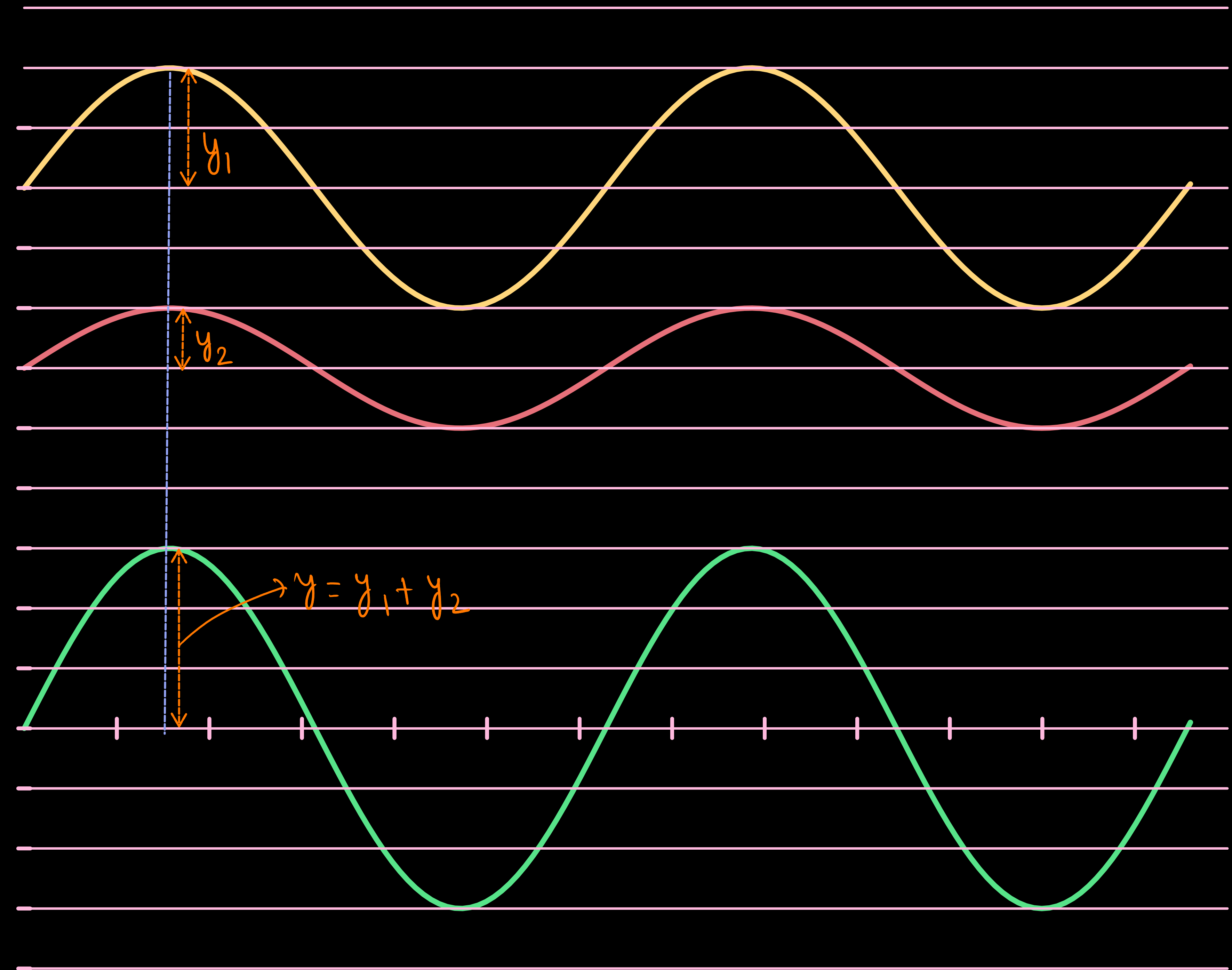


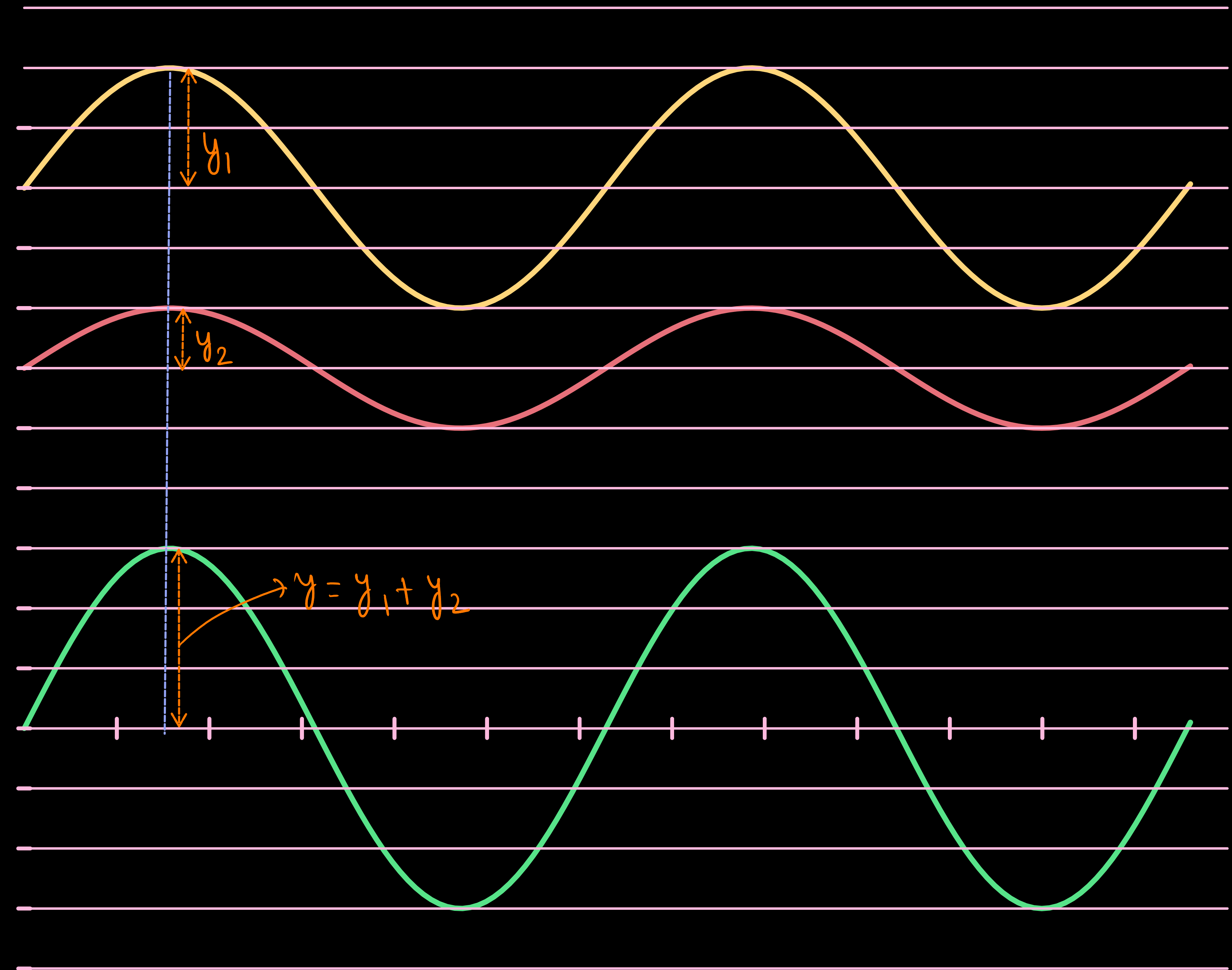
Principle of Superposition of Waves

Principle of Superposition of Waves



Principle of Superposition of Waves

It states that when two or more waves of light travelling in a medium superpose upon each other, then the resultant displacement at any instant is equal to the vector sum of the displacements due to individual waves.



Sustained Interference of Light

Sustained Interference of Light

Sustained Interference

The interference pattern in which the position of maximum and minimum intensity of light remains fixed all along on the screen, is called sustained or permanent interference pattern.

Conditions for sustained Interference →

- ① Sources must be coherent →
The light waves should be of same wavelength and have zero/constant phase difference
- ② The light waves emitted by the two sources should preferably be of the same amplitude.
- ③ The two sources of light must be close to each other.
- ④ The two sources emit the light waves continuously.

Superposition of Two Waves and Condition for Constructive and Destructive Interference

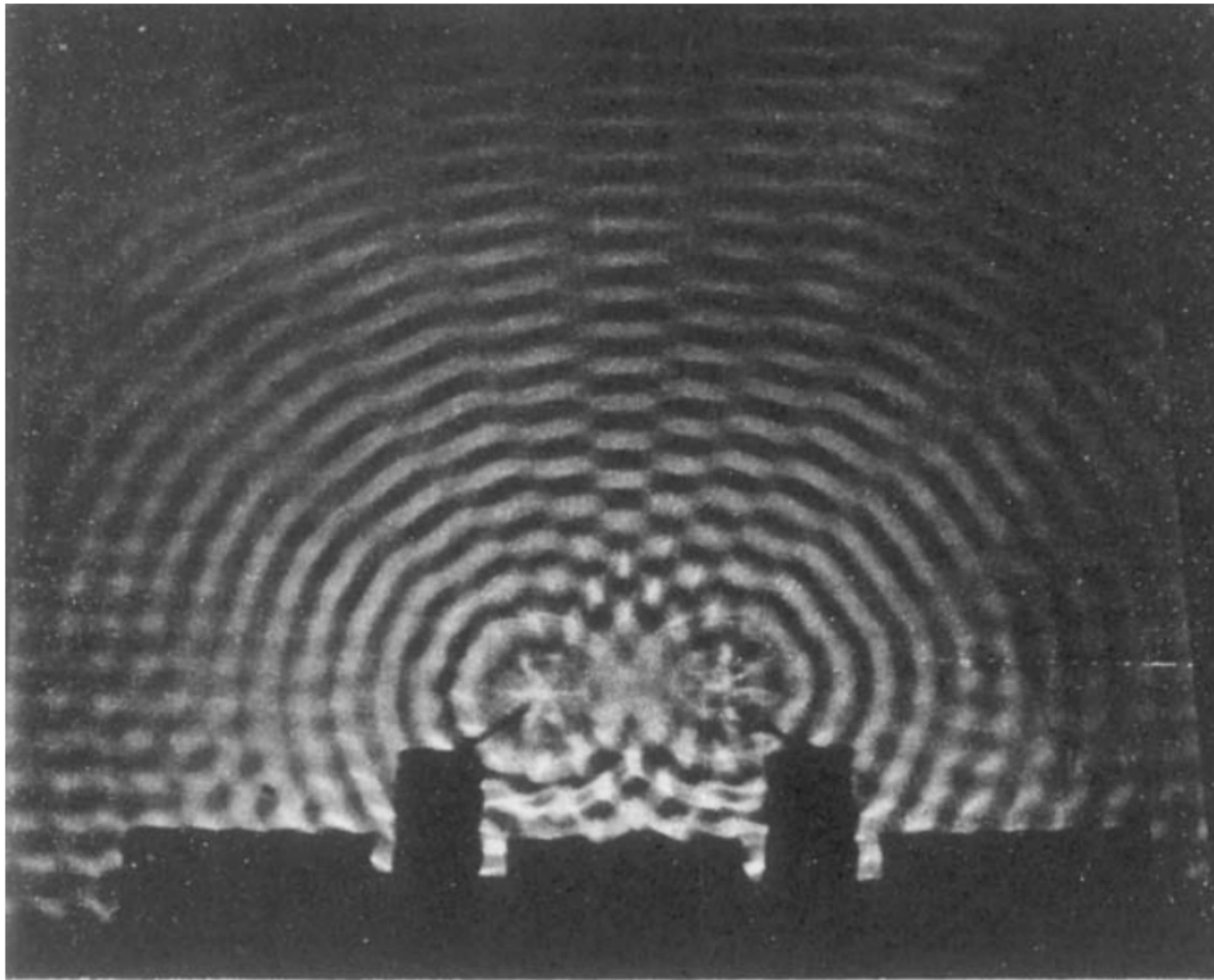


Fig. 14.2 The actual interference pattern produced from two point sources vibrating in phase in a ripple tank (After Ref. 9, used with permission).

Superposition of Two Waves and Condition for Constructive and Destructive Interference

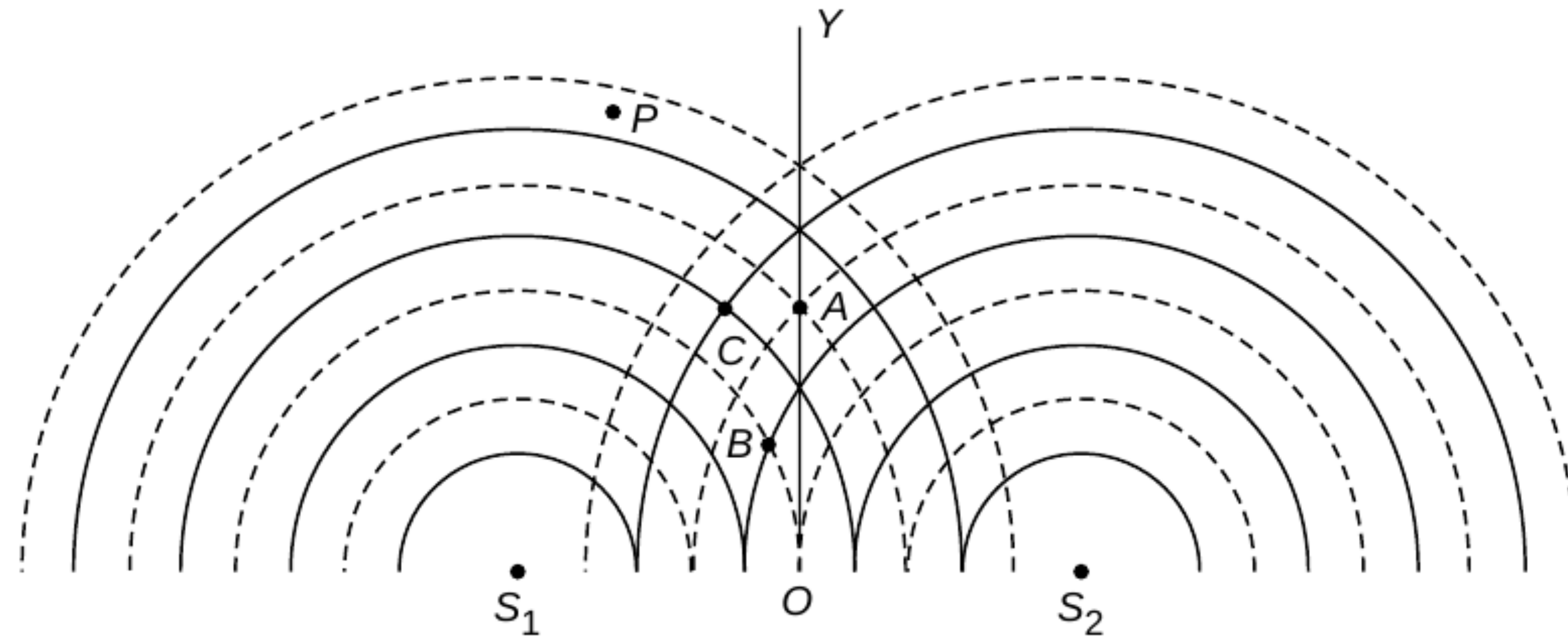


Fig. 14.1 Waves emanating from two point sources S_1 and S_2 vibrating in phase. The solid and the dashed curves represent the positions of the crests and troughs, respectively.

Superposition of Two Waves and Condition for Constructive and Destructive Interference

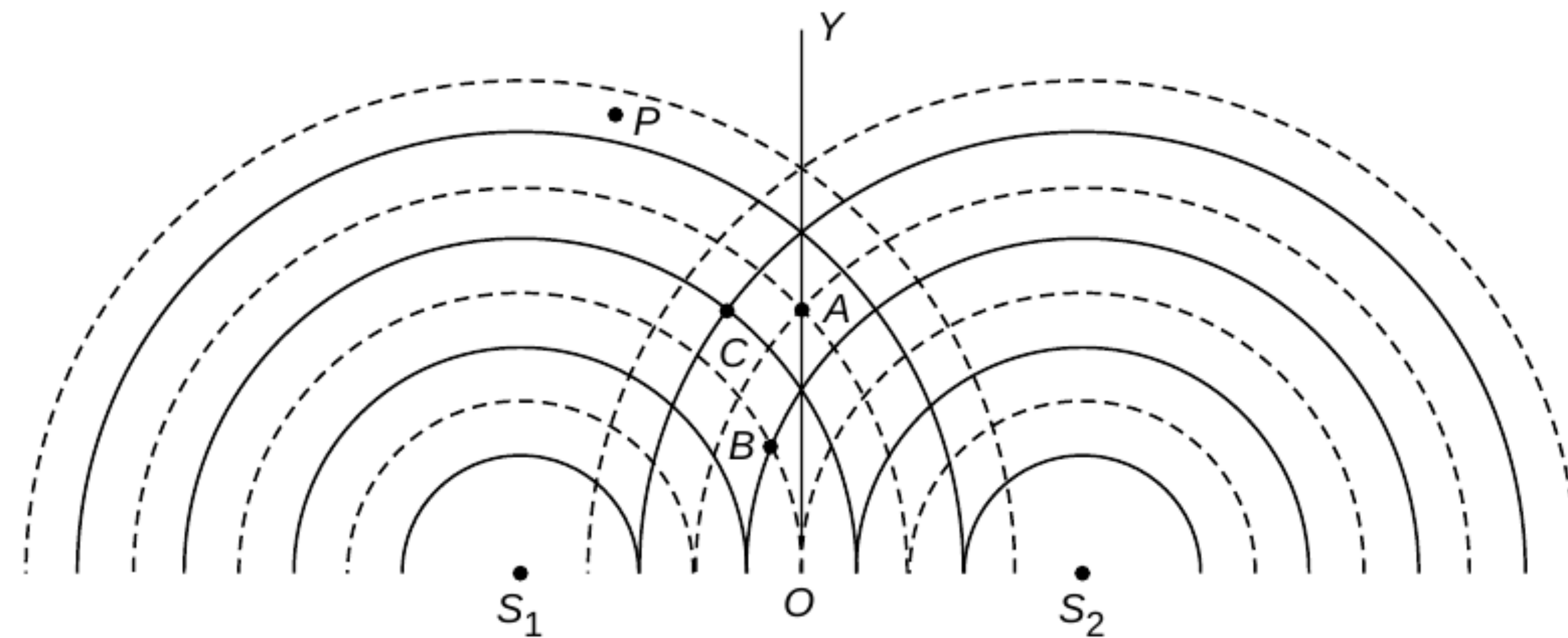


Fig. 14.1 Waves emanating from two point sources S_1 and S_2 vibrating in phase. The solid and the dashed curves represent the positions of the crests and troughs, respectively.

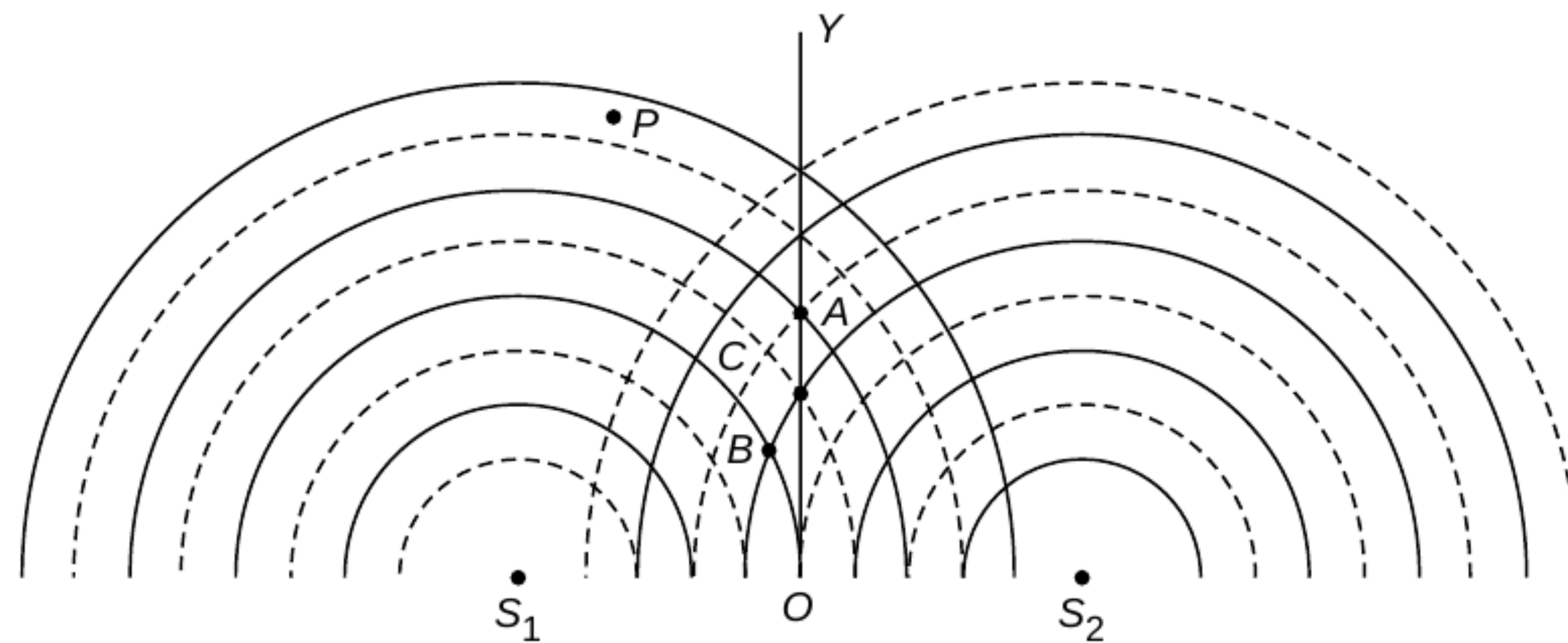


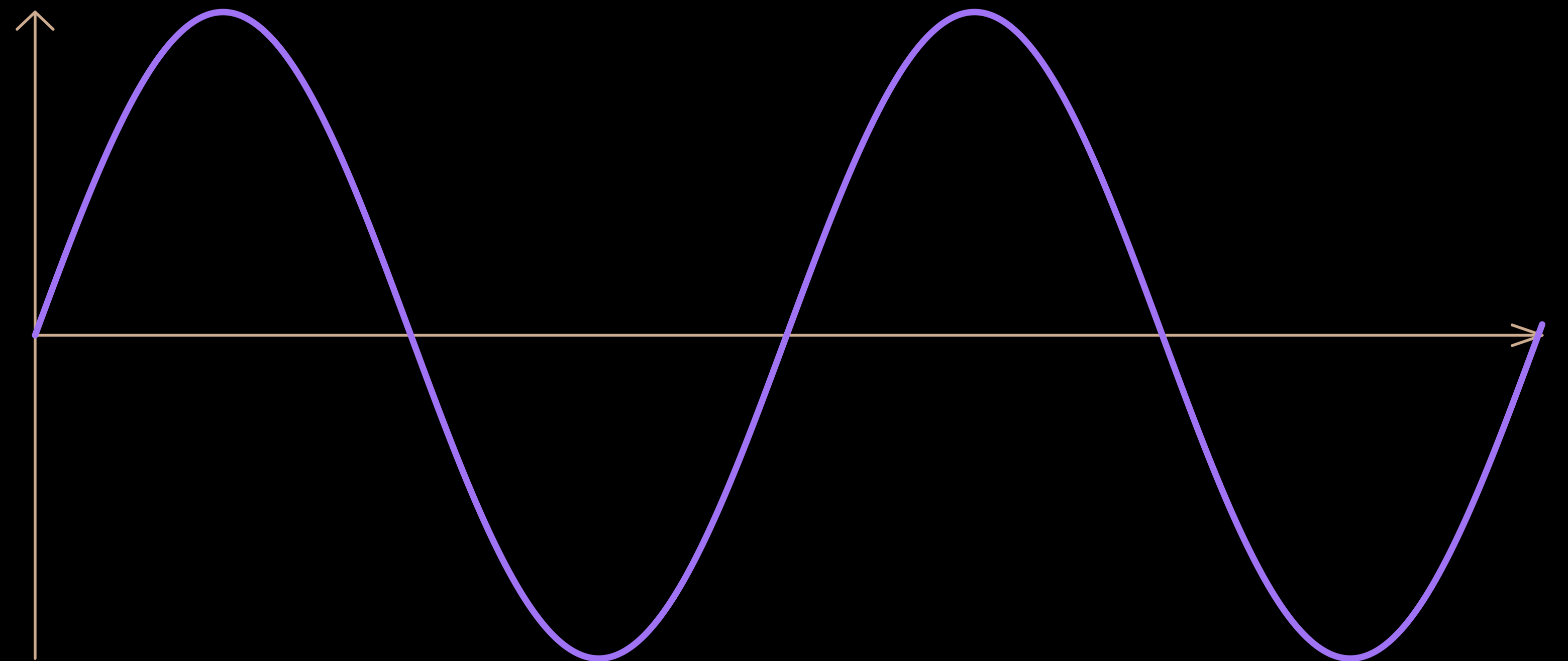
Fig. 14.4 Waves emanating from two point sources S_1 and S_2 vibrating out of phase.

Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :

Superposition of Two Waves and Condition for Constructive and Destructive Interference

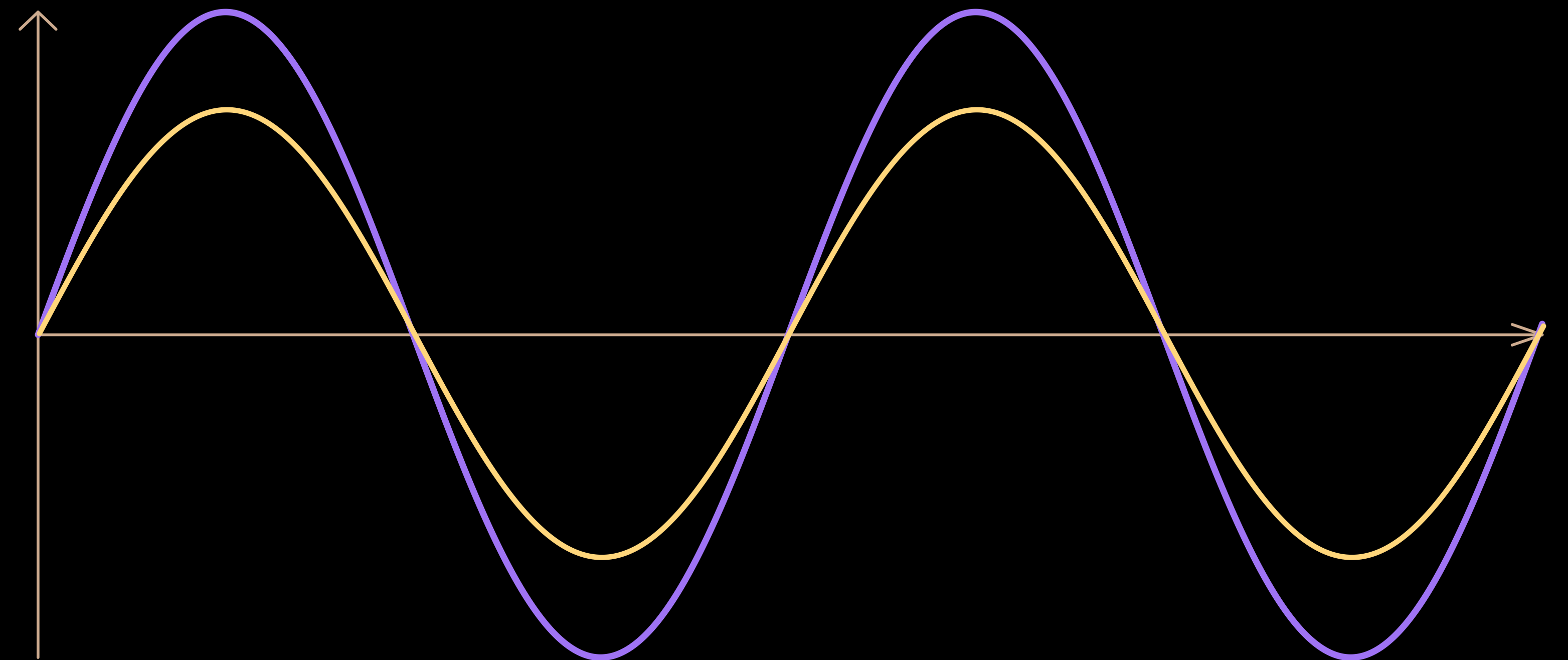
Relation Between Path Difference (x) and Phase Difference (Φ) :



Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :

Phase Difference $\Phi = 0$

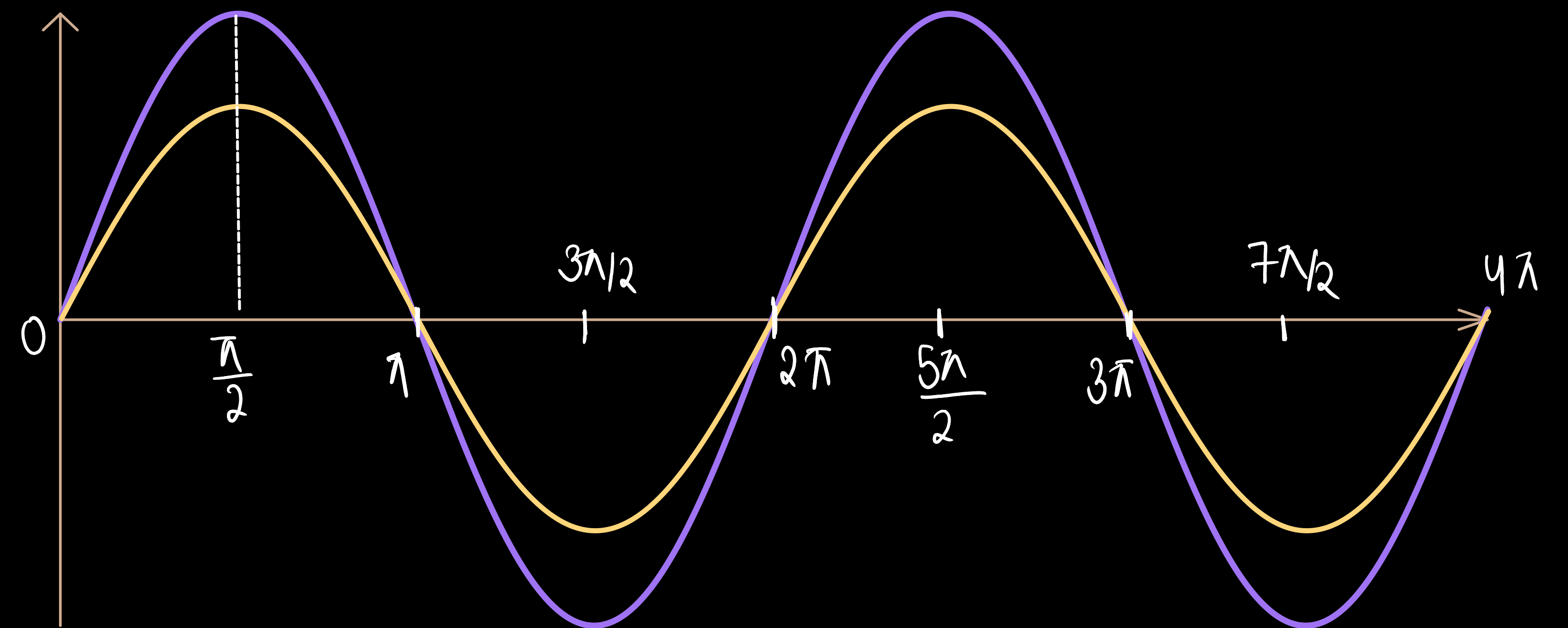


Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :

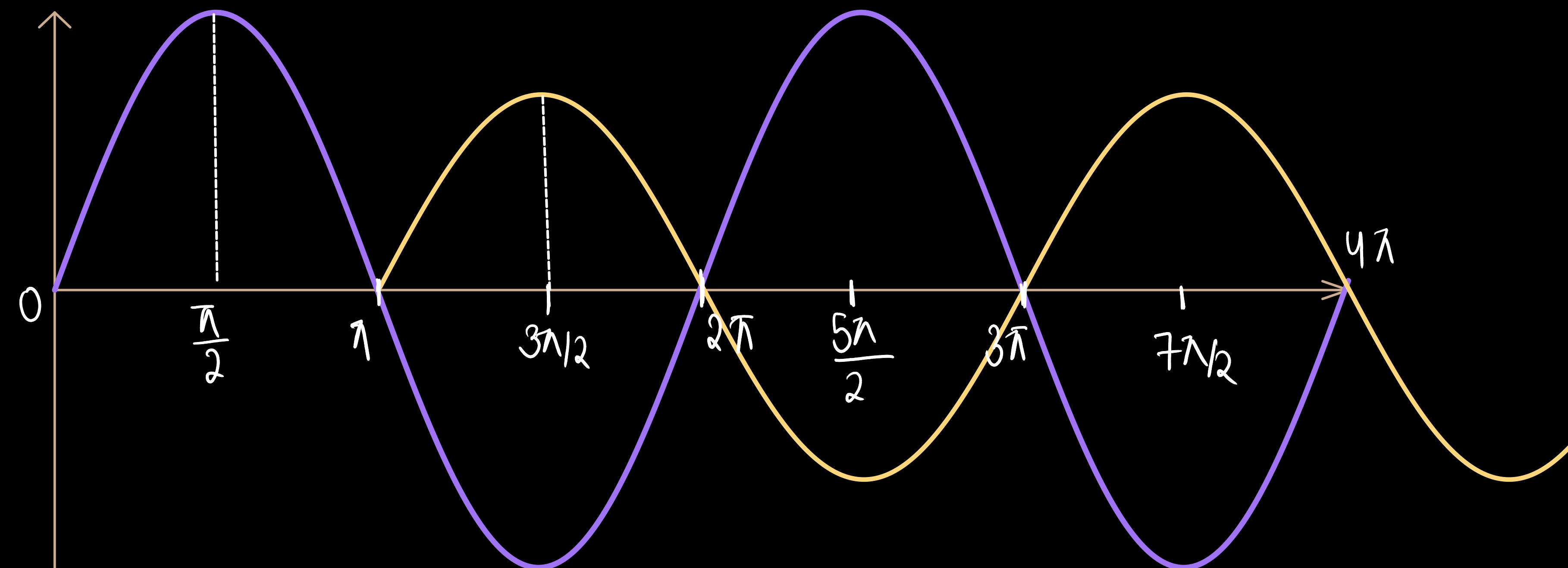
Phase Difference $\Phi = 0$

Path Difference $x = 0$



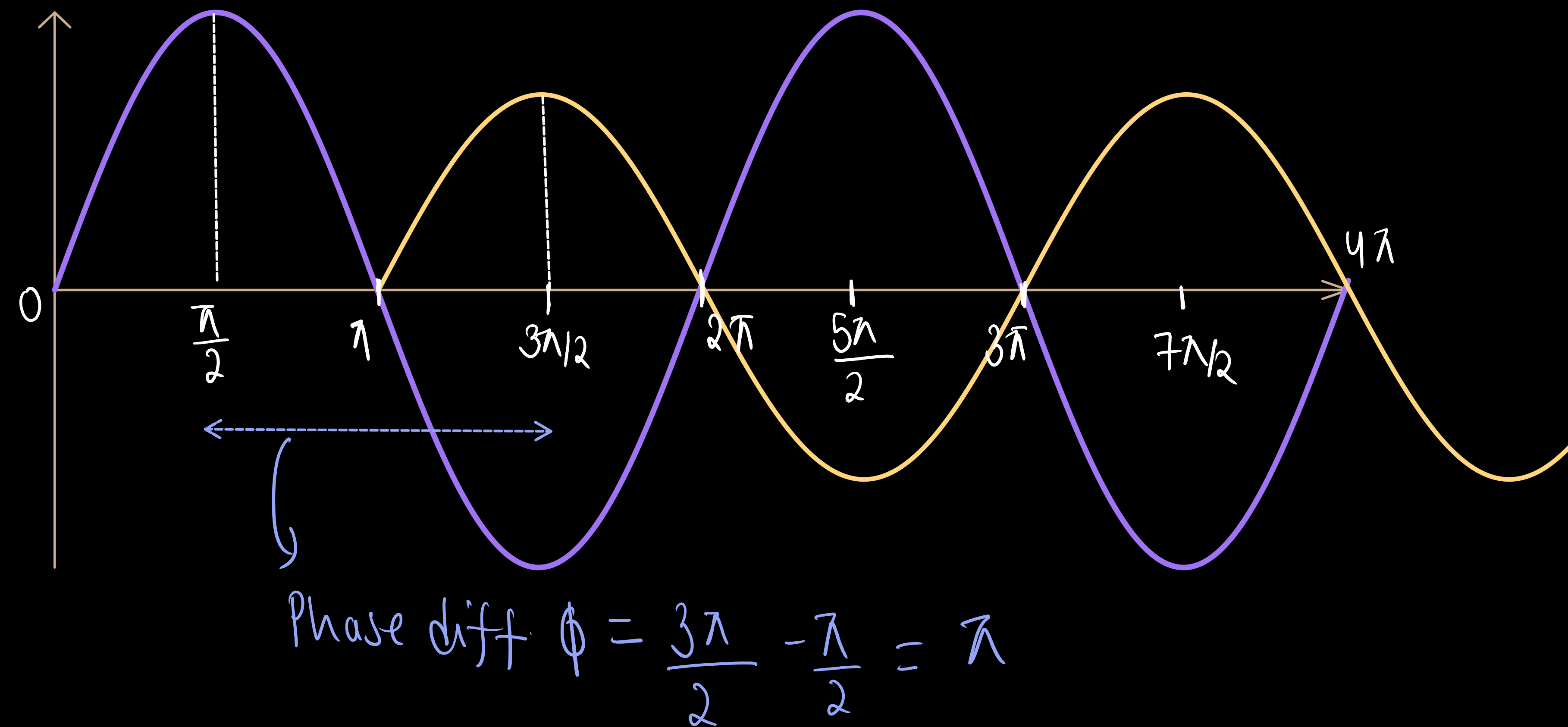
Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :



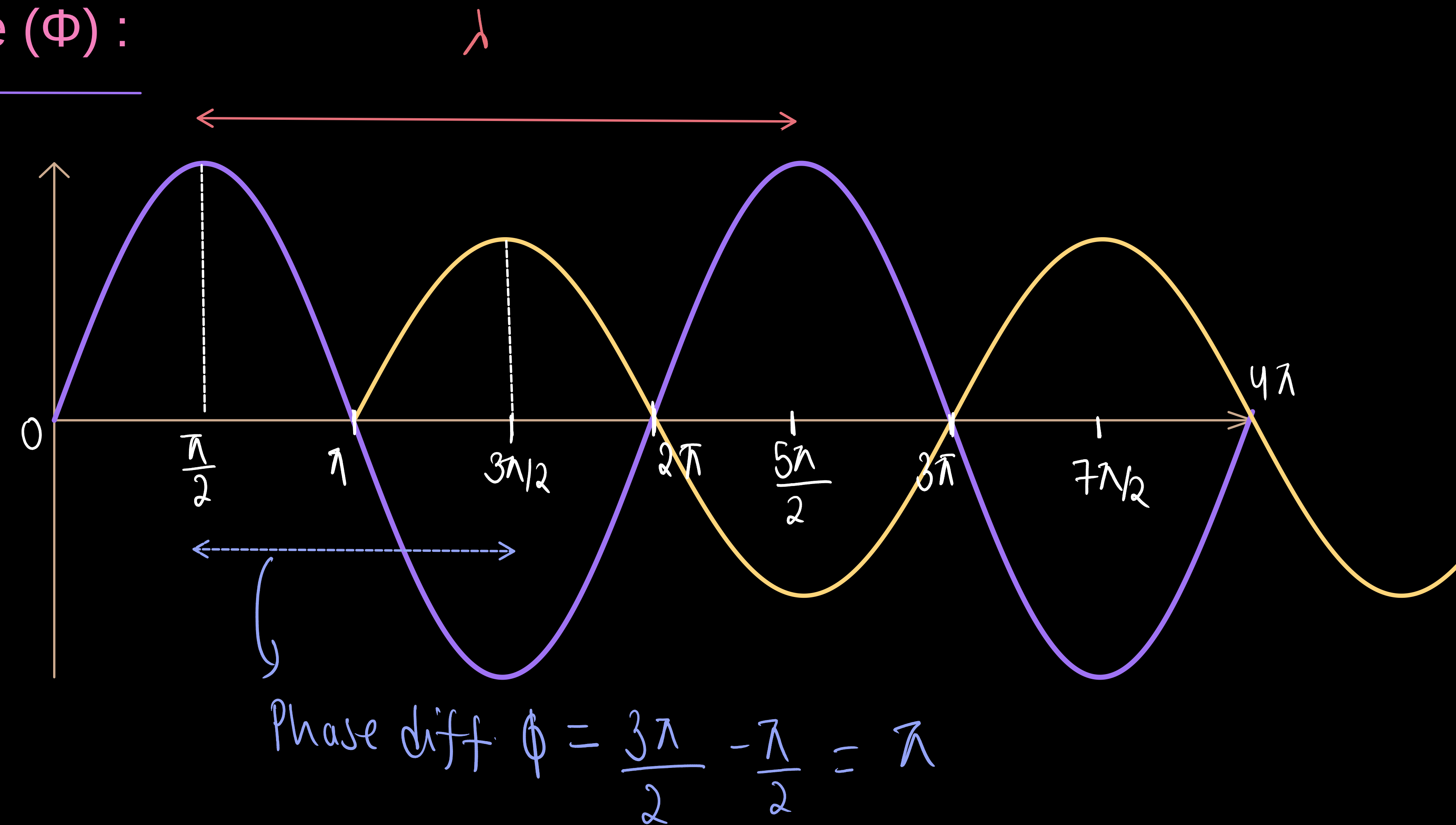
Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :



Superposition of Two Waves and Condition for Constructive and Destructive Interference

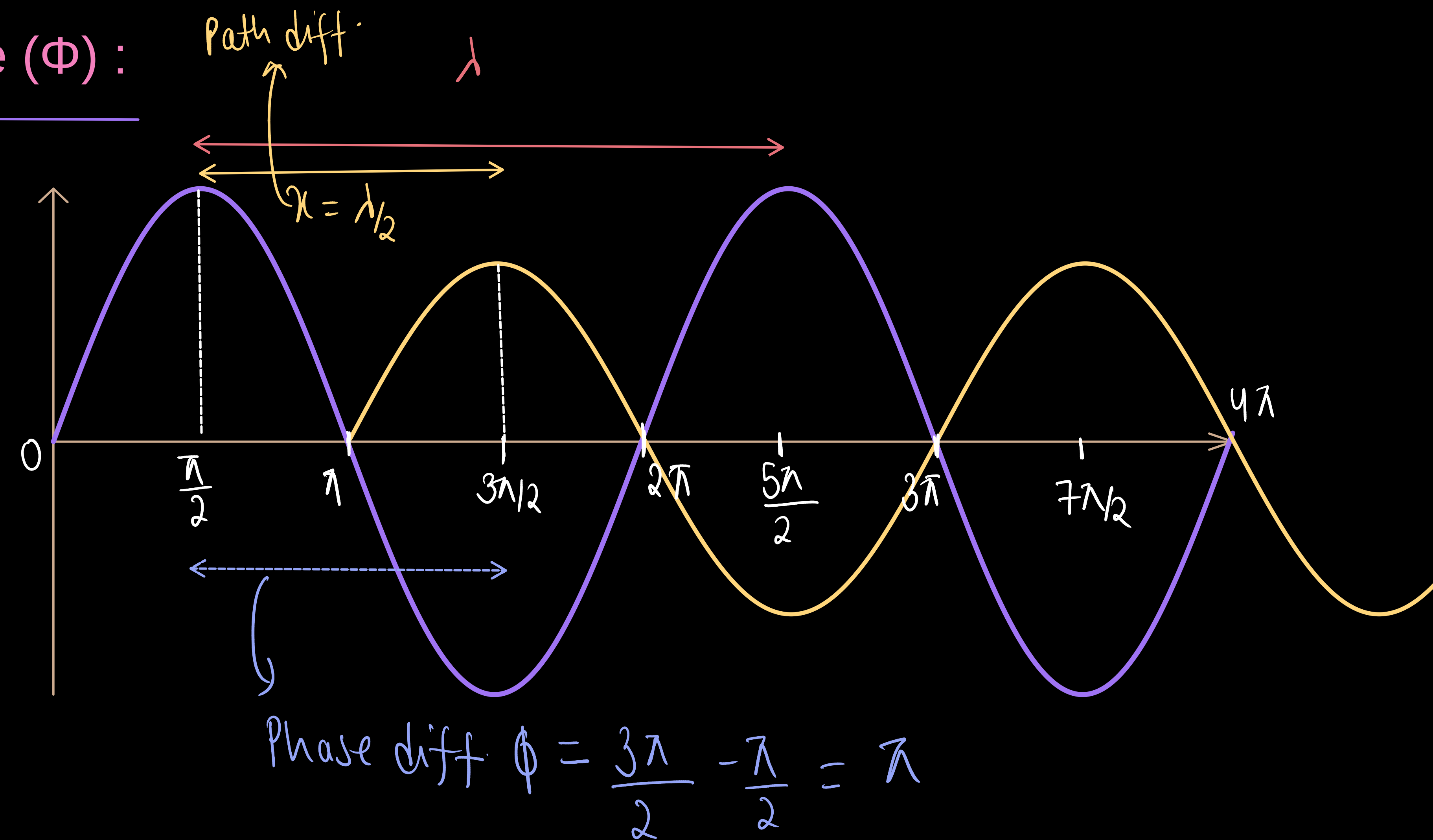
Relation Between Path Difference (x) and Phase Difference (Φ) :



Superposition of Two Waves and Condition for Constructive and Destructive Interference

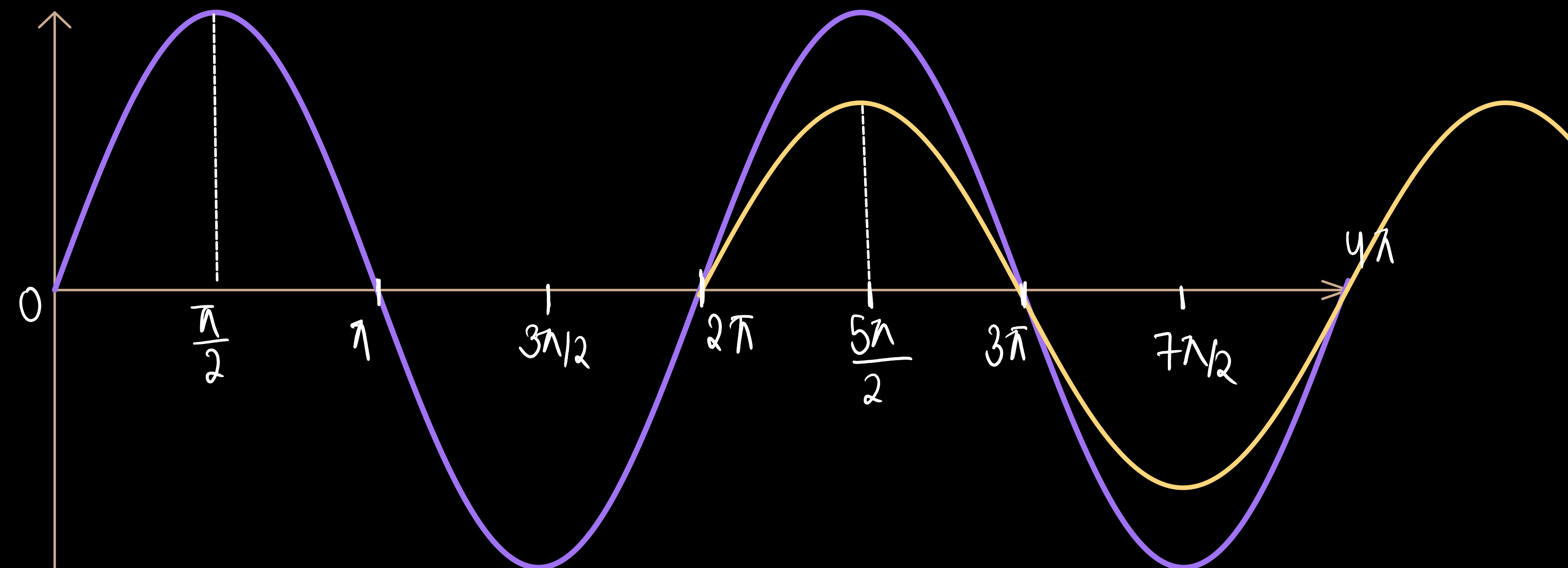
Relation Between Path Difference (x) and Phase Difference (Φ) :

When $\phi = \pi$
 $x = \lambda/2$



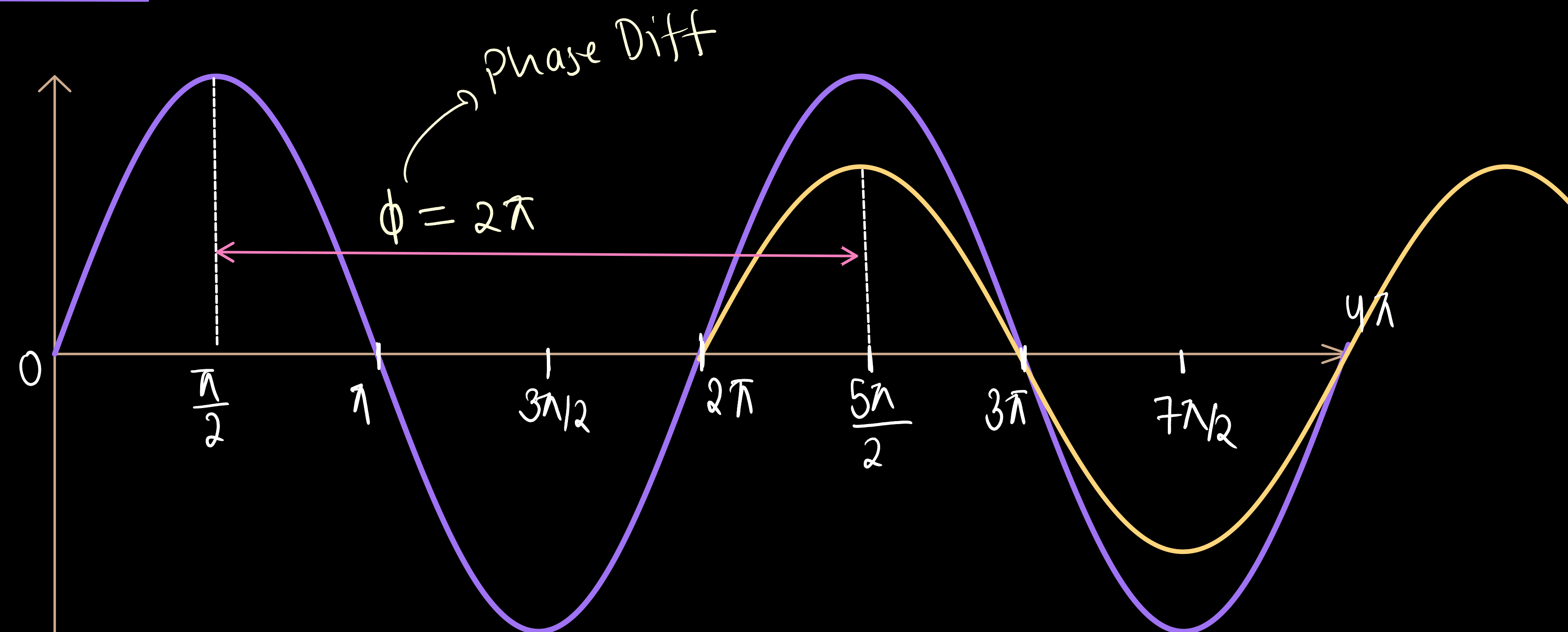
Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :



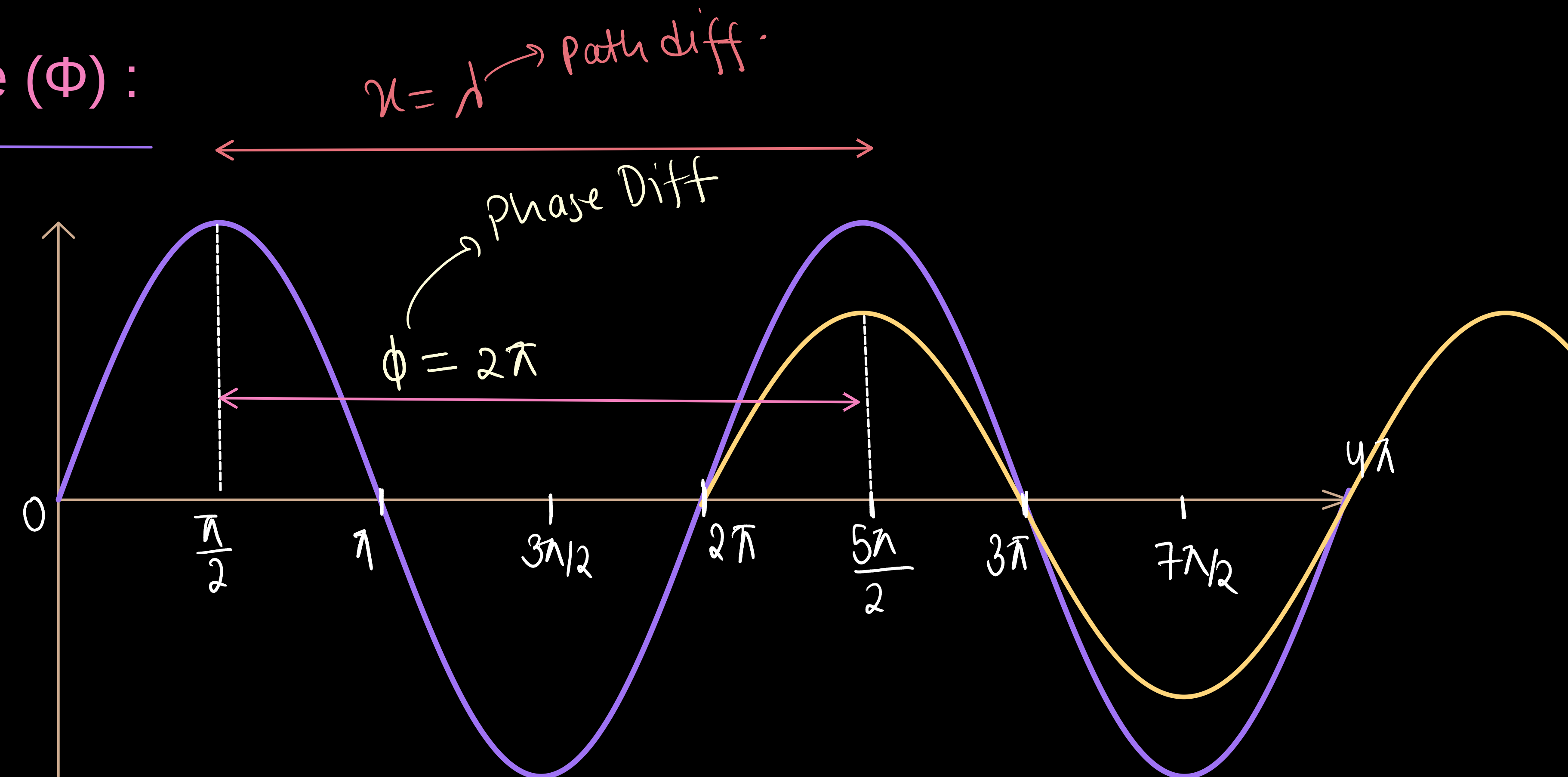
Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :



Superposition of Two Waves and Condition for Constructive and Destructive Interference

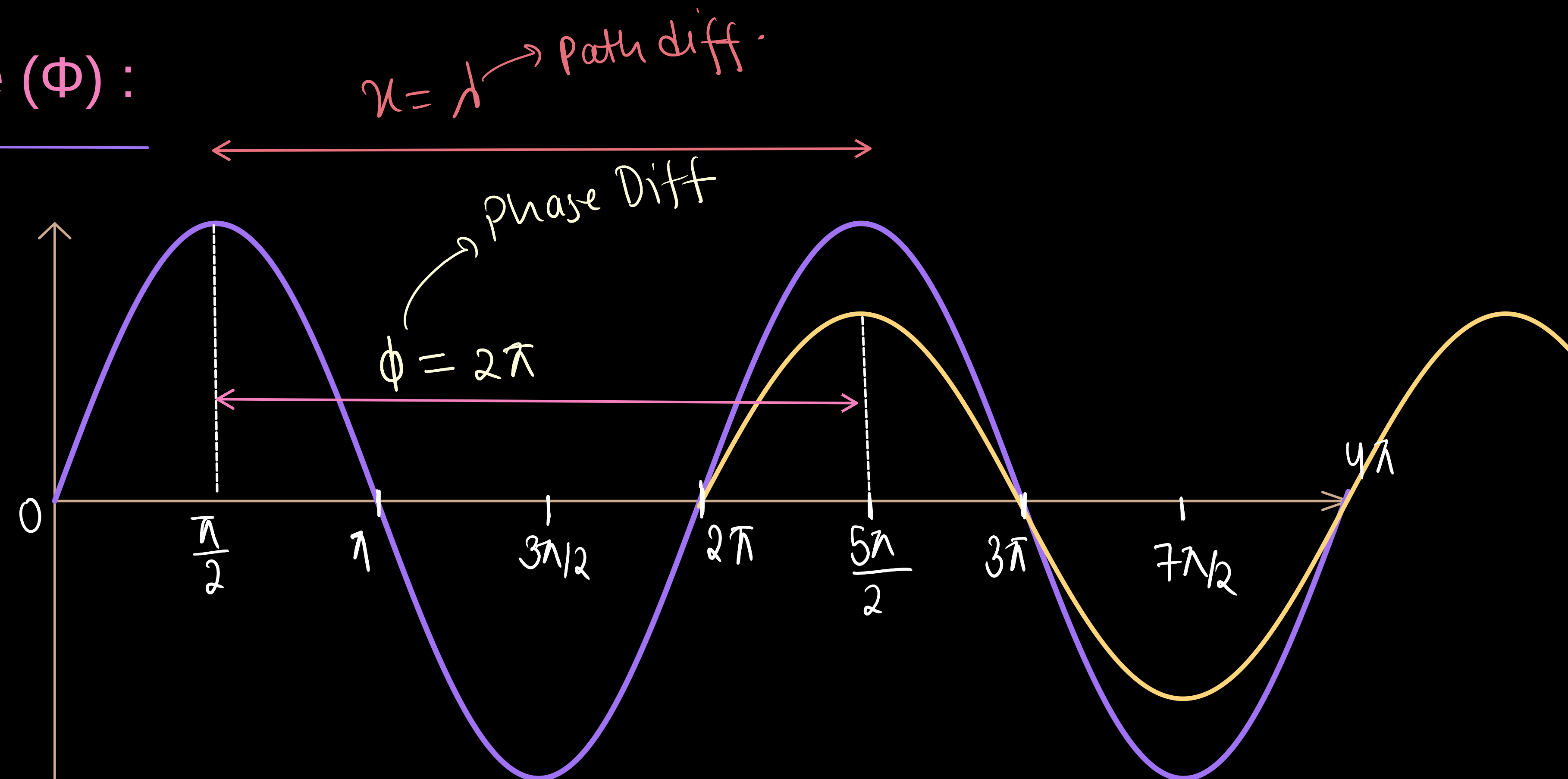
Relation Between Path Difference (x) and Phase Difference (Φ) :



Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :

when $\phi = 2\pi$
 $x = \lambda$



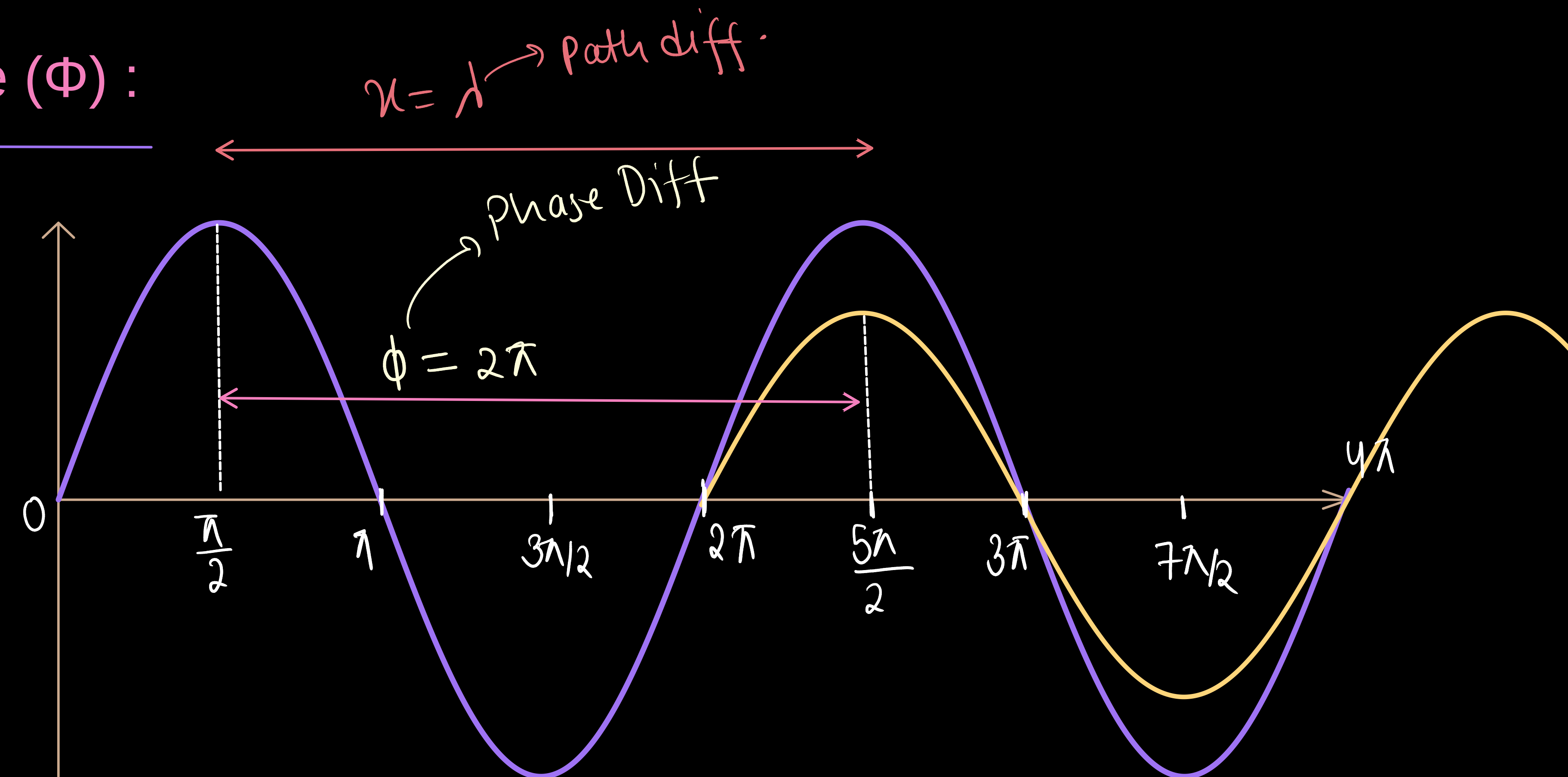
Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :

when $\phi = 2\pi$

$$x = \lambda$$

$$\frac{\phi}{x} = \frac{2\pi}{\lambda}$$



Superposition of Two Waves and Condition for Constructive and Destructive Interference

Relation Between Path Difference (x) and Phase Difference (Φ) :

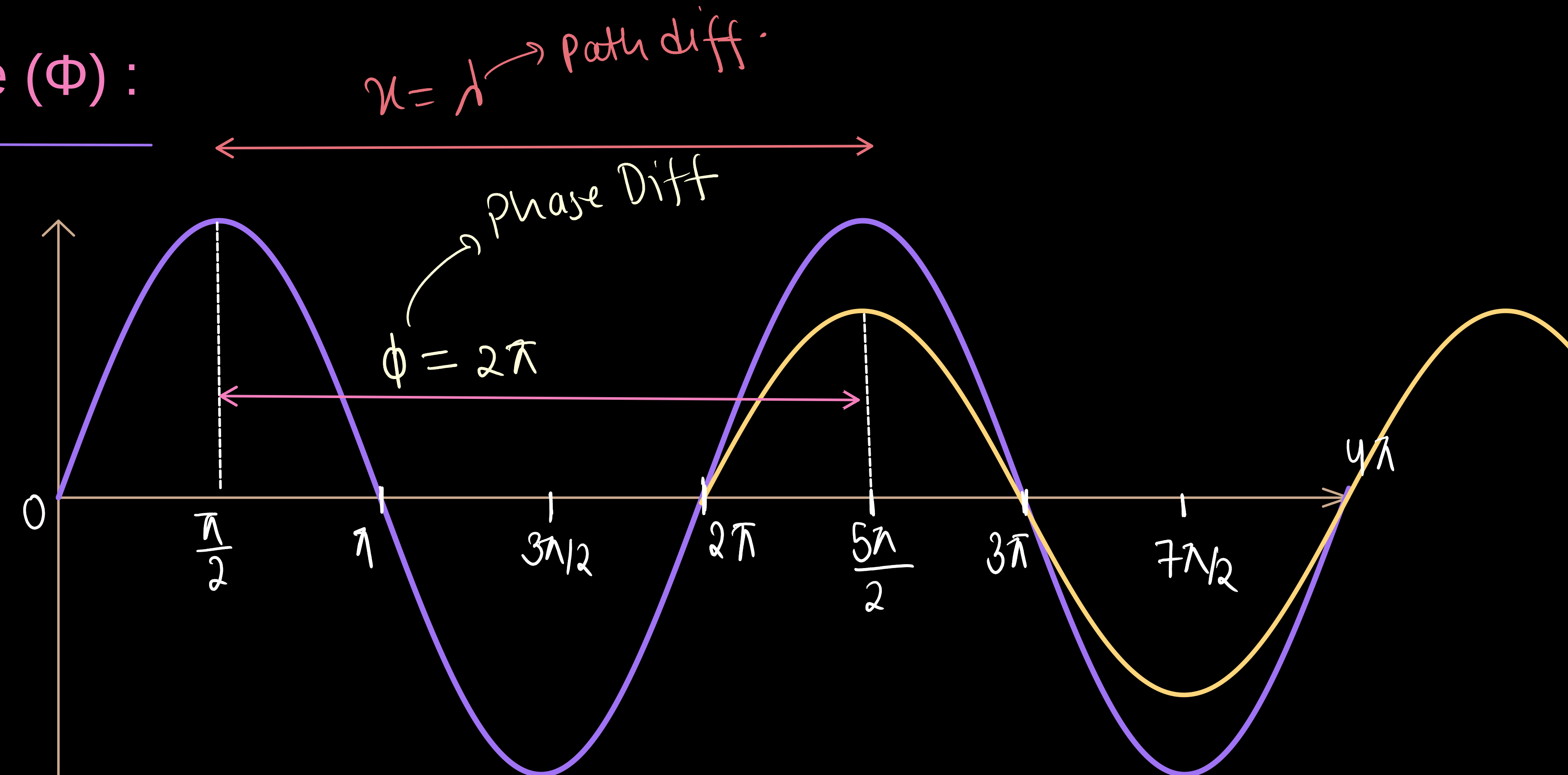
when $\phi = 2\pi$

$$x = \lambda$$

$$\frac{\phi}{x} = \frac{2\pi}{\lambda}$$

$$\phi = \frac{2\pi}{\lambda} x$$

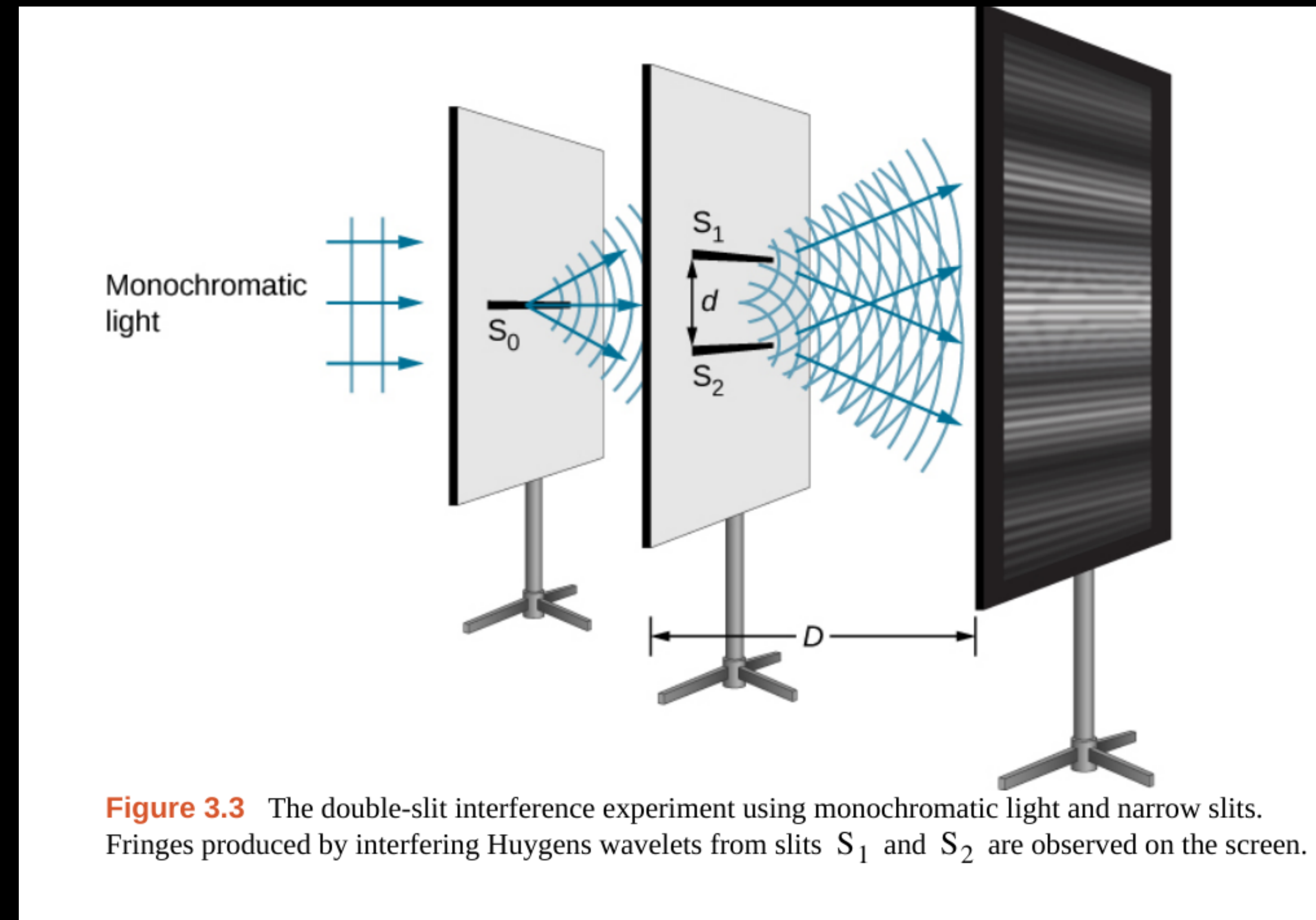
→ Relation b/w
Phase diff.
and path diff.



Superposition of Two Waves and Condition for Constructive and Destructive Interference

Young's Double Slit Experiment

Young's Double Slit Experiment



Young's Double Slit Experiment

Young's Double Slit

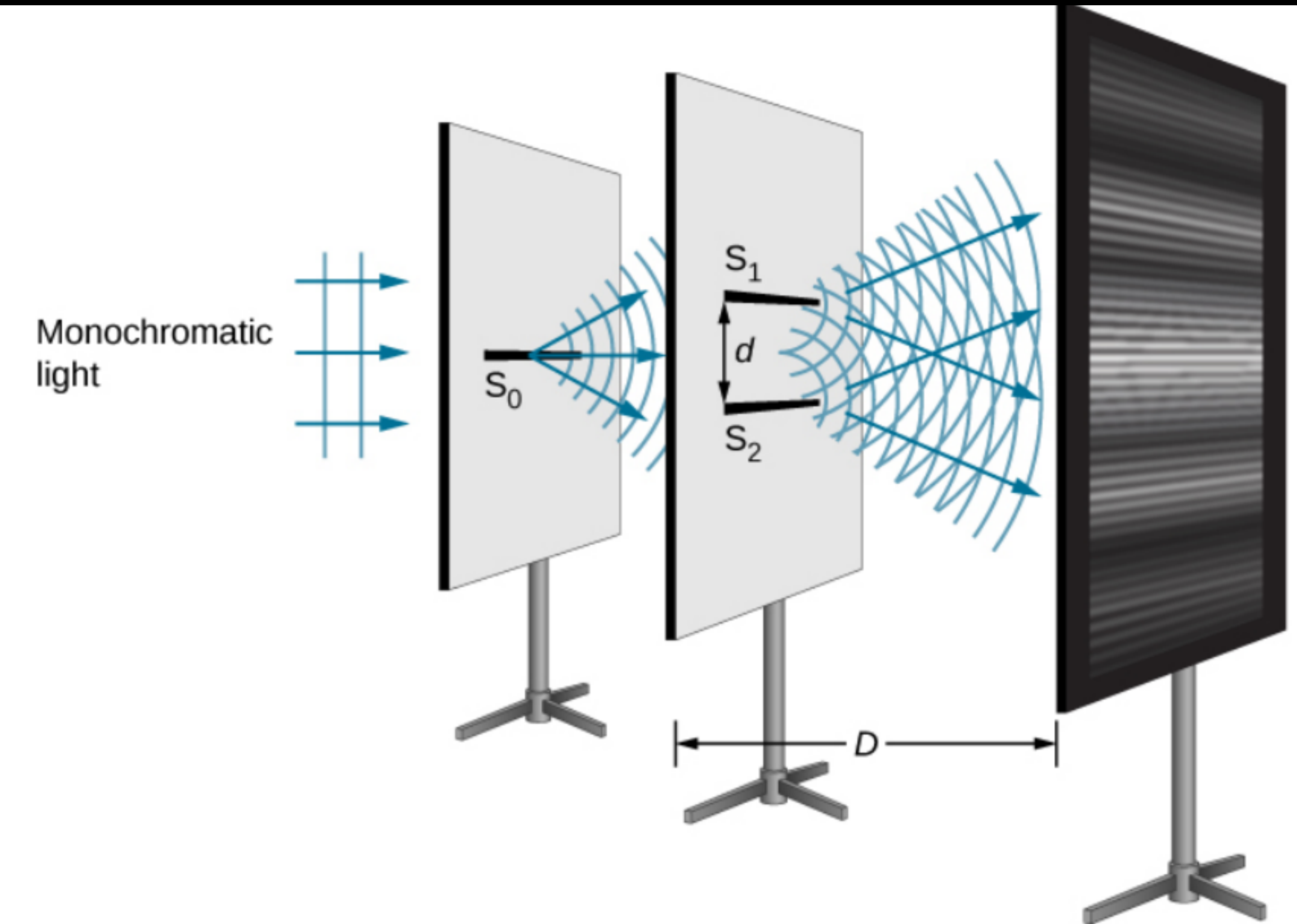
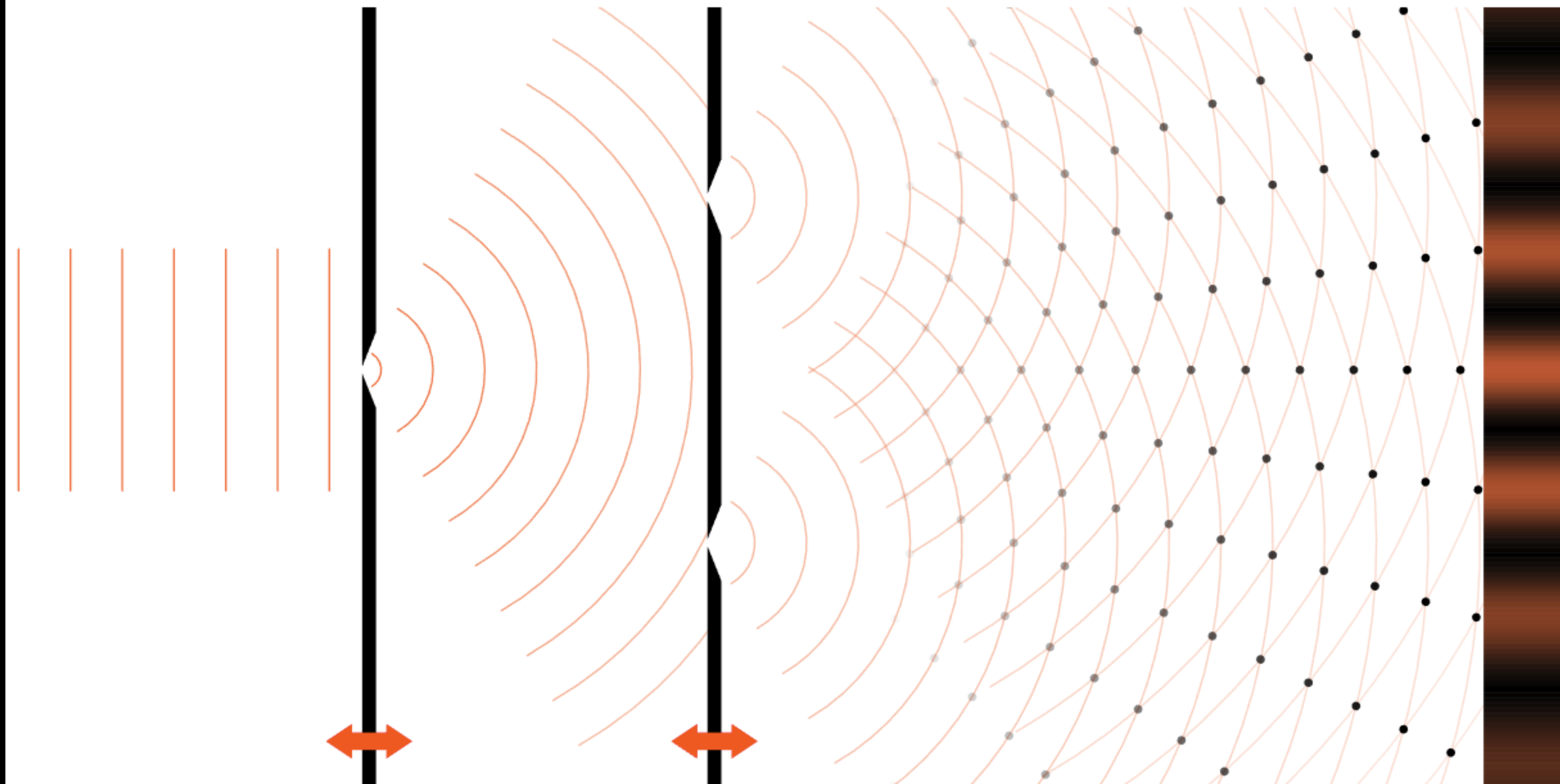
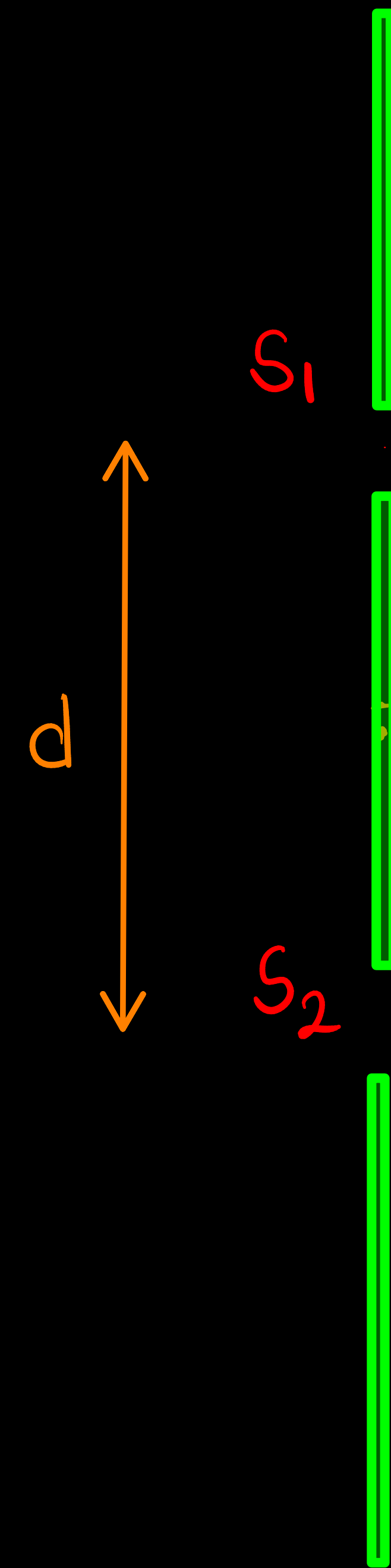


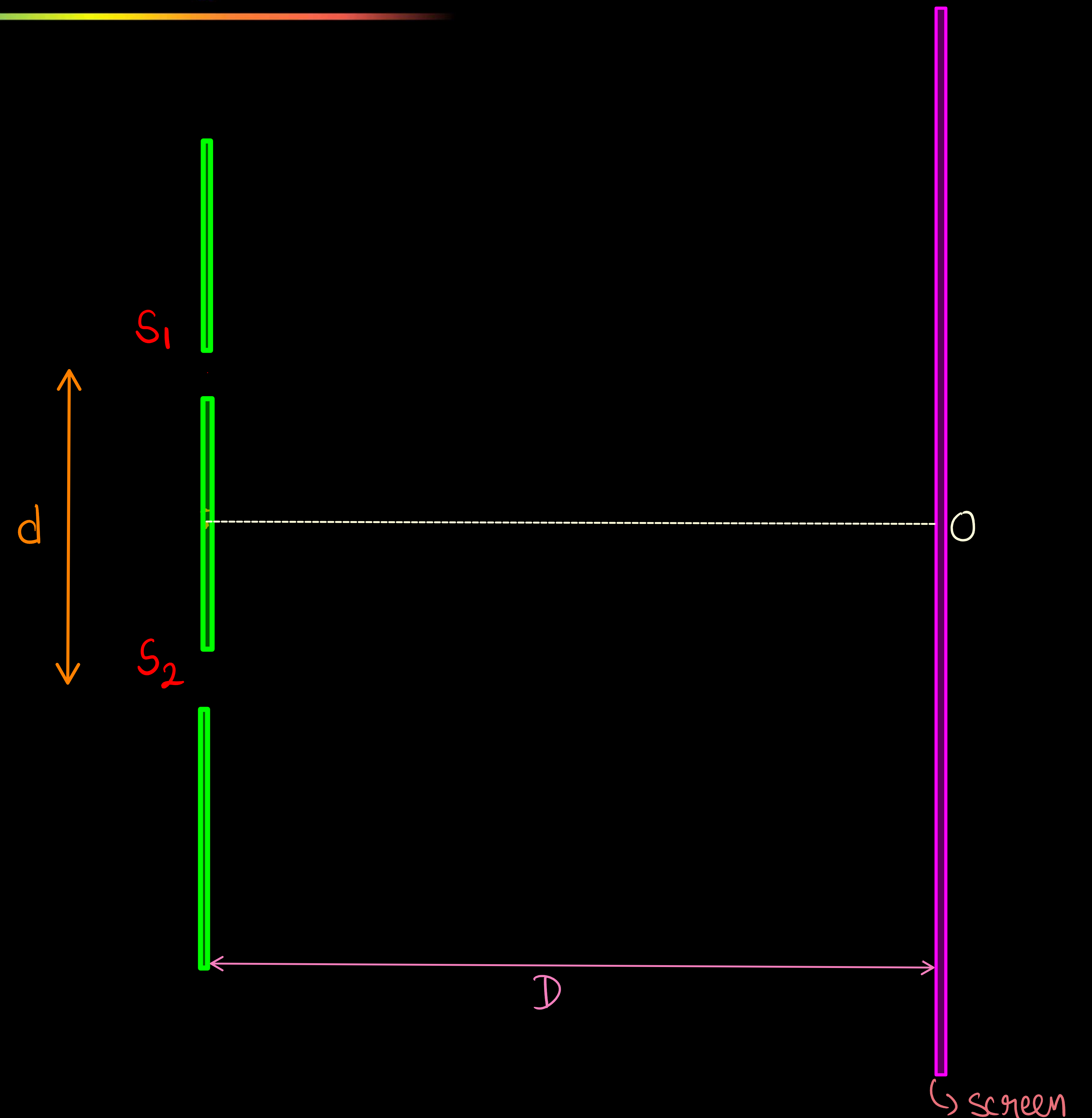
Figure 3.3 The double-slit interference experiment using monochromatic light and narrow slits. Fringes produced by interfering Huygens wavelets from slits S_1 and S_2 are observed on the screen.

Young's Double Slit Experiment

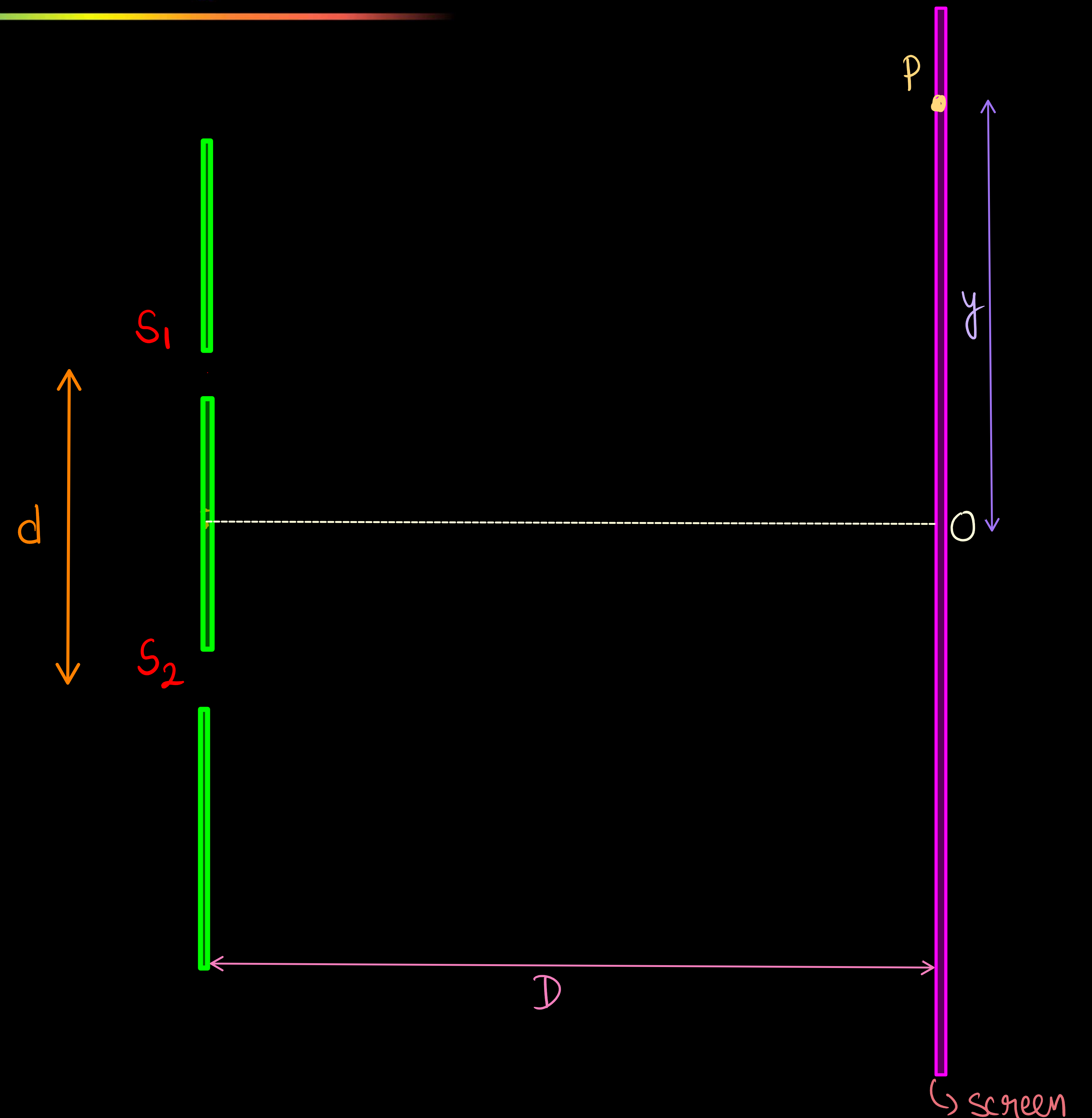
Young's Double Slit Experiment (Derivation Not Required)



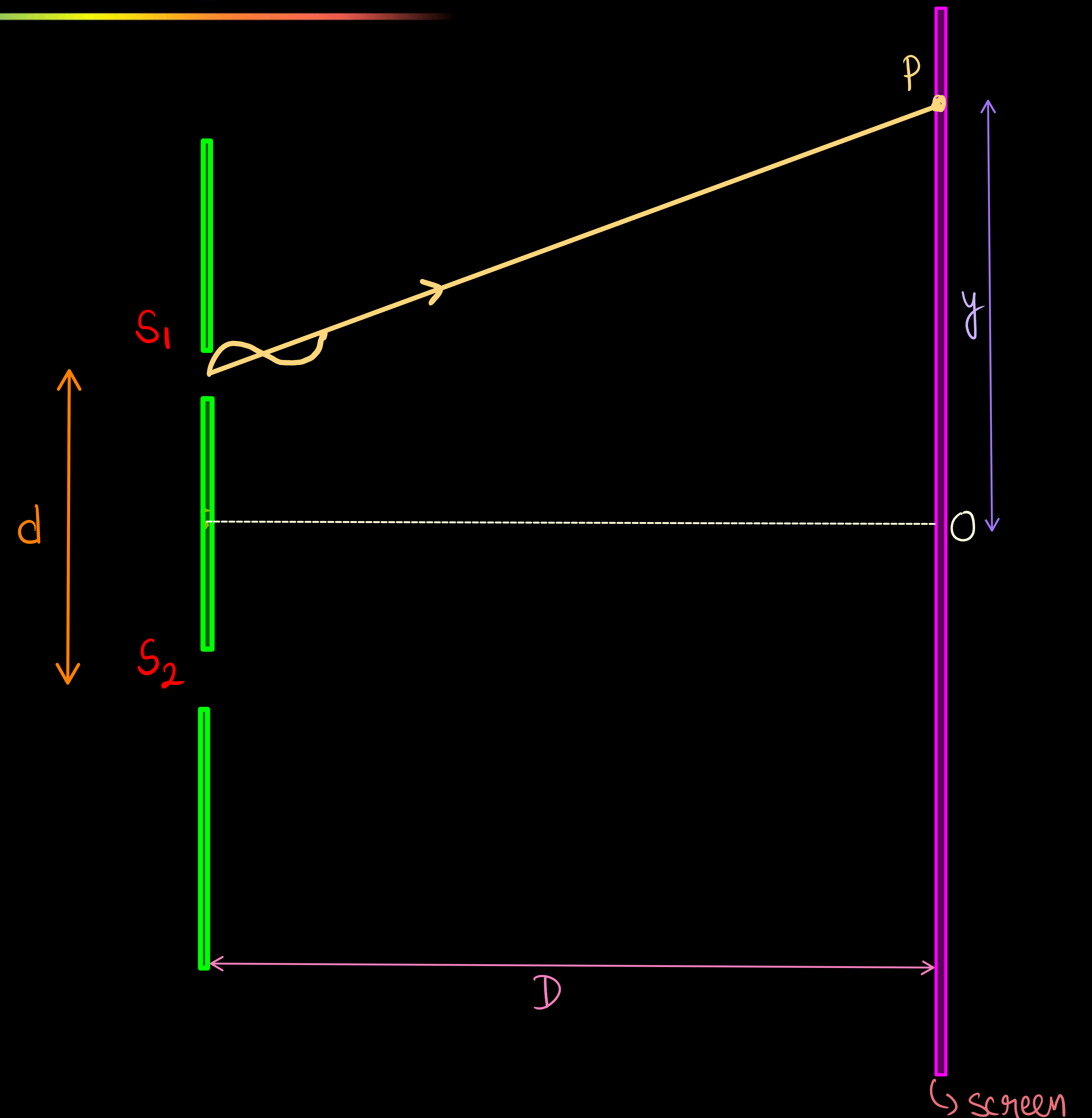
Young's Double Slit Experiment (Derivation Not Required)



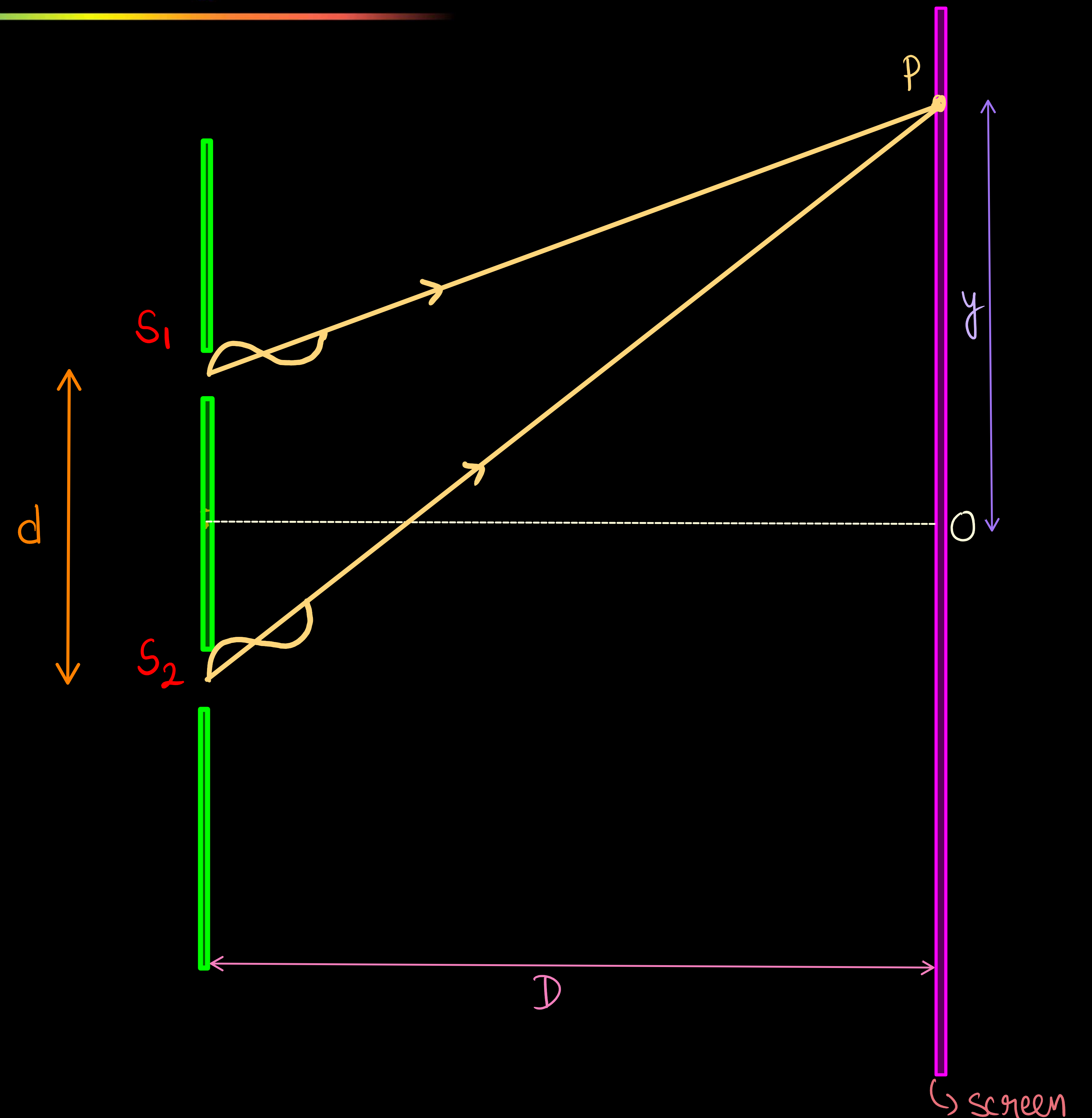
Young's Double Slit Experiment (Derivation Not Required)



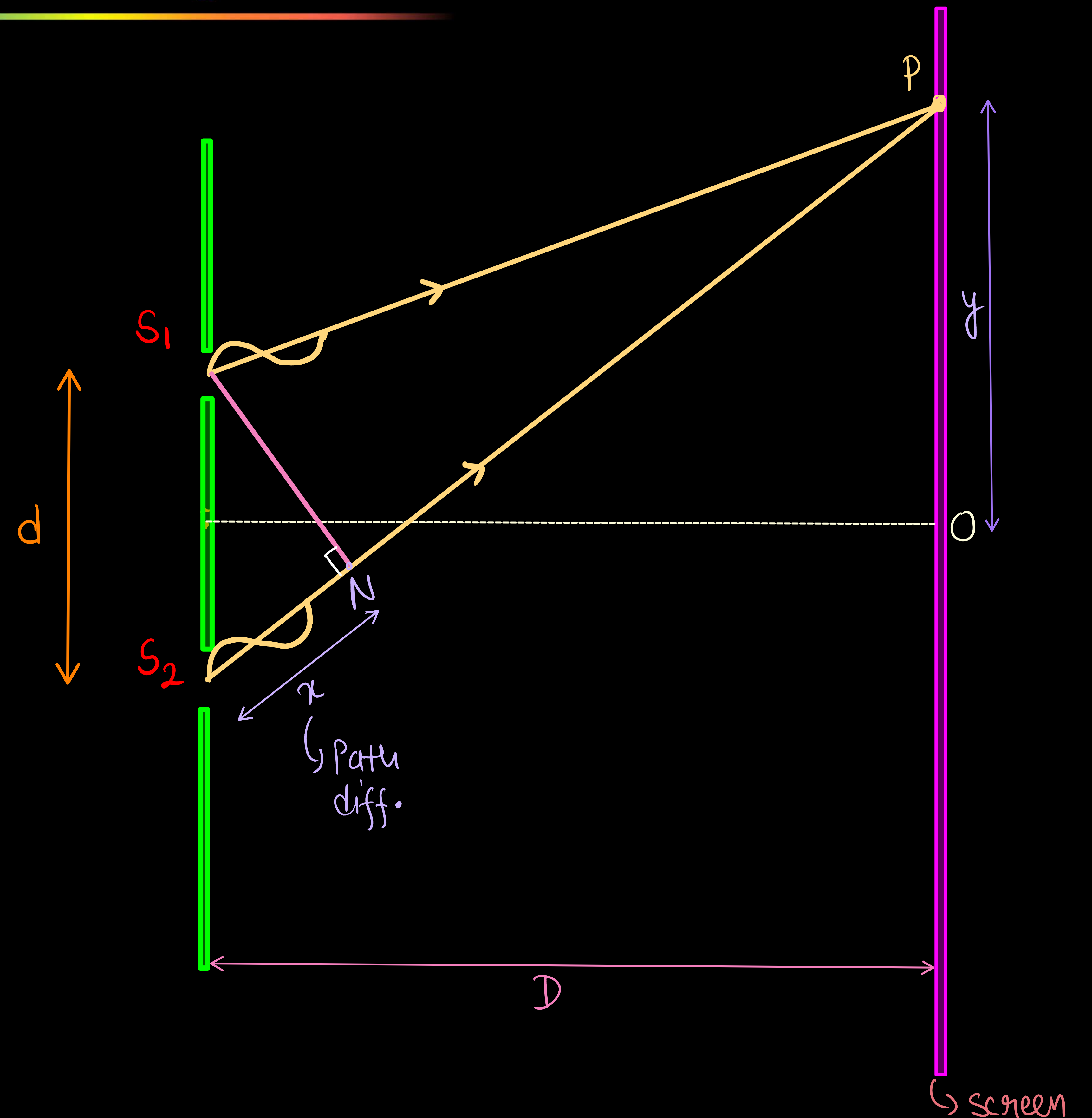
Young's Double Slit Experiment (Derivation Not Required)



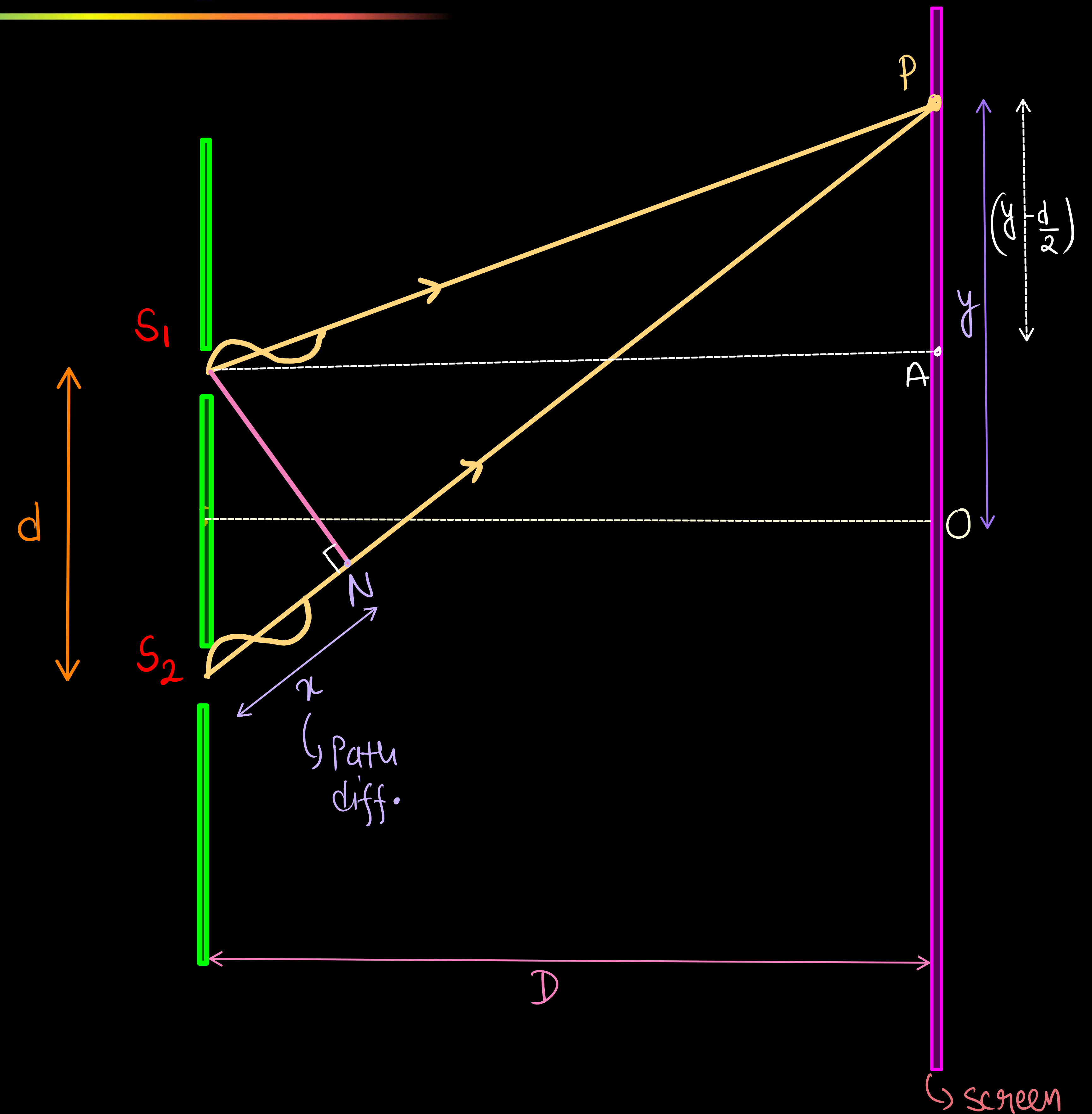
Young's Double Slit Experiment (Derivation Not Required)



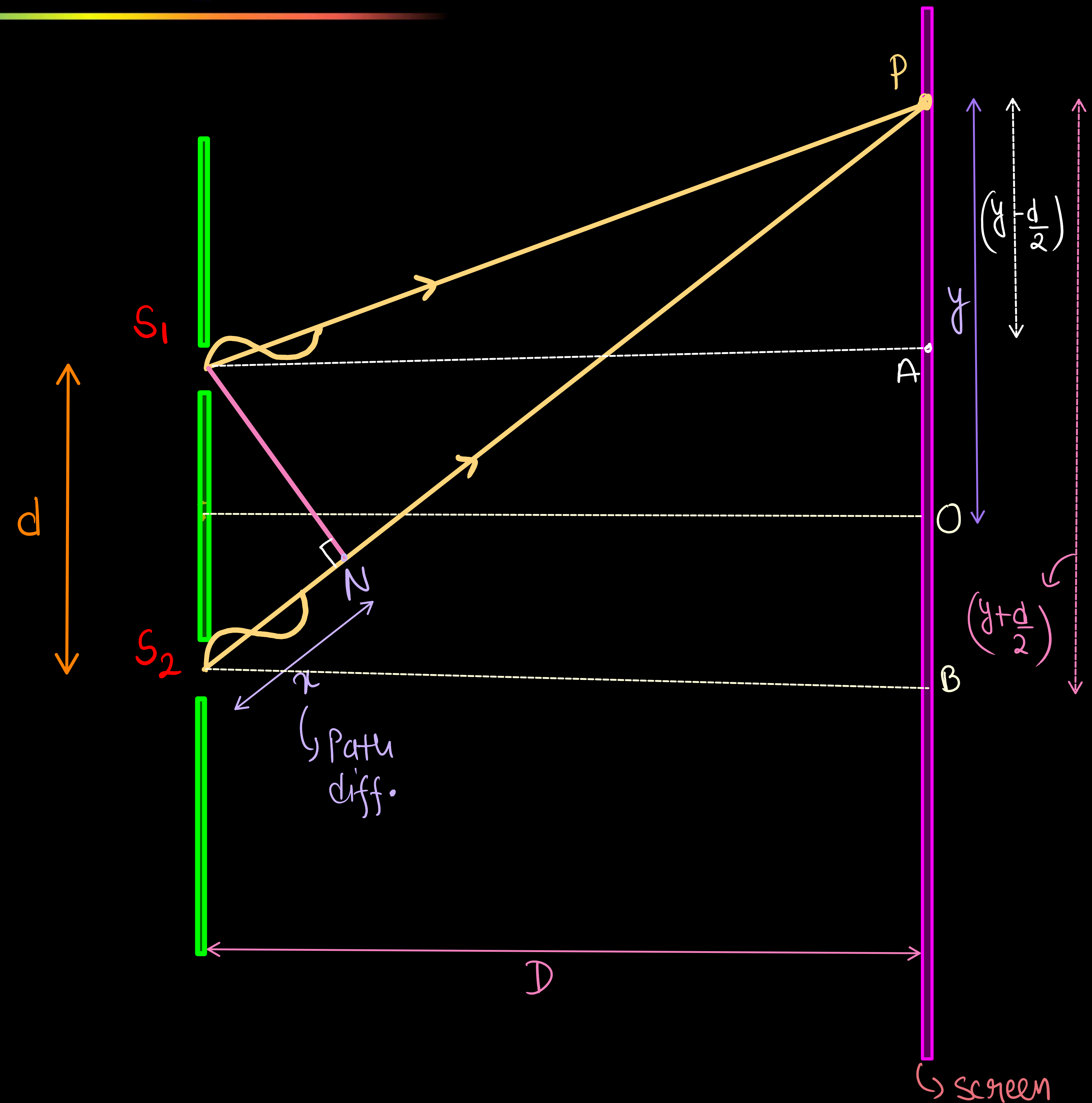
Young's Double Slit Experiment (Derivation Not Required)



Young's Double Slit Experiment (Derivation Not Required)

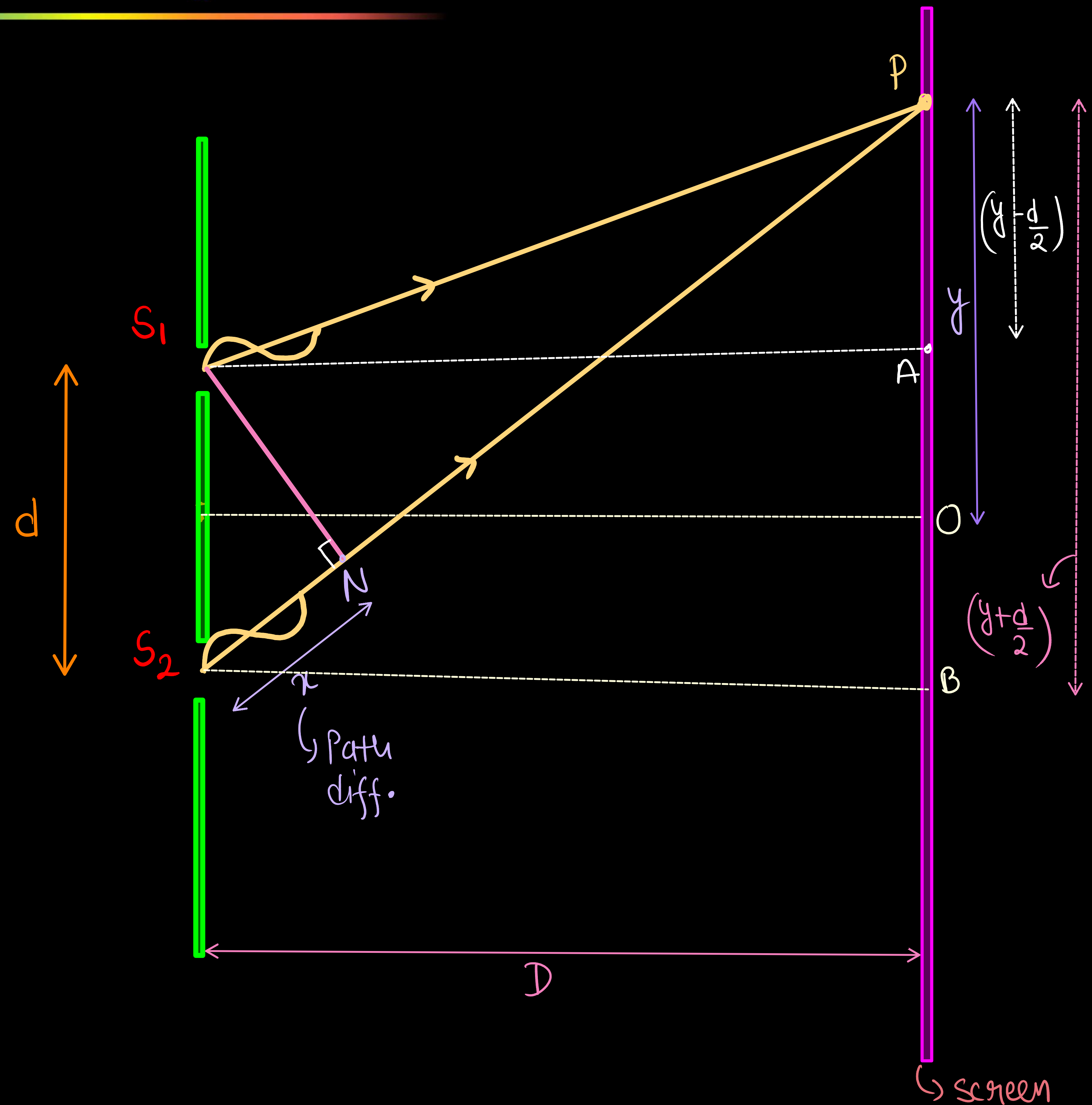


Young's Double Slit Experiment (Derivation Not Required)



Young's Double Slit Experiment (Derivation Not Required)

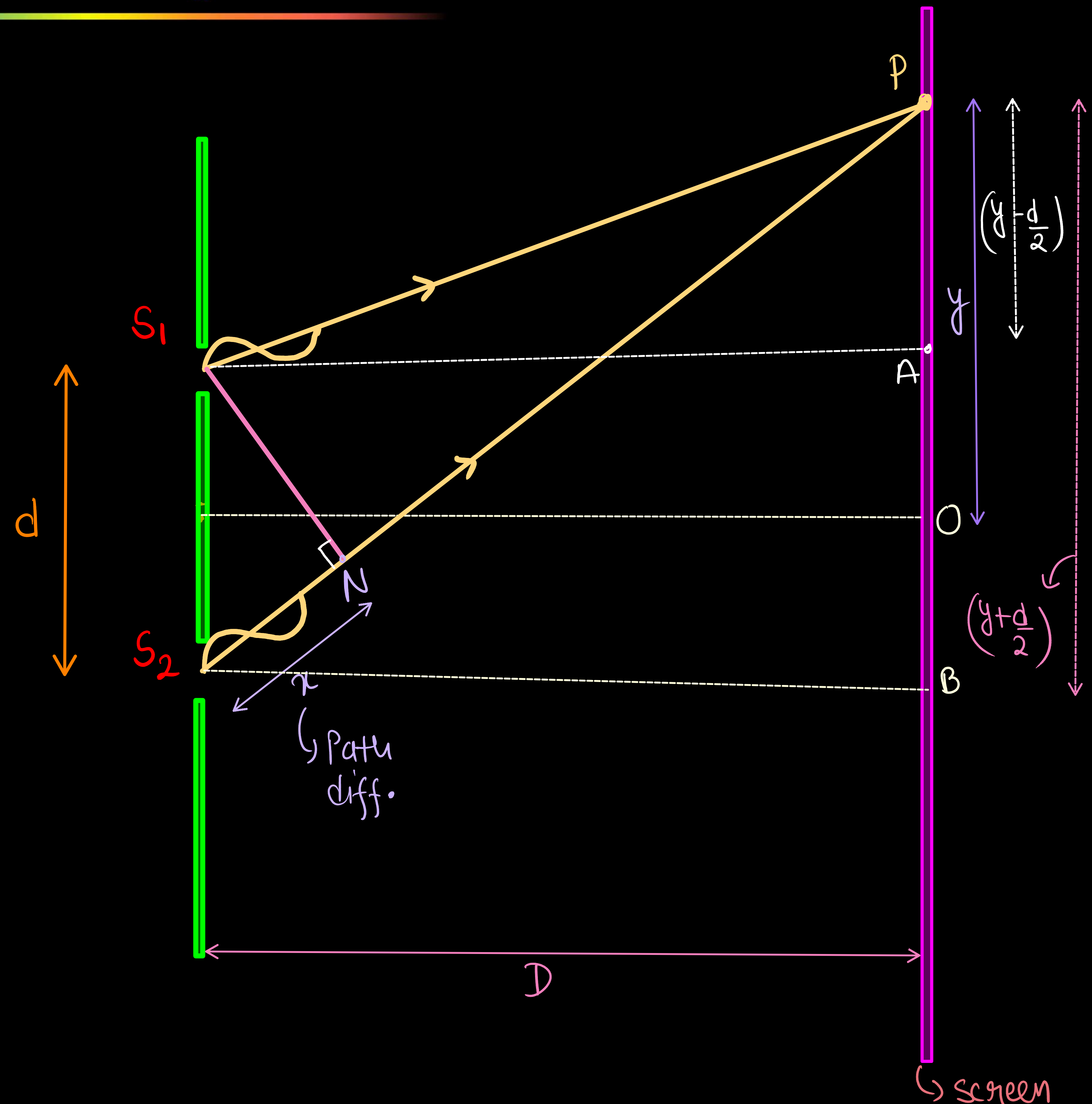
$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- ①}$$



Young's Double Slit Experiment (Derivation Not Required)

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

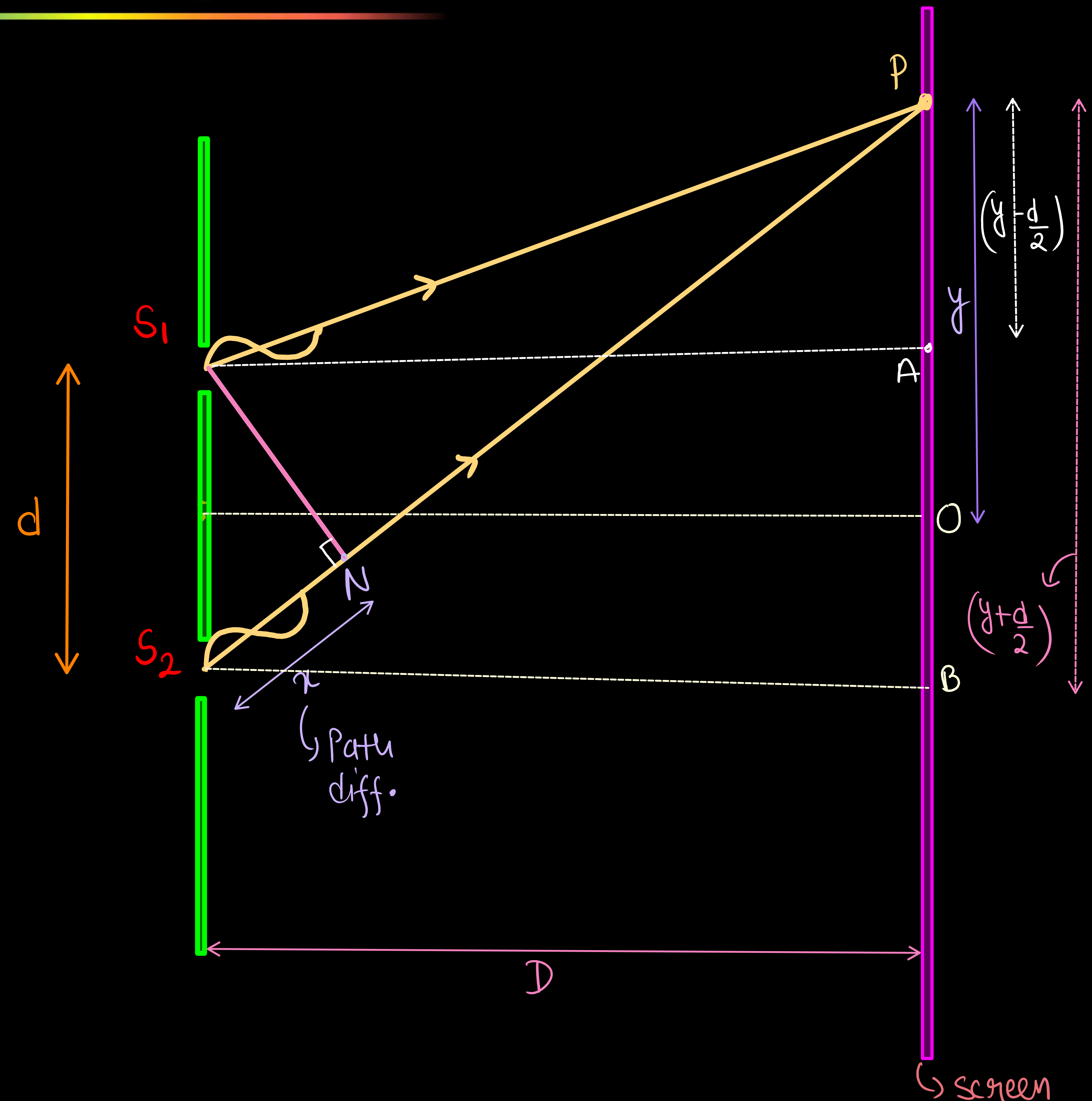
$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$



Young's Double Slit Experiment (Derivation Not Required)

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$

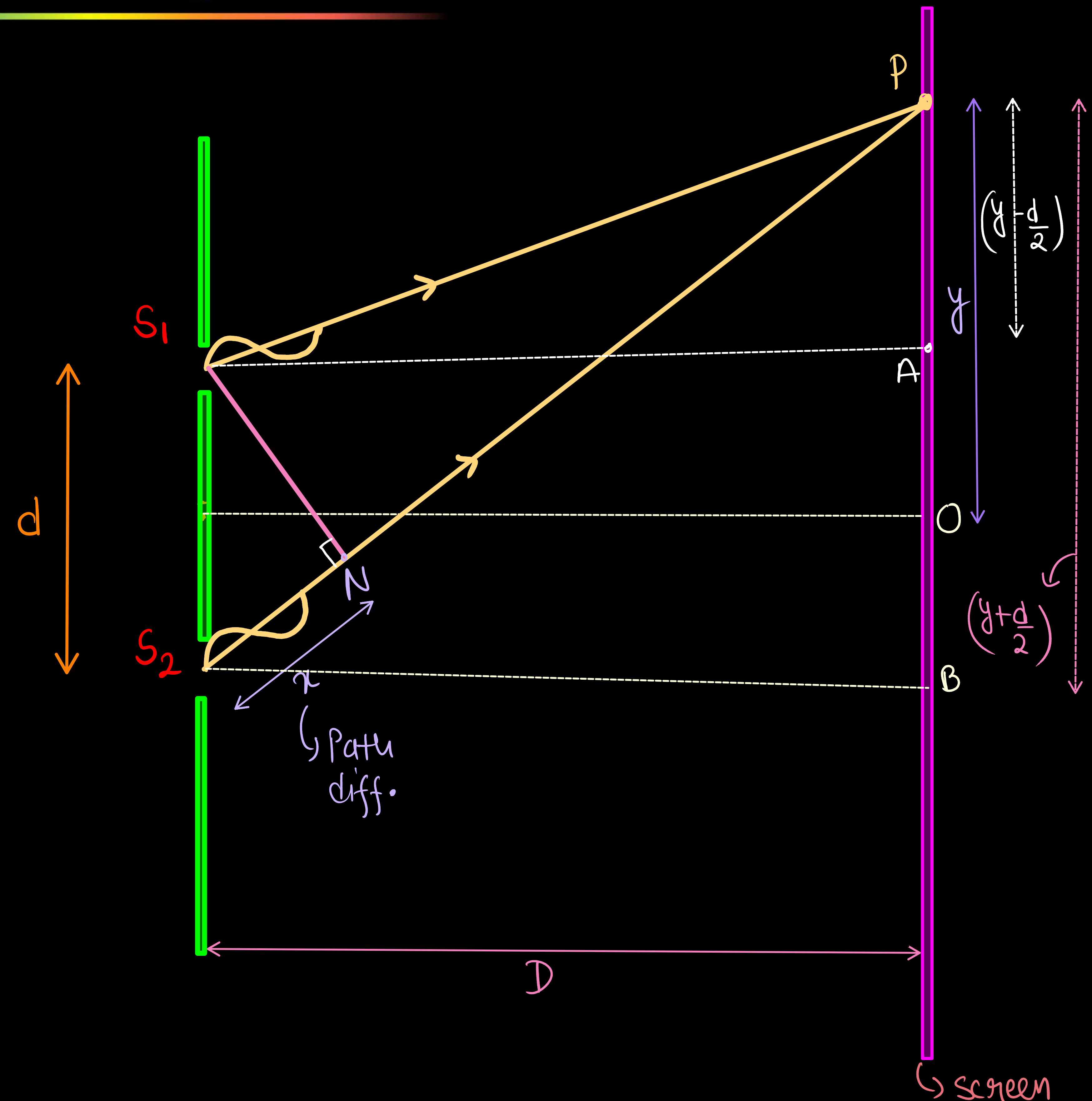


Young's Double Slit Experiment (Derivation Not Required)

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$



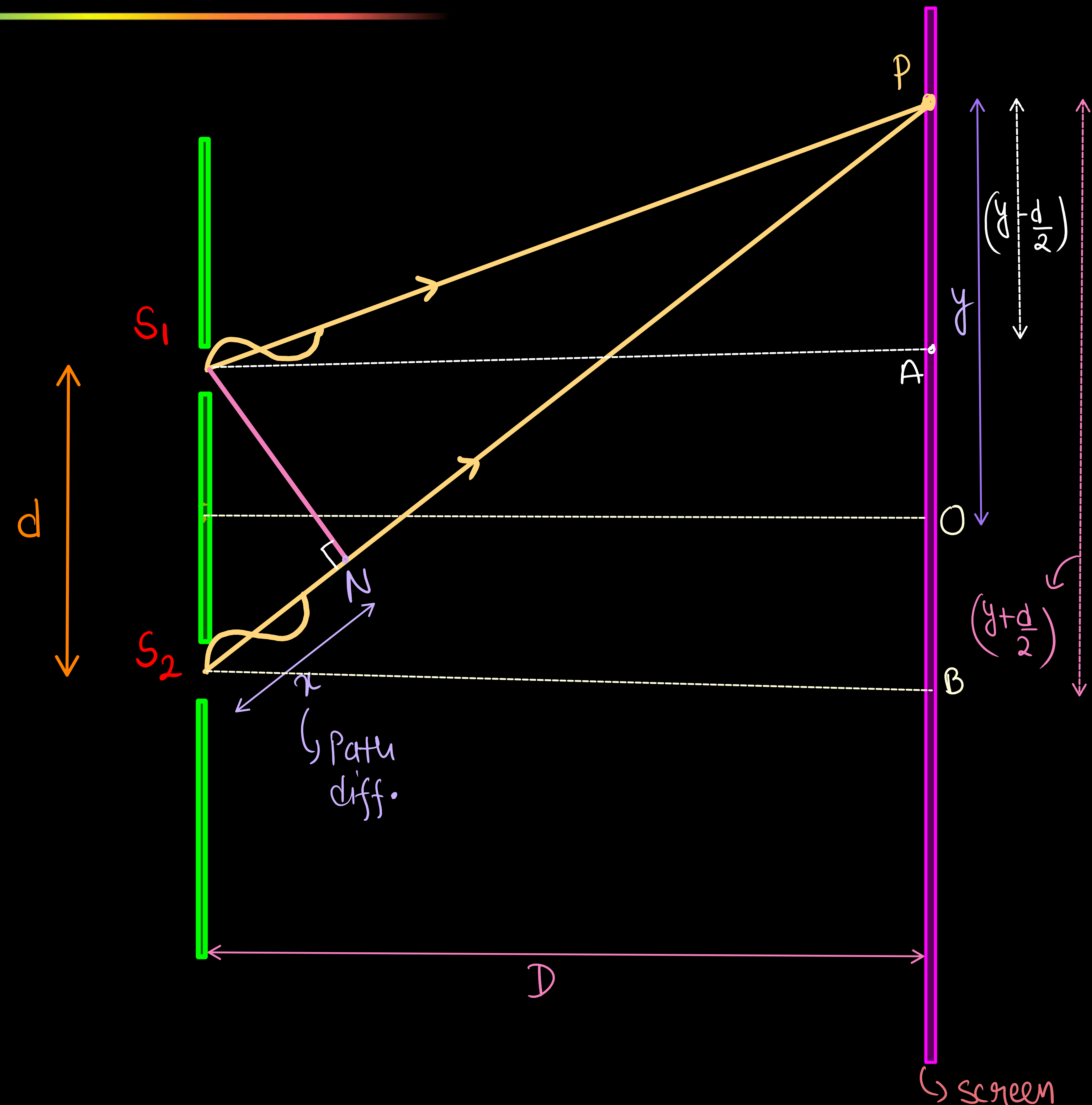
Young's Double Slit Experiment (Derivation Not Required)

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2} + y - \frac{d}{2}\right) \left(y + \frac{d}{2} - y + \frac{d}{2}\right)$$



Young's Double Slit Experiment (Derivation Not Required)

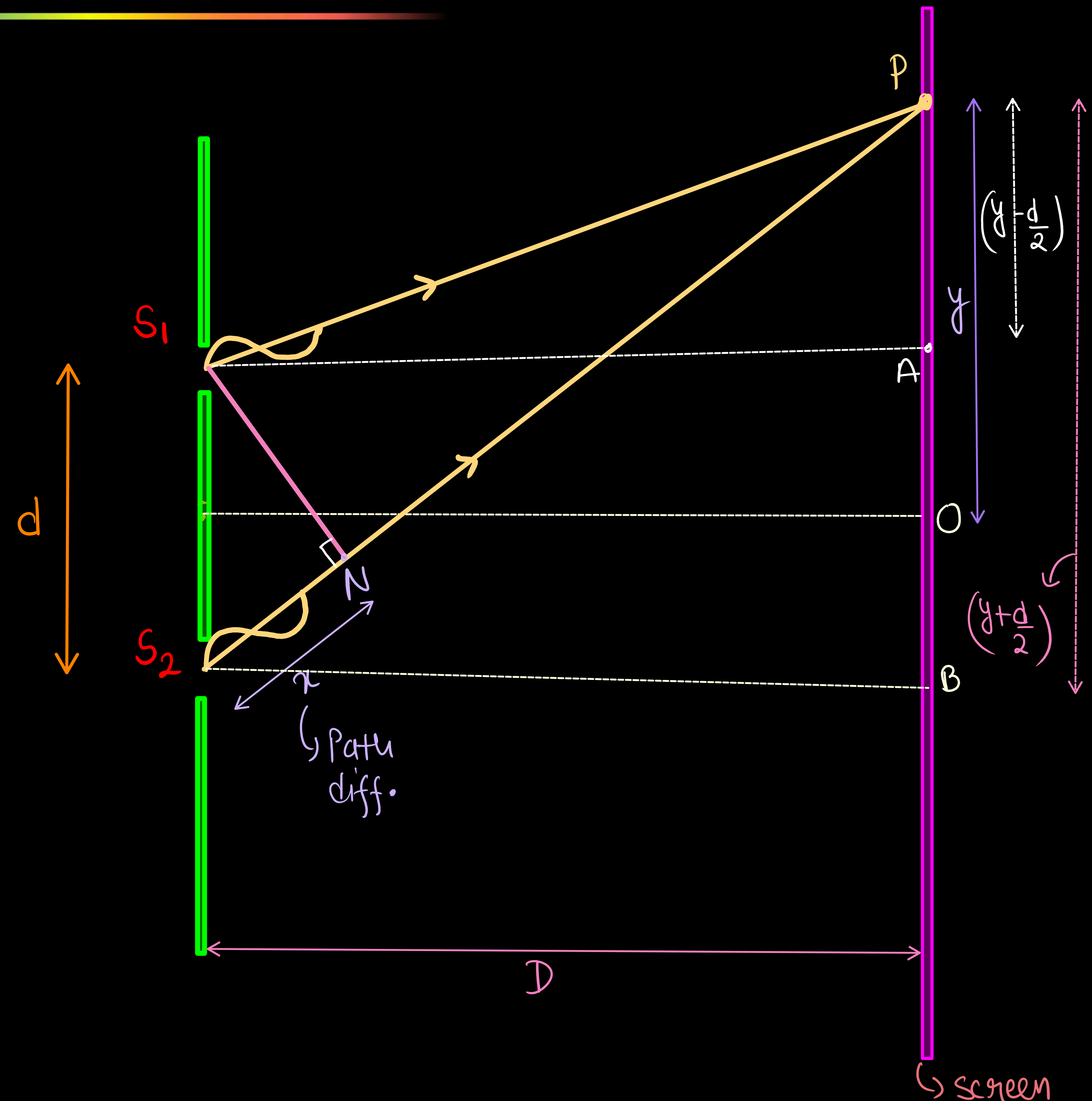
$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2} + y - \frac{d}{2}\right) \left(y + \frac{d}{2} - y + \frac{d}{2}\right)$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd$$



Young's Double Slit Experiment (Derivation Not Required)

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

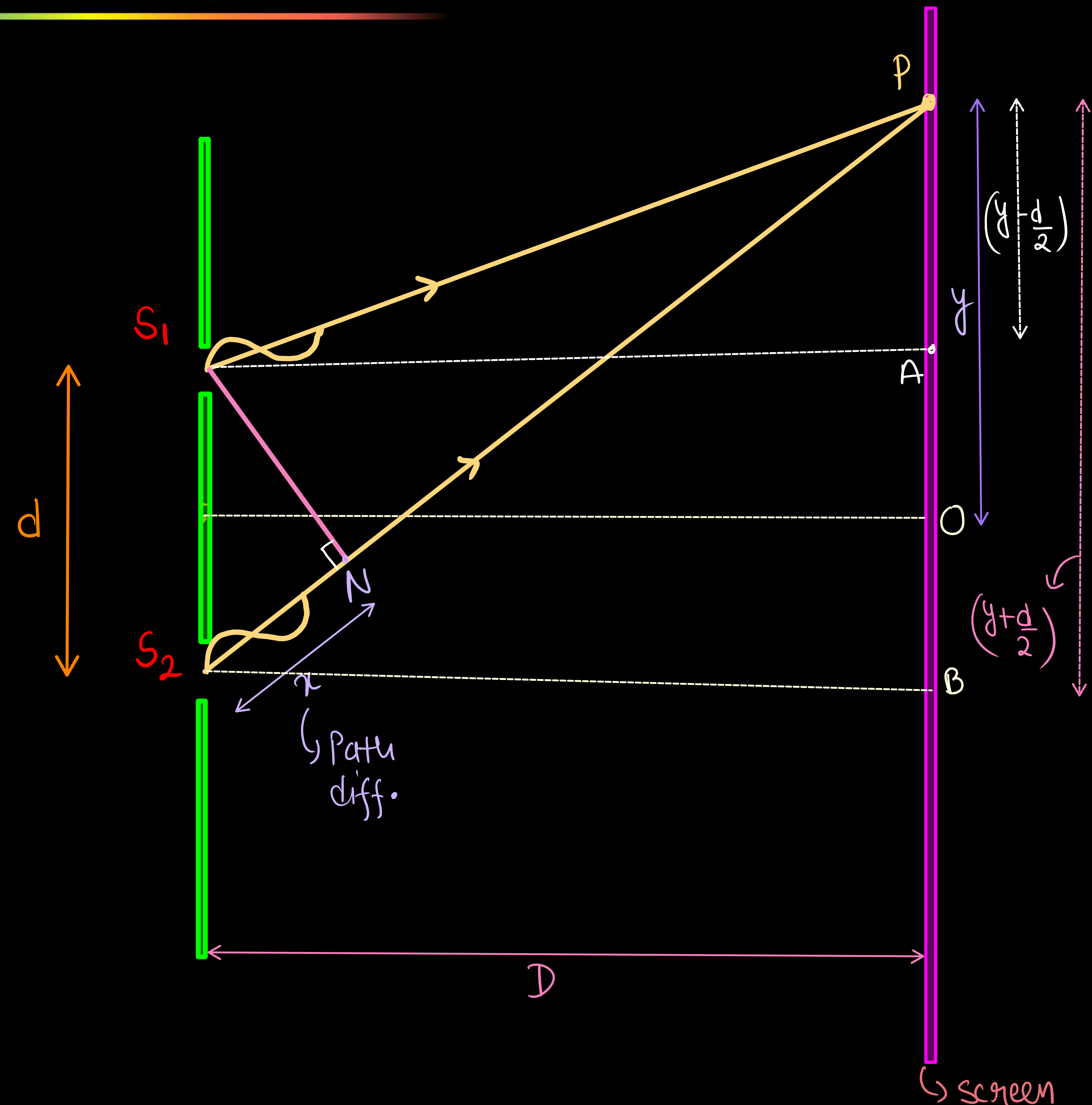
$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2} + y - \frac{d}{2}\right) \left(y + \frac{d}{2} - y + \frac{d}{2}\right)$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd$$

$$\chi = (S_2P - S_1P) = \frac{2yd}{S_2P + S_1P}$$



Young's Double Slit Experiment (Derivation Not Required)

$$S_1 P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad - (1)$$

$$S_2 p^2 = D^2 + \left(y + \frac{d}{2}\right)^2 - (2)$$

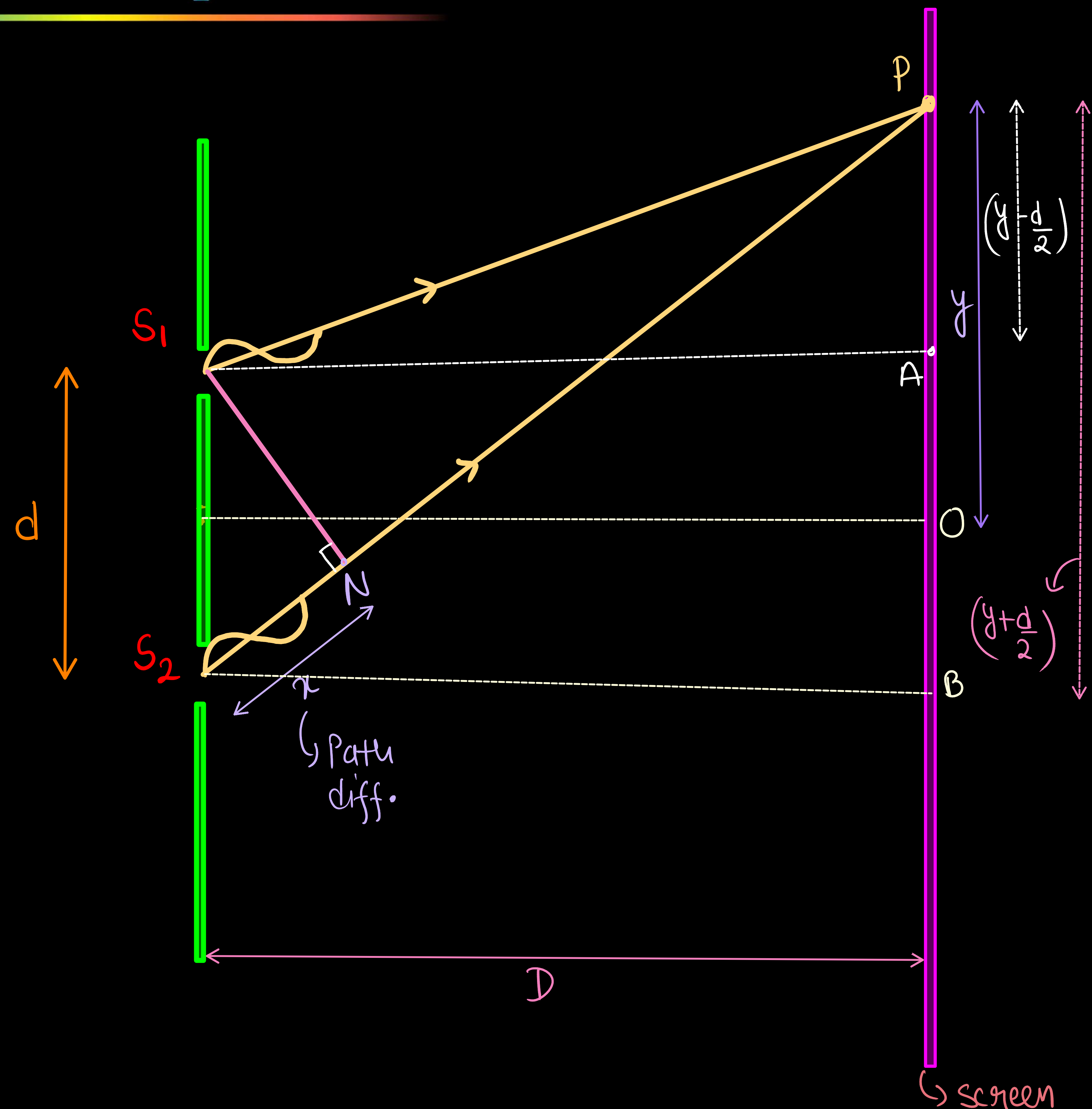
$$S_2 p^2 - S_1 p^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$S_2 p^2 - S_1 p^2 = \left(y + \frac{d}{2} + y - \frac{d}{2}\right) \left(y + \frac{d}{2} - y + \frac{d}{2}\right)$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd$$

$$x = (s_2 p - s_1 p) = \frac{2 y d}{s_2 p + s_1 p}$$

if $d \ll D$, then $S_2 P + S_1 P \approx 2D$



Young's Double Slit Experiment (Derivation Not Required)

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2} + y - \frac{d}{2}\right) \left(y + \frac{d}{2} - y + \frac{d}{2}\right)$$

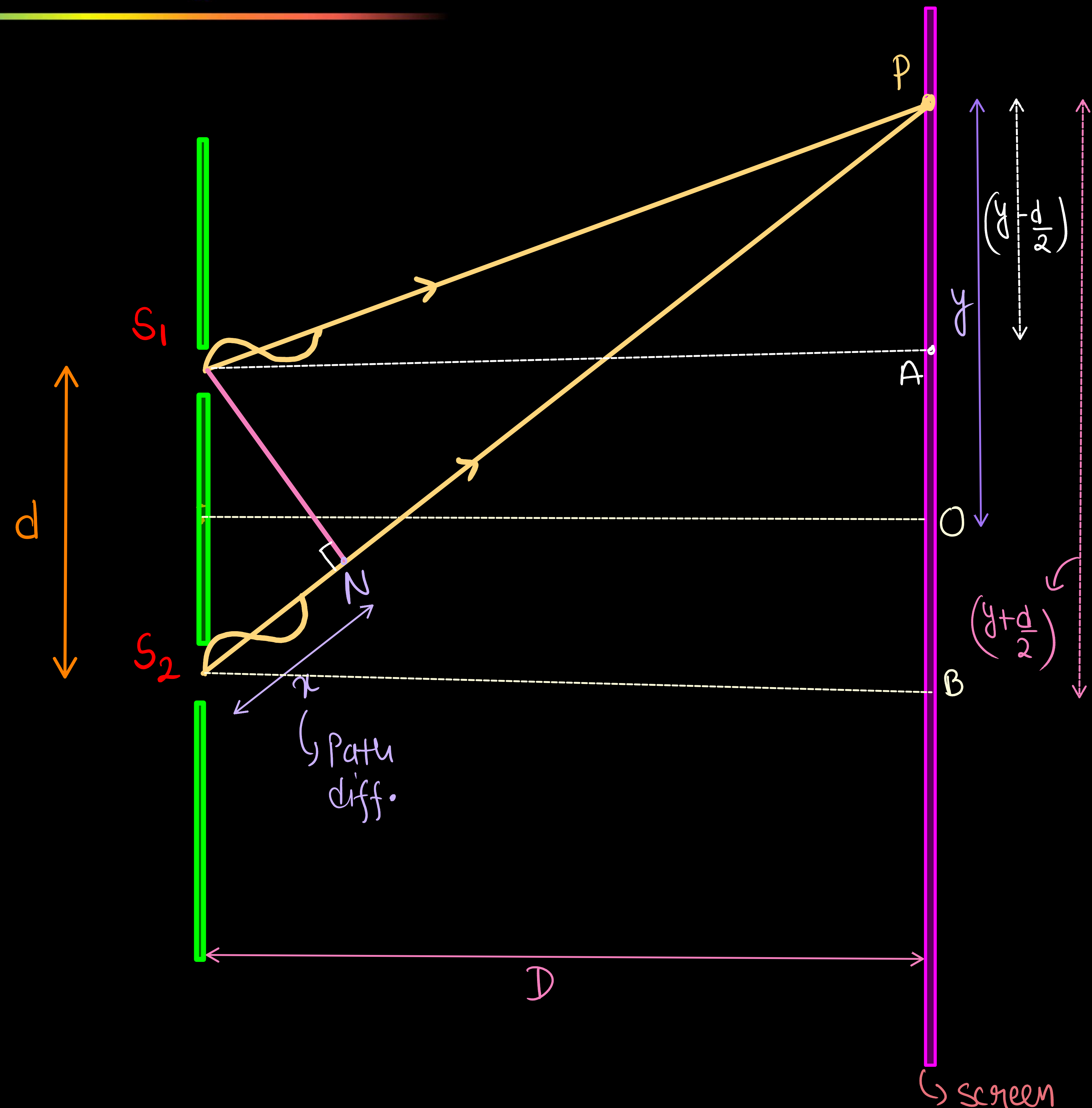
$$(S_2P - S_1P)(S_2P + S_1P) = 2yd$$

$$\downarrow$$

$$\chi = (S_2P - S_1P) = \frac{2yd}{S_2P + S_1P}$$

if $d \ll D$, then $S_2P + S_1P \approx 2D$

$$\chi = \frac{2yd}{2D}$$



Young's Double Slit Experiment (Derivation Not Required)

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2 \quad \text{--- (1)}$$

$$S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2 \quad \text{--- (2)}$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2} + y - \frac{d}{2}\right) \left(y + \frac{d}{2} - y + \frac{d}{2}\right)$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd$$

$$\downarrow$$

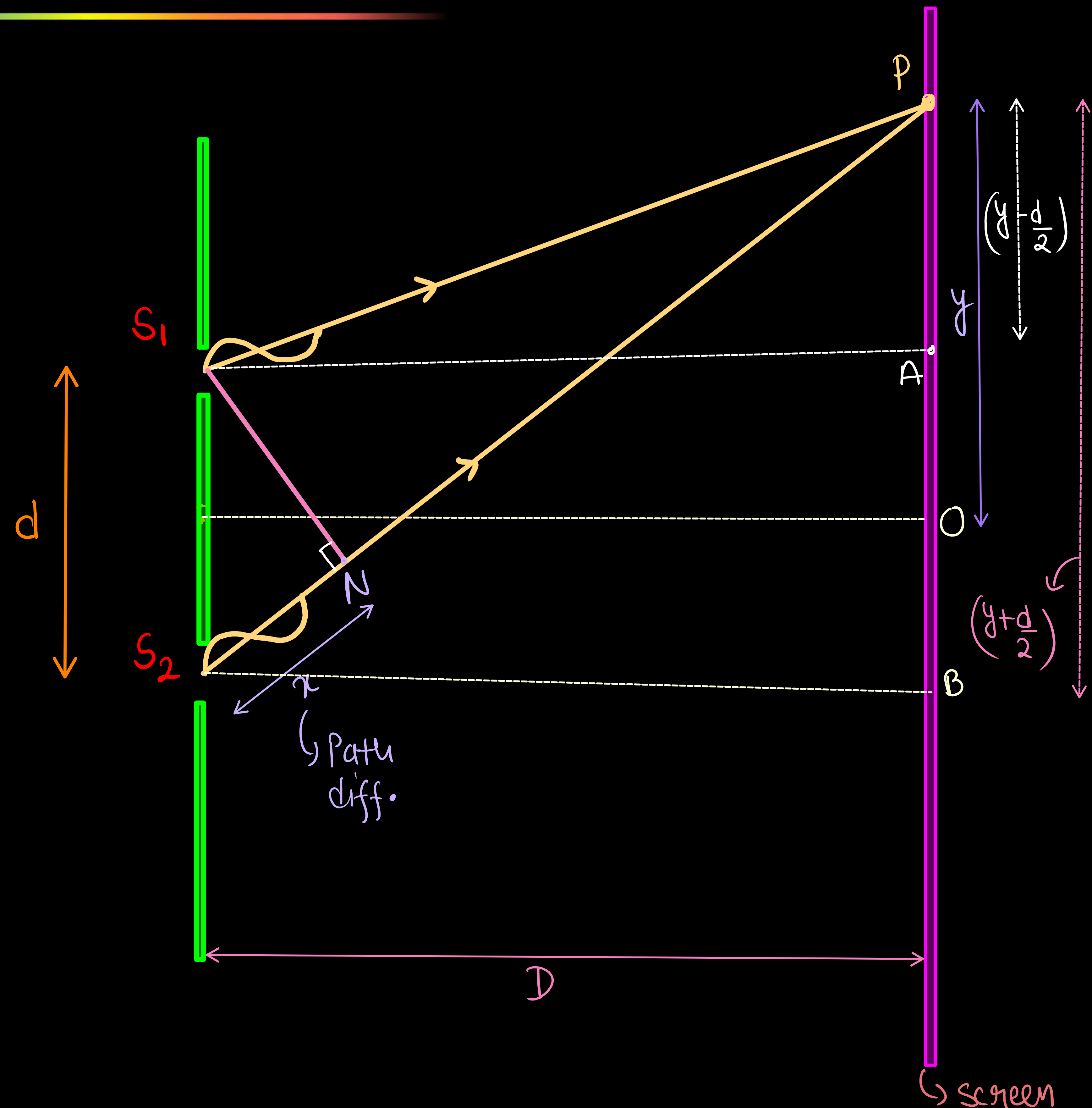
$$\chi = (S_2P - S_1P) = \frac{2yd}{S_2P + S_1P}$$

if $d \ll D$, then $S_2P + S_1P \approx 2D$

$$\chi = \frac{2yd}{2D}$$

→ Path difference.

$$\chi = \frac{yd}{D} \quad \text{--- (3)}$$

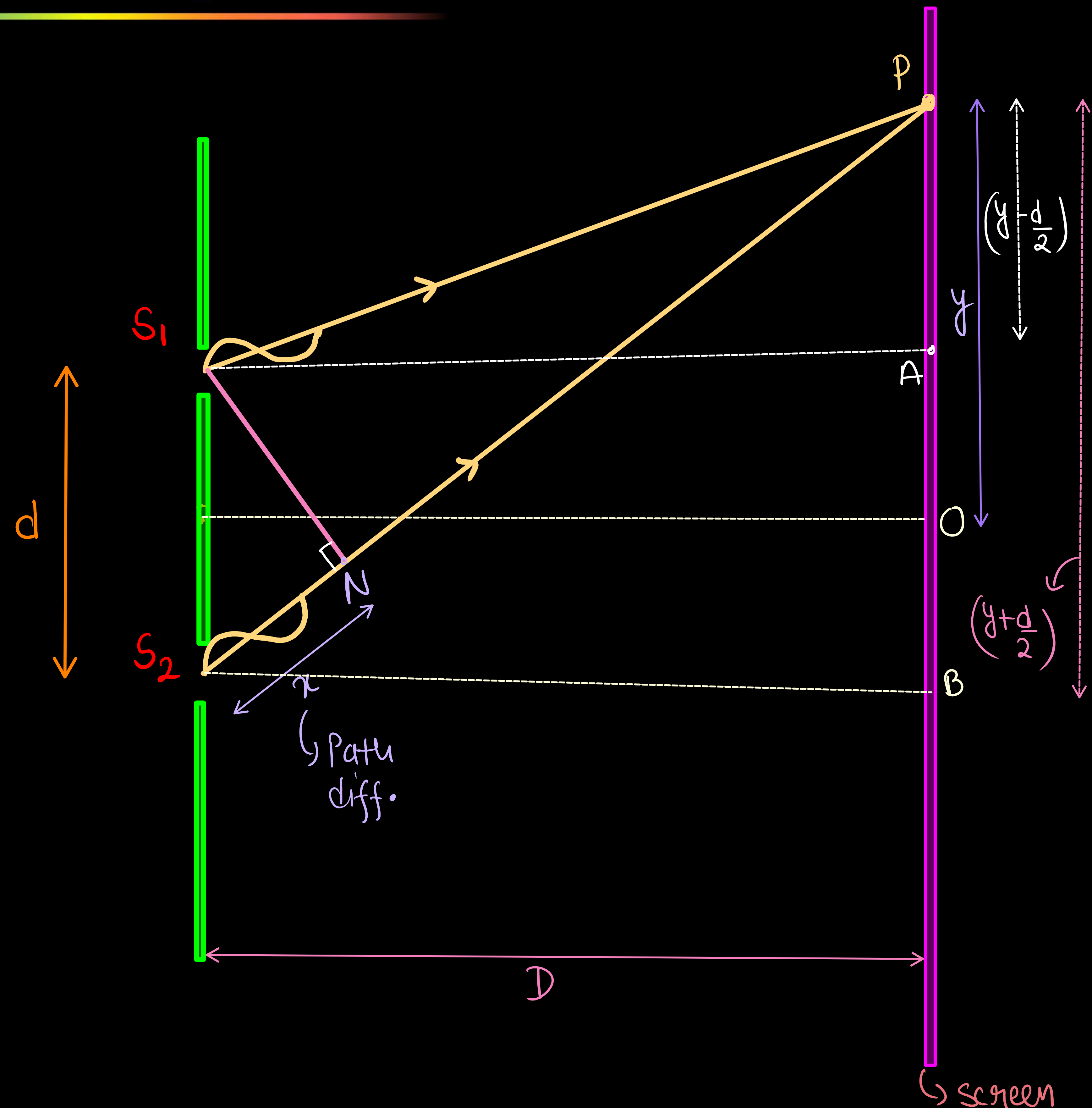


Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \text{Path difference}$$

* Position of maxima and minima on the screen -

The point P will be bright fringe (maxima) if -
 $x = n\lambda$



Young's Double Slit Experiment

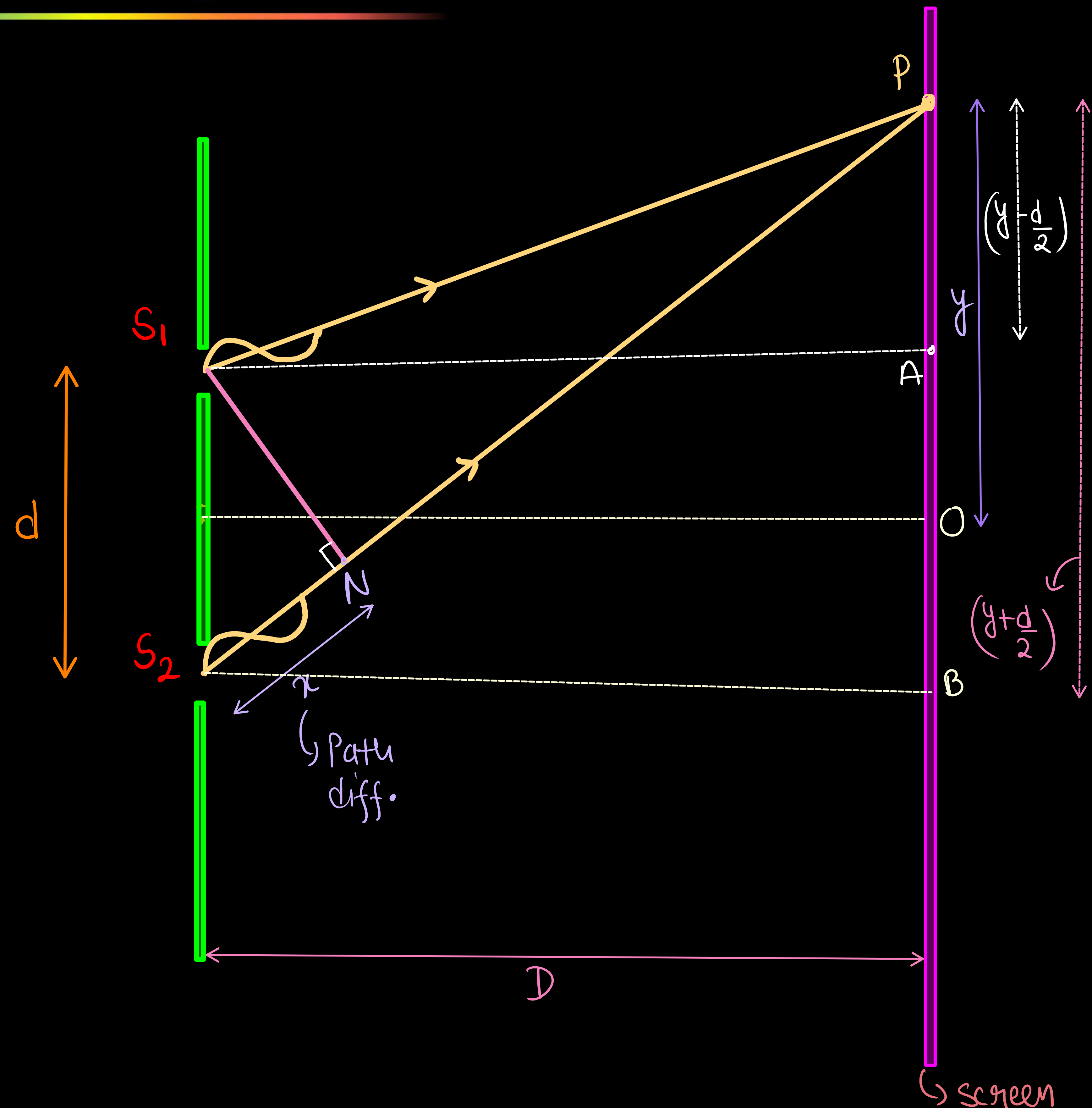
$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \rightarrow \text{Path difference}$$

* Position of maxima and minima on the screen -

The point P will be bright fringe (maxima) if -

$$x = n\lambda$$

$$\frac{y d}{D} = n\lambda$$



Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \rightarrow \text{Path difference.}$$

* Position of maxima and minima on the screen -

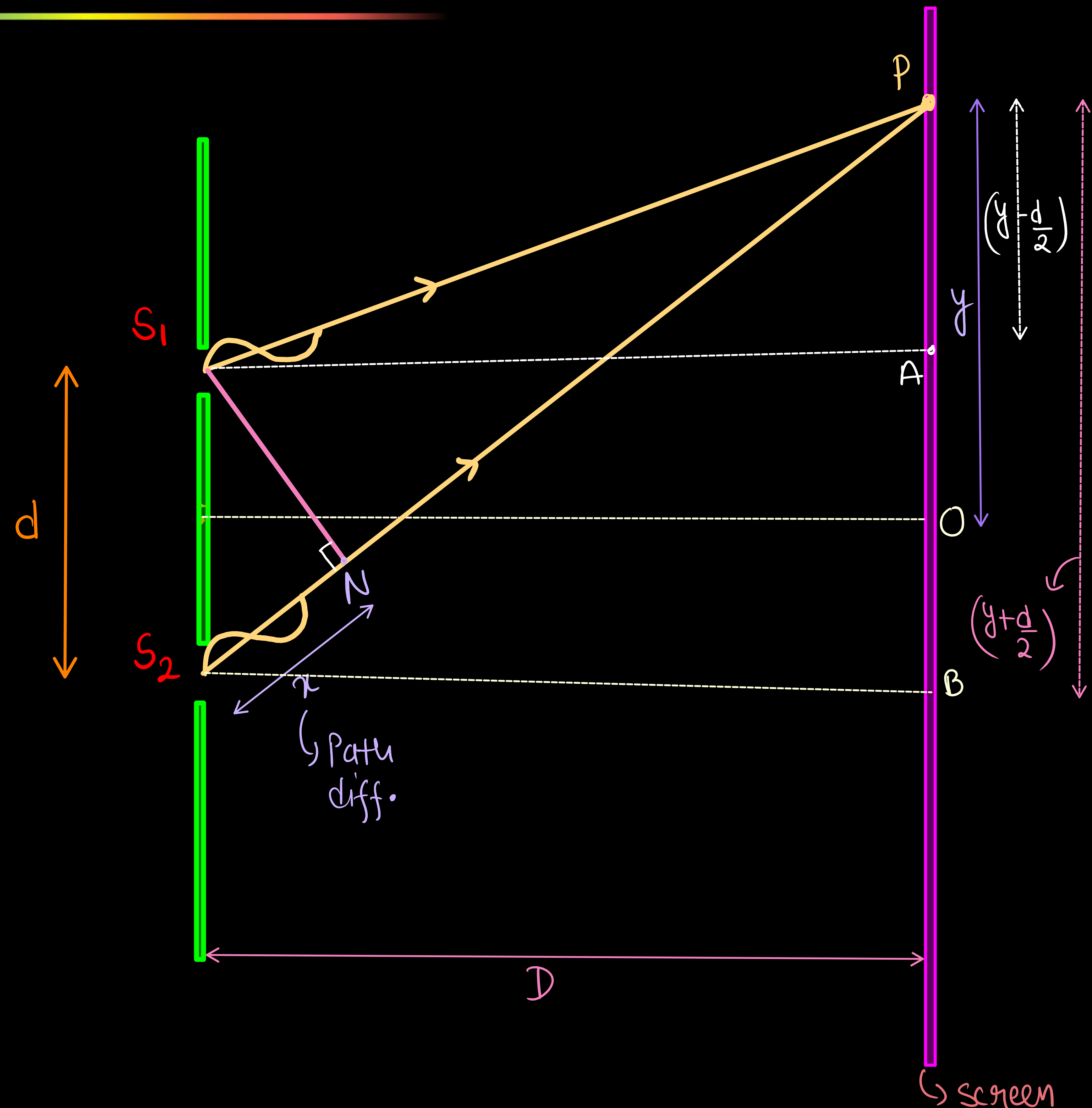
The point P will be bright fringe (maxima) if -

$$x = n\lambda$$

$$\frac{y d}{D} = n\lambda$$

$$\boxed{y_n = \frac{n\lambda D}{d}}$$

Distance of n^{th} Bright fringe from O.
($n = 0, 1, 2, 3, \dots$)



Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \rightarrow \text{Path difference}$$

* Position of maxima and minima on the screen -

The point P will be bright fringe (maxima) if -

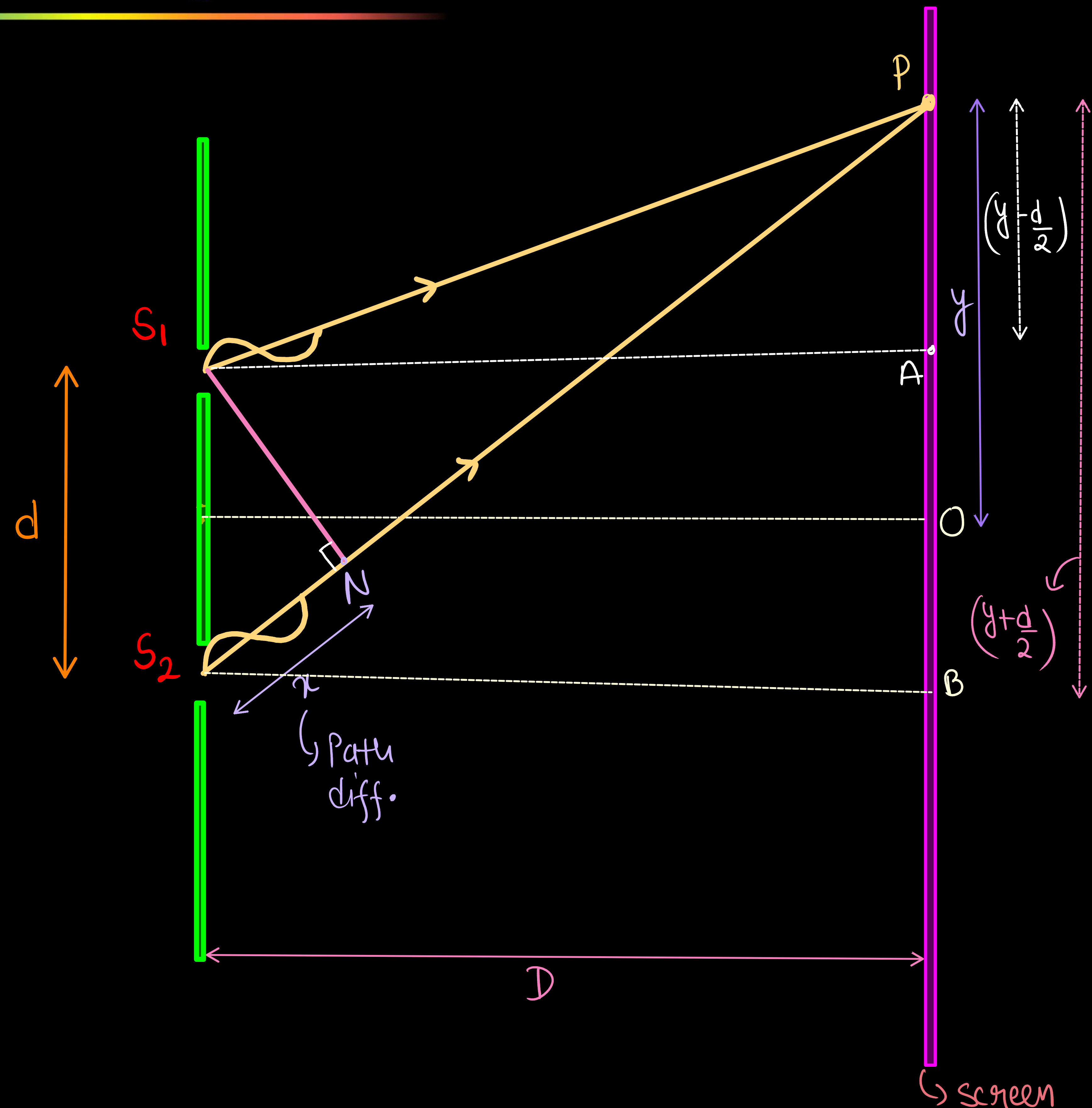
$$x = n\lambda$$

$$\frac{y d}{D} = n\lambda$$

$$\boxed{y_n = \frac{n\lambda D}{d}}$$

Distance of n^{th} Bright fringe from O.
($n = 0, 1, 2, 3, \dots$)

for $n = 0$, $y = 0 \Rightarrow$ central bright fringe



Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \text{Path difference}$$

* Position of maxima and minima on the screen -

The point P will be bright fringe (maxima) if -

$$x = n\lambda$$

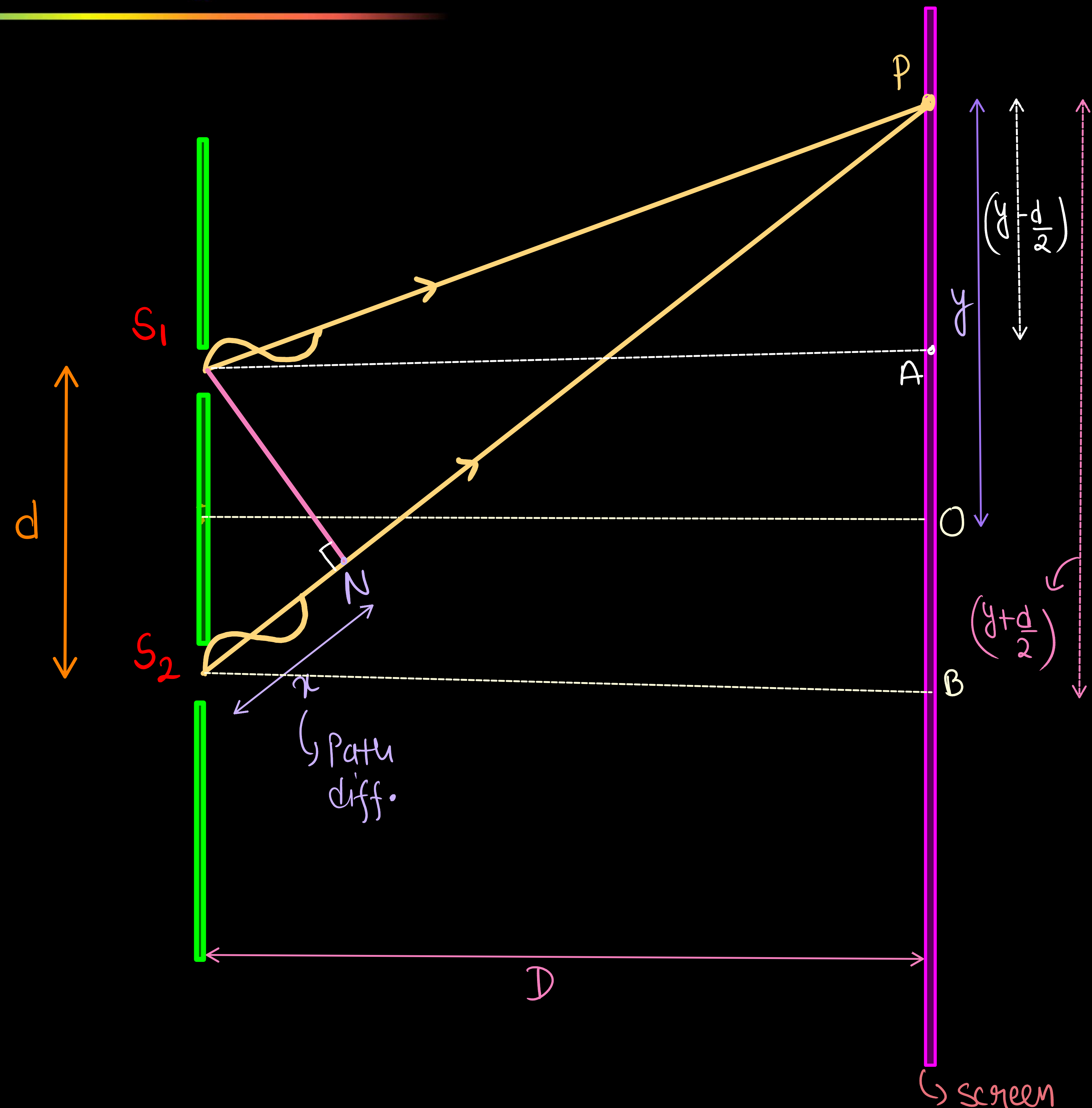
$$\frac{y d}{D} = n\lambda$$

$$\boxed{y_n = \frac{n\lambda D}{d}}$$

Distance of n^{th} Bright fringe from O.
($n = 0, 1, 2, 3, \dots$)

for $n = 0$, $y = 0 \Rightarrow$ central bright fringe

$n = 1$, $y = \frac{\lambda D}{d} \Rightarrow$ Distance of 1st bright fringe



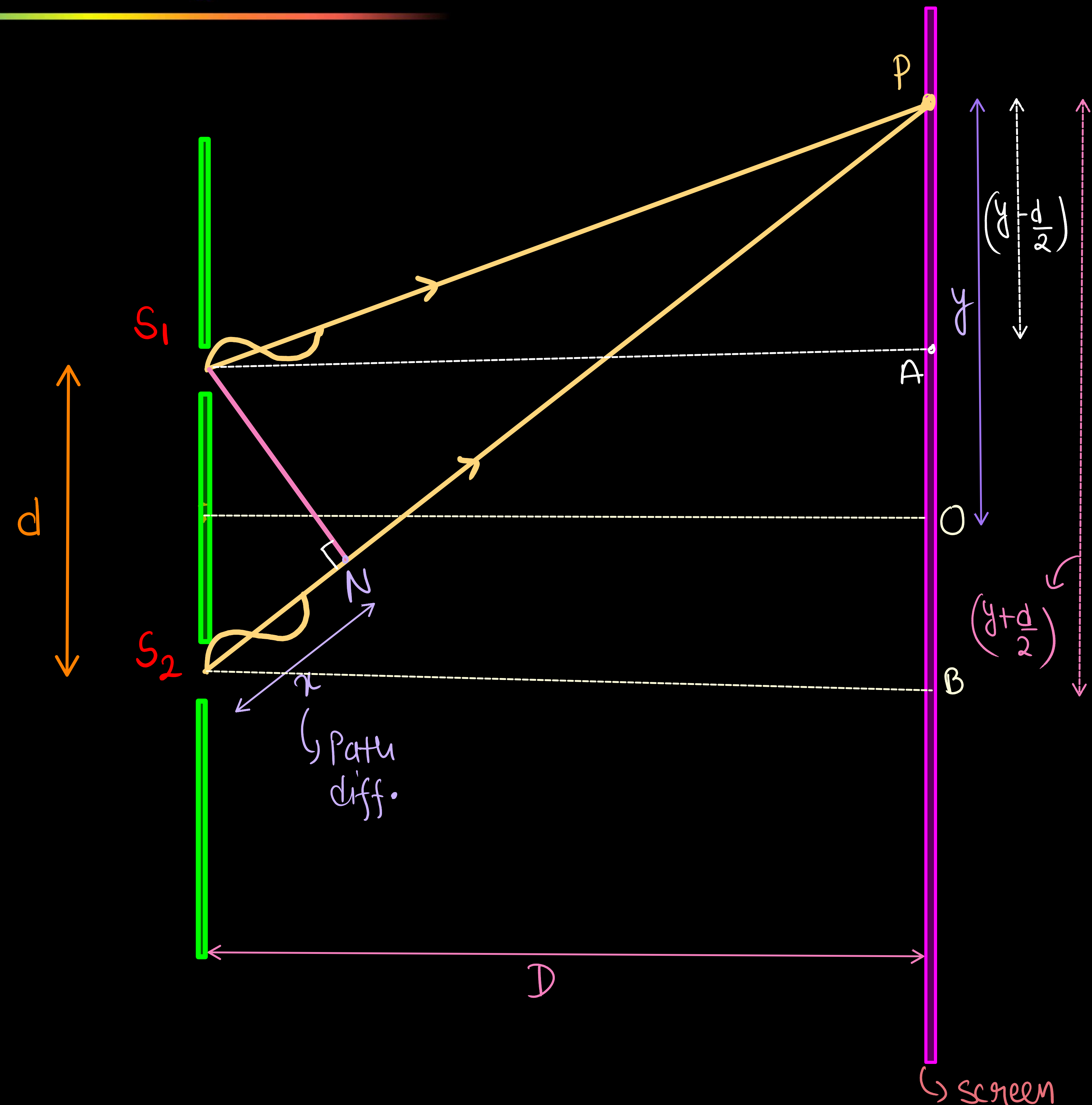
Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \text{Path difference}$$

* Position of maxima and minima on the screen -

The point P will be dark fringe (minima) if -

$$x = (2n-1) \frac{\lambda}{2} \quad (n = 1, 2, 3, \dots)$$



Young's Double Slit Experiment

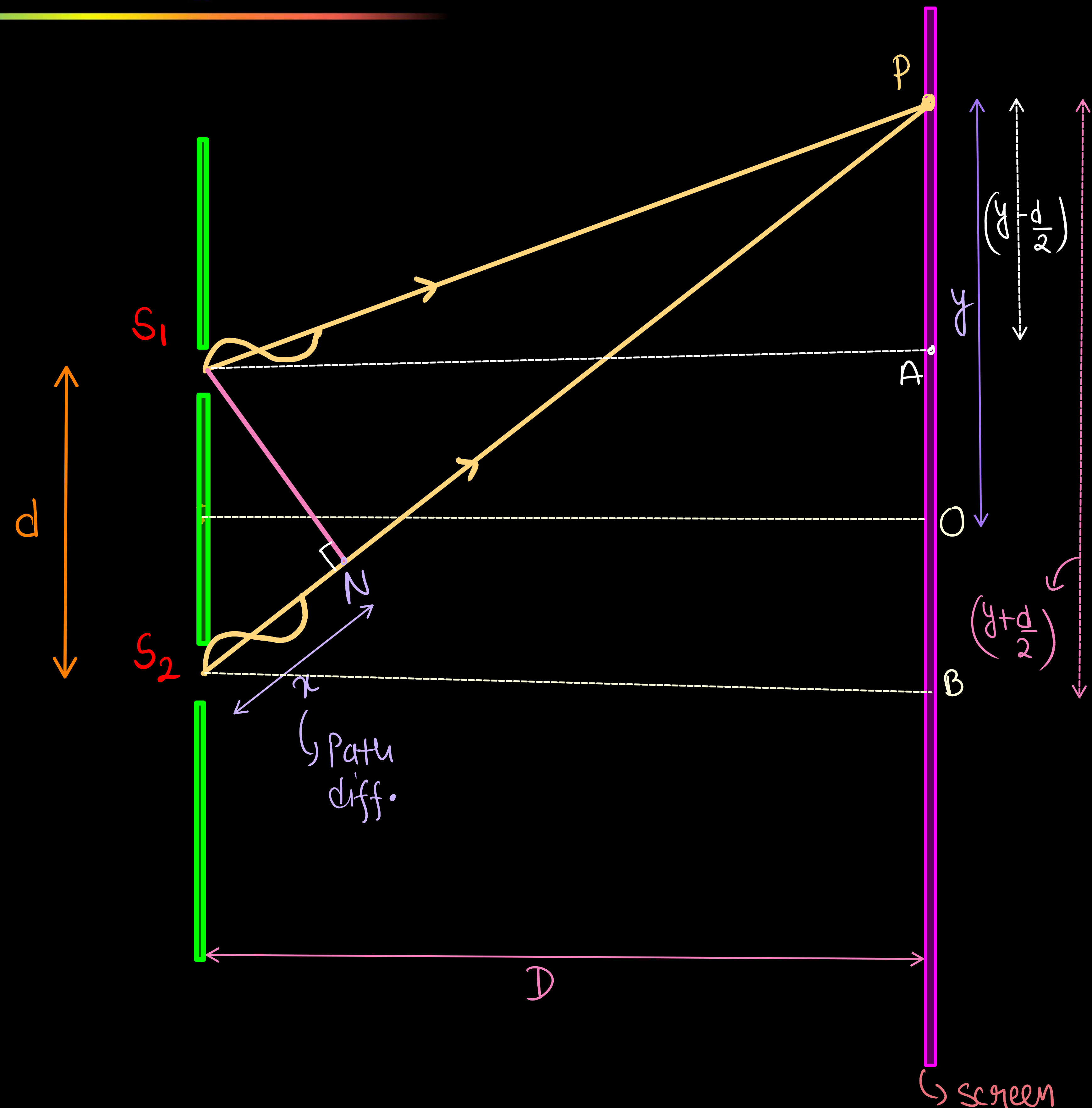
$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \rightarrow \text{Path difference.}$$

* Position of maxima and minima on the screen -

The point P will be dark fringe (minima) if -

$$x = (2n-1) \frac{\lambda}{2} \quad (n=1, 2, 3, \dots)$$

$$\frac{y d}{D} = (2n-1) \frac{\lambda}{2}$$



Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \text{Path difference}$$

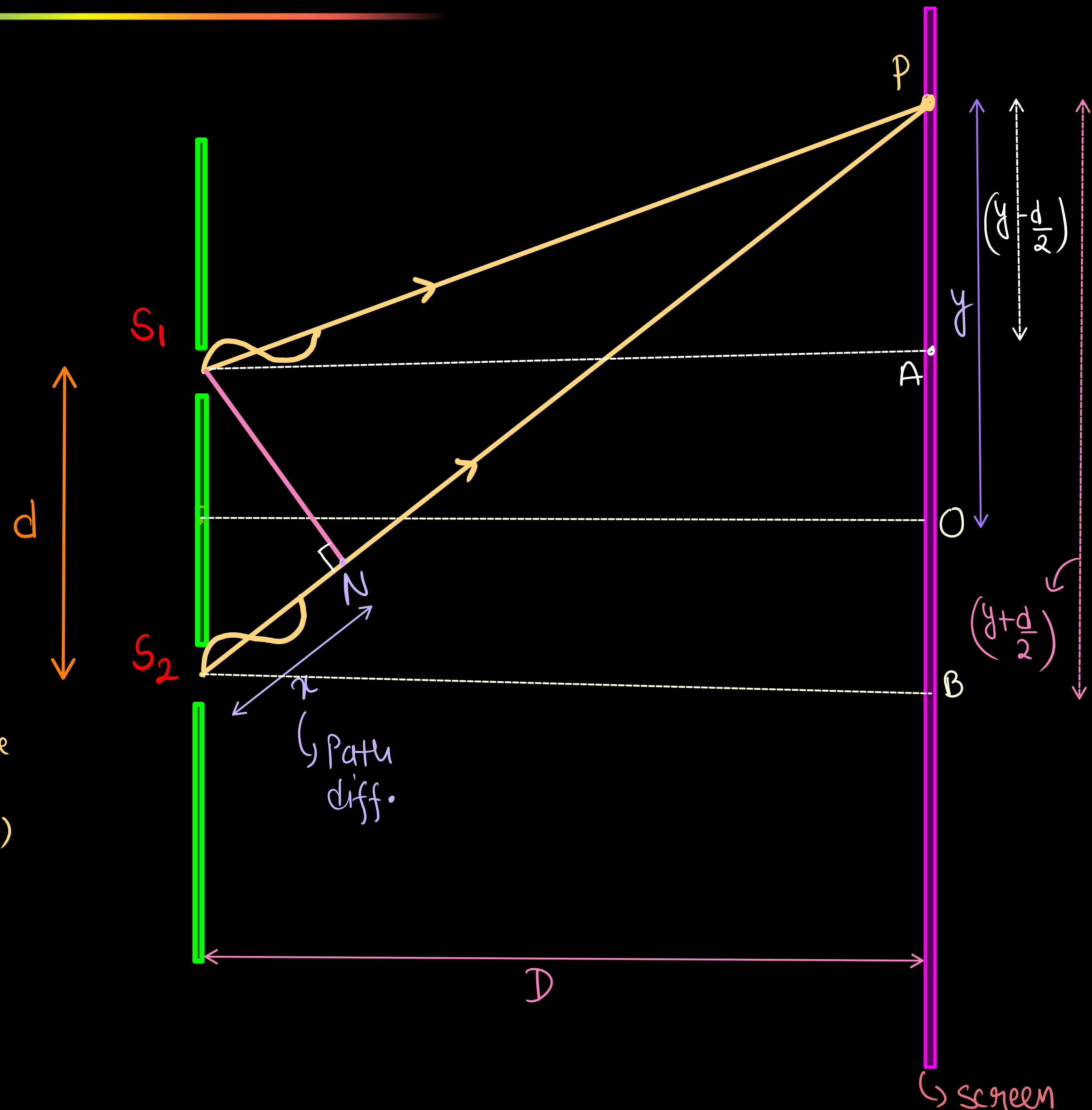
* Position of maxima and minima on the screen -

The point P will be dark fringe (minima) if -

$$x = (2n-1) \frac{\lambda}{2} \quad (n=1, 2, 3, \dots)$$

$$\frac{y d}{D} = (2n-1) \frac{\lambda}{2}$$

$$\boxed{y_n = \frac{(2n-1) \lambda D}{2d}} \quad \text{⑤} \quad \text{Distance of } n^{\text{th}} \text{ Dark fringe from O.} \\ (n=1, 2, 3, \dots)$$



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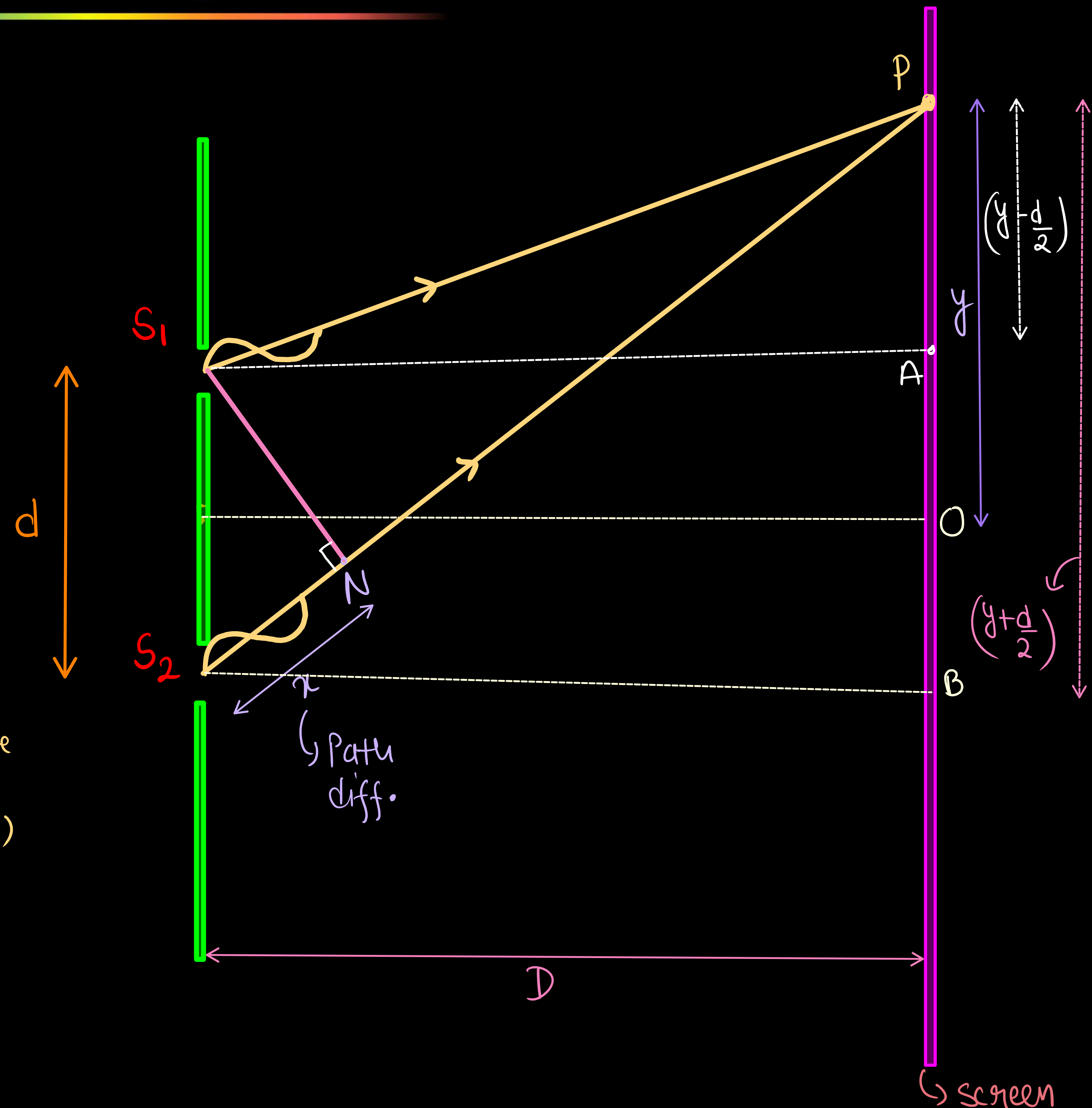
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Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{--- (3) --- Path difference.}$$

* Position of maxima and minima on the screen -

The point P will be dark fringe (minima) if -

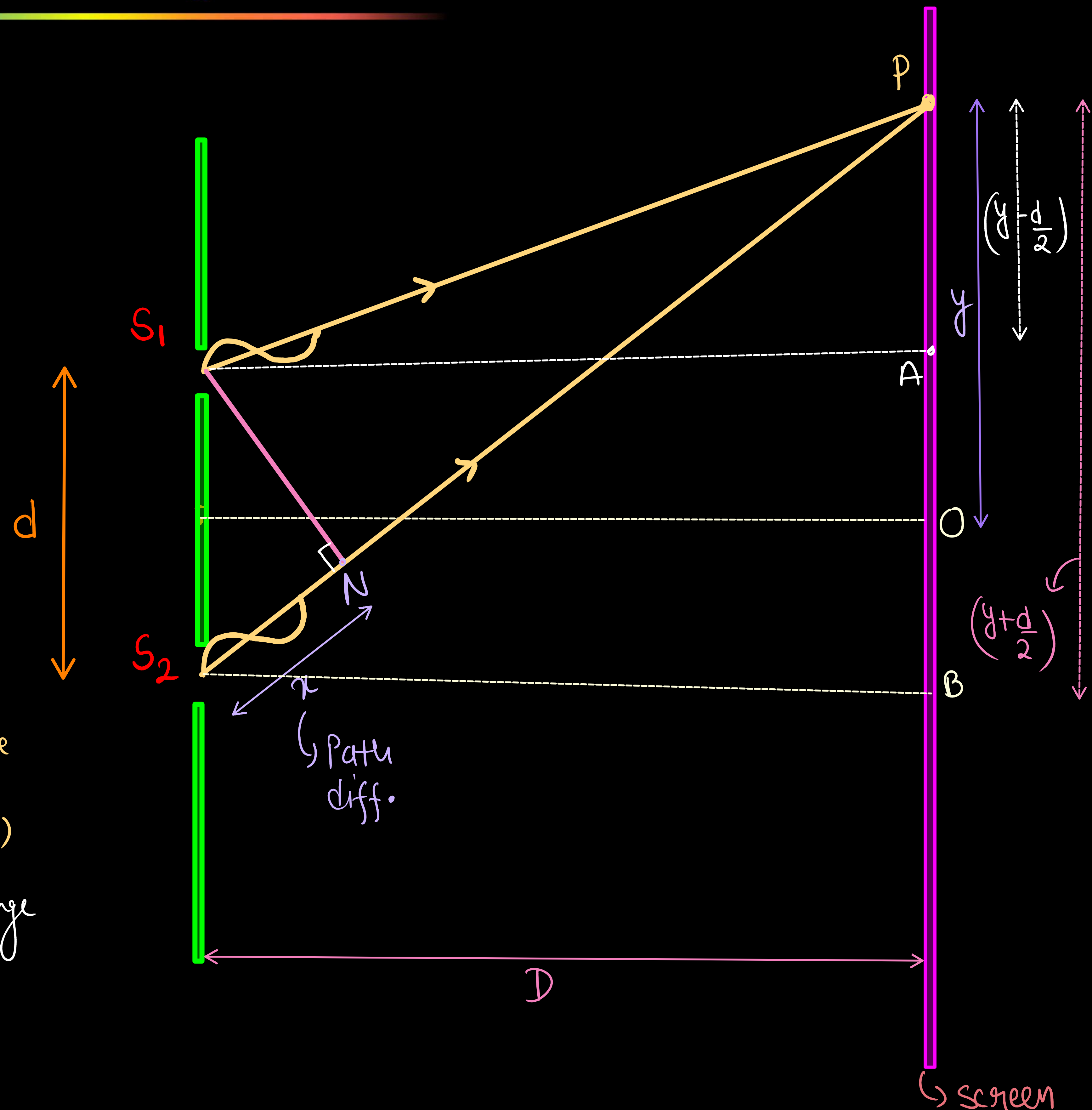
$$x = (2n-1) \frac{\lambda}{2} \quad (n=1, 2, 3, \dots)$$

$$\frac{y d}{D} = (2n-1) \frac{\lambda}{2}$$

$$\boxed{y_n = \frac{(2n-1) \lambda D}{2d}} \quad \text{--- (5) --- Distance of } n^{\text{th}} \text{ Dark fringe from O. } (n=1, 2, 3, \dots)$$

for $n=1$

$$y_1 = \frac{\lambda D}{2d} \rightarrow \text{Distance of first dark fringe}$$



Young's Double Slit Experiment

$$\boxed{x = \frac{y d}{D}} \quad \text{③} \quad \text{Path difference}$$

* Position of maxima and minima on the screen -

The point P will be dark fringe (minima) if -

$$x = (2n-1) \frac{\lambda}{2} \quad (n=1, 2, 3, \dots)$$

$$\frac{y d}{D} = (2n-1) \frac{\lambda}{2}$$

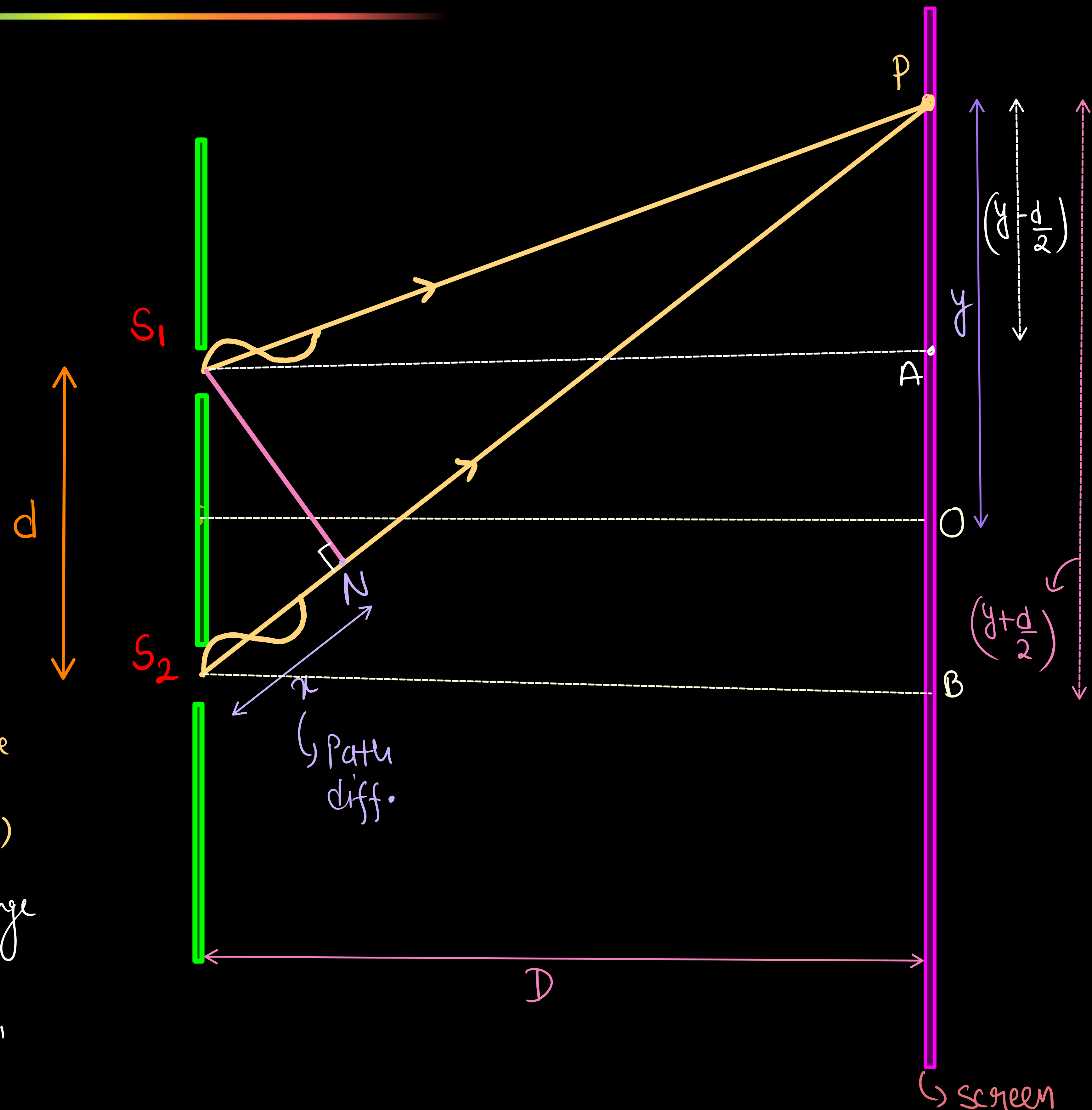
$$\boxed{y_n = \frac{(2n-1) \lambda D}{2d}} \quad \text{⑤} \quad \text{Distance of } n^{\text{th}} \text{ Dark fringe from O.} \\ (n=1, 2, 3, \dots)$$

for $n=1$

$$y_1 = \frac{\lambda D}{2d} \rightarrow \text{Distance of first dark fringe}$$

$n=2$

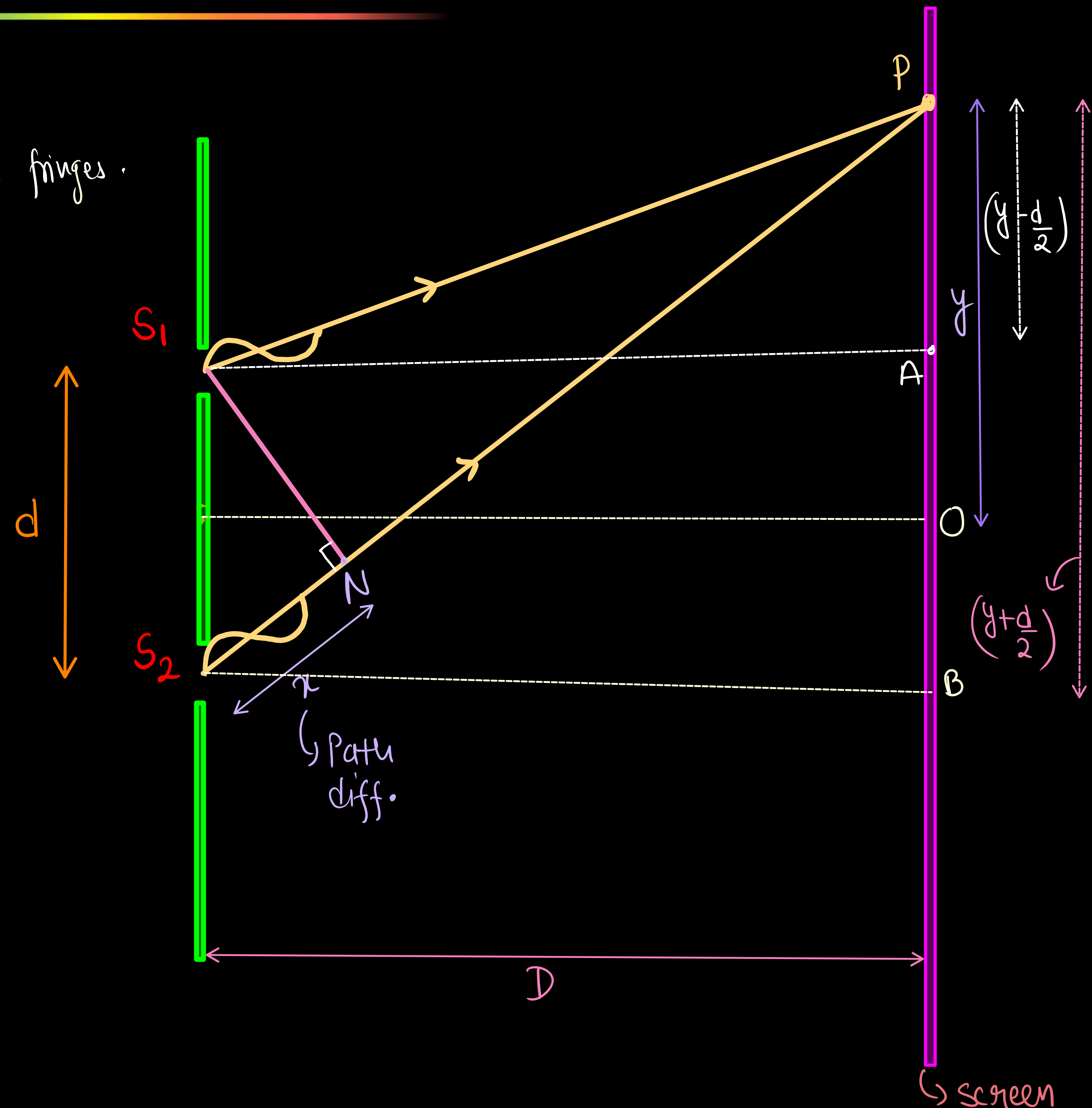
$$y_2 = 3 \frac{\lambda D}{2d} \rightarrow \text{'' '' 2}^{\text{nd}} \text{ '' ''}$$



Young's Double Slit Experiment

* Fringe width (β)

The spacing b/w any two consecutive bright fringes or dark fringes.

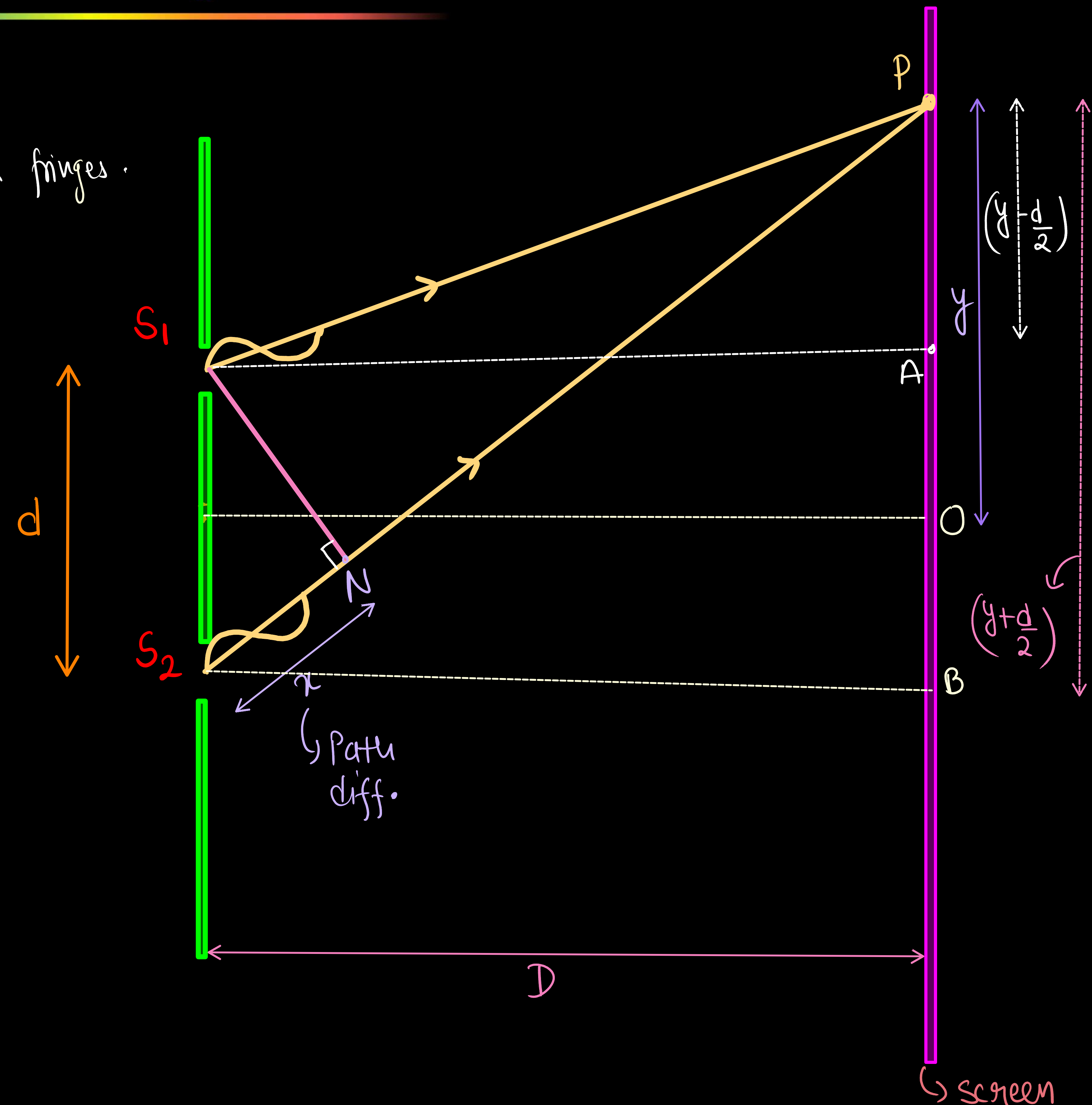


Young's Double Slit Experiment

* Fringe width (β)

The spacing b/w any two consecutive bright fringes or dark fringes.

$$\beta = y_{n+1} - y_n$$



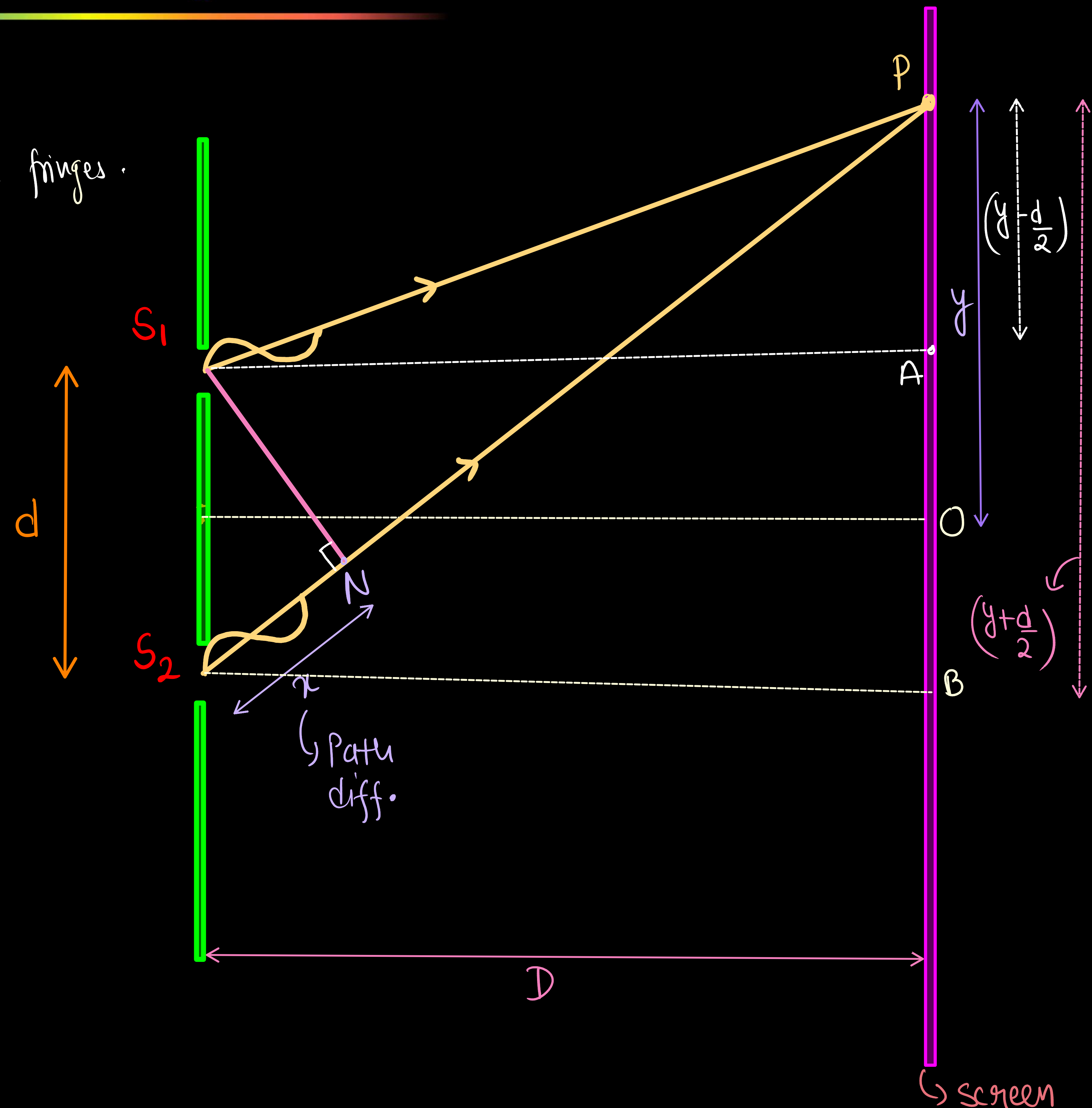
Young's Double Slit Experiment

* Fringe width (β)

The spacing b/w any two consecutive bright fringes or dark fringes.

$$\beta = y_{n+1} - y_n$$

$$\beta = (n+1) \frac{\lambda D}{d} - n \frac{\lambda D}{d}$$



Young's Double Slit Experiment

* Fringe width (β)

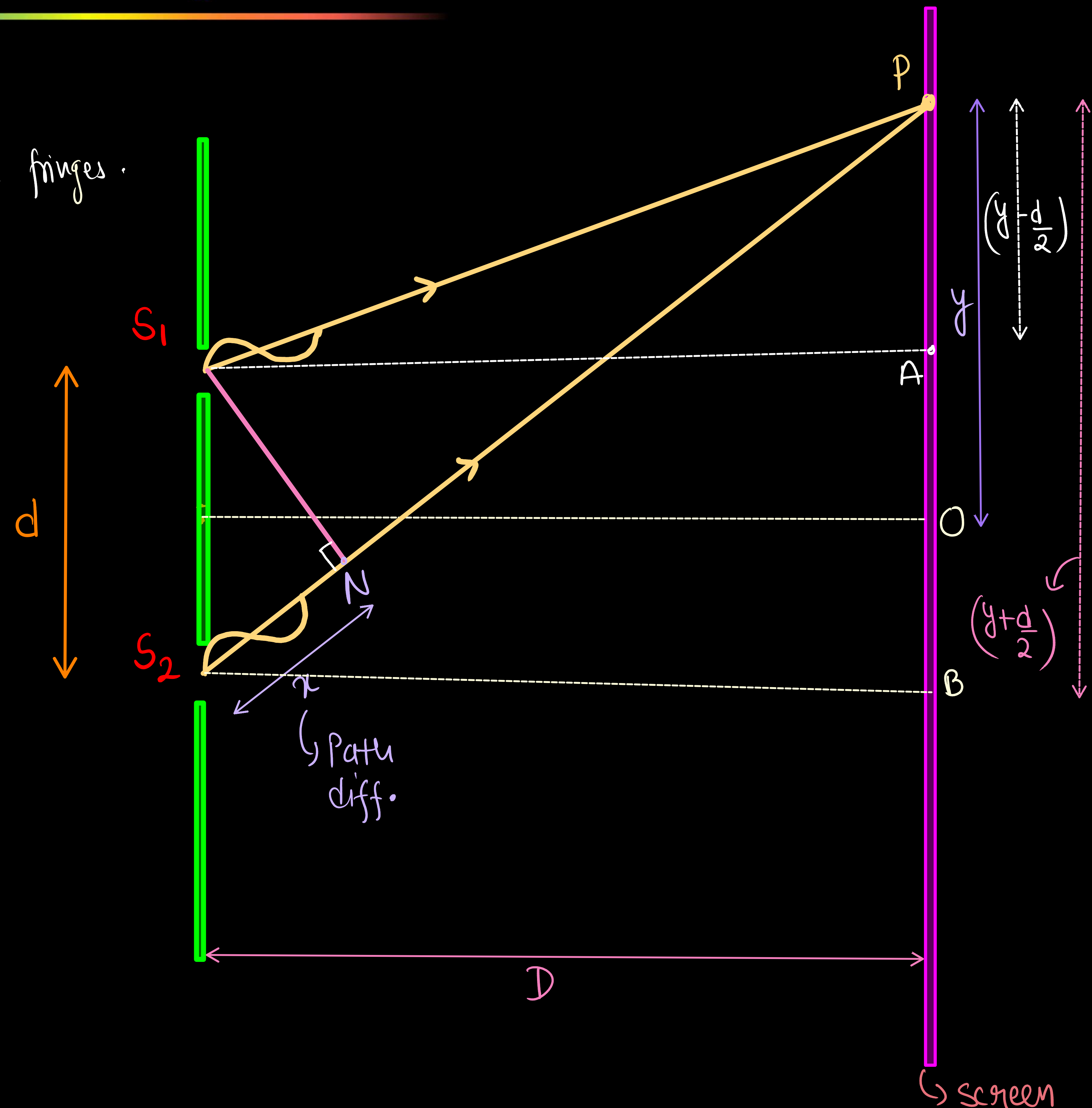
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$$\boxed{\beta = \frac{\lambda D}{d}} \quad \text{--- (6)}$$

* Angular fringe width (θ)



Young's Double Slit Experiment

* Fringe width (β)

The spacing b/w any two consecutive bright fringes or dark fringes.

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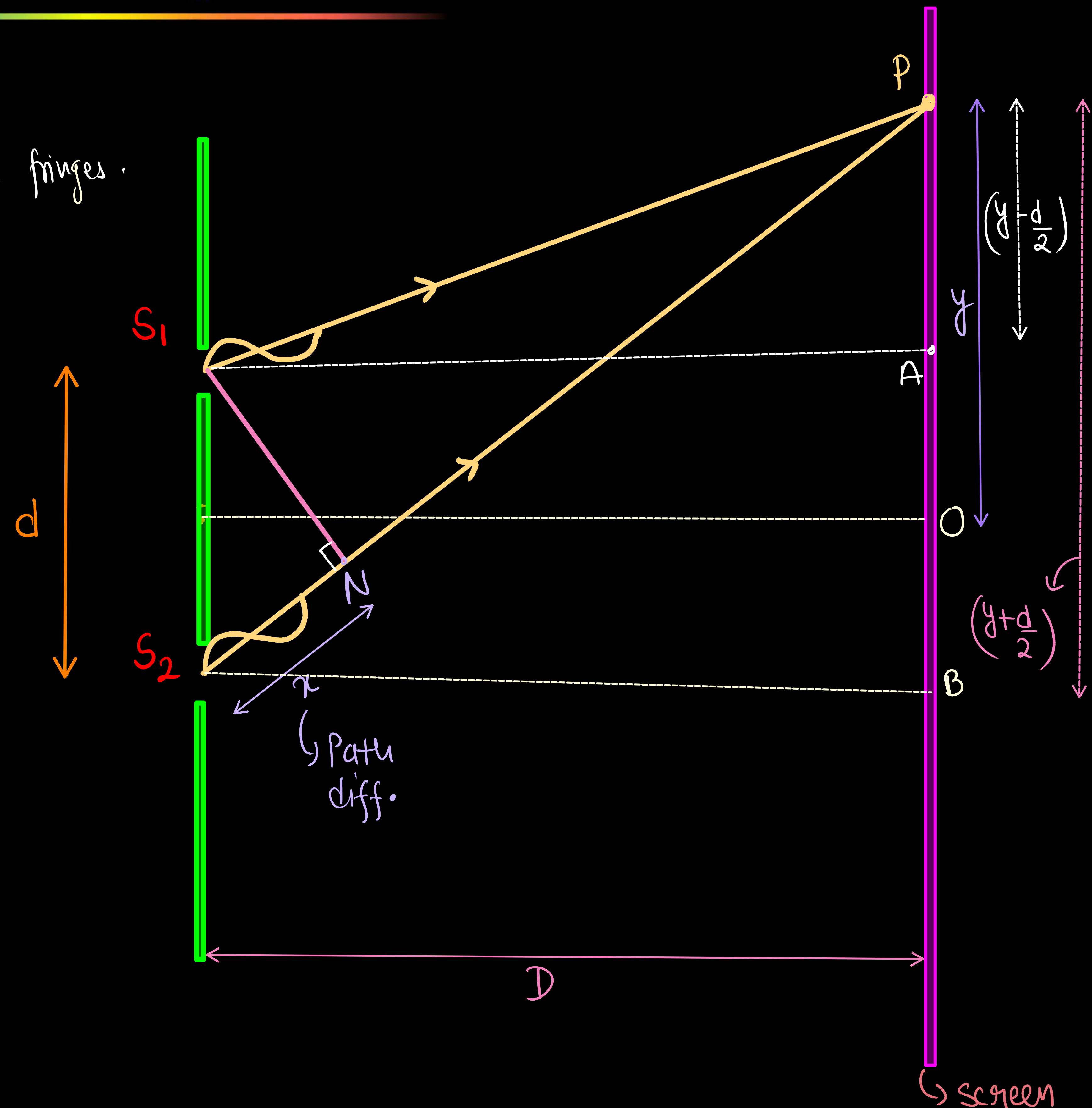
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$$\boxed{\beta = \frac{\lambda D}{d}} \quad \text{--- (6)}$$

* Angular fringe width (θ)

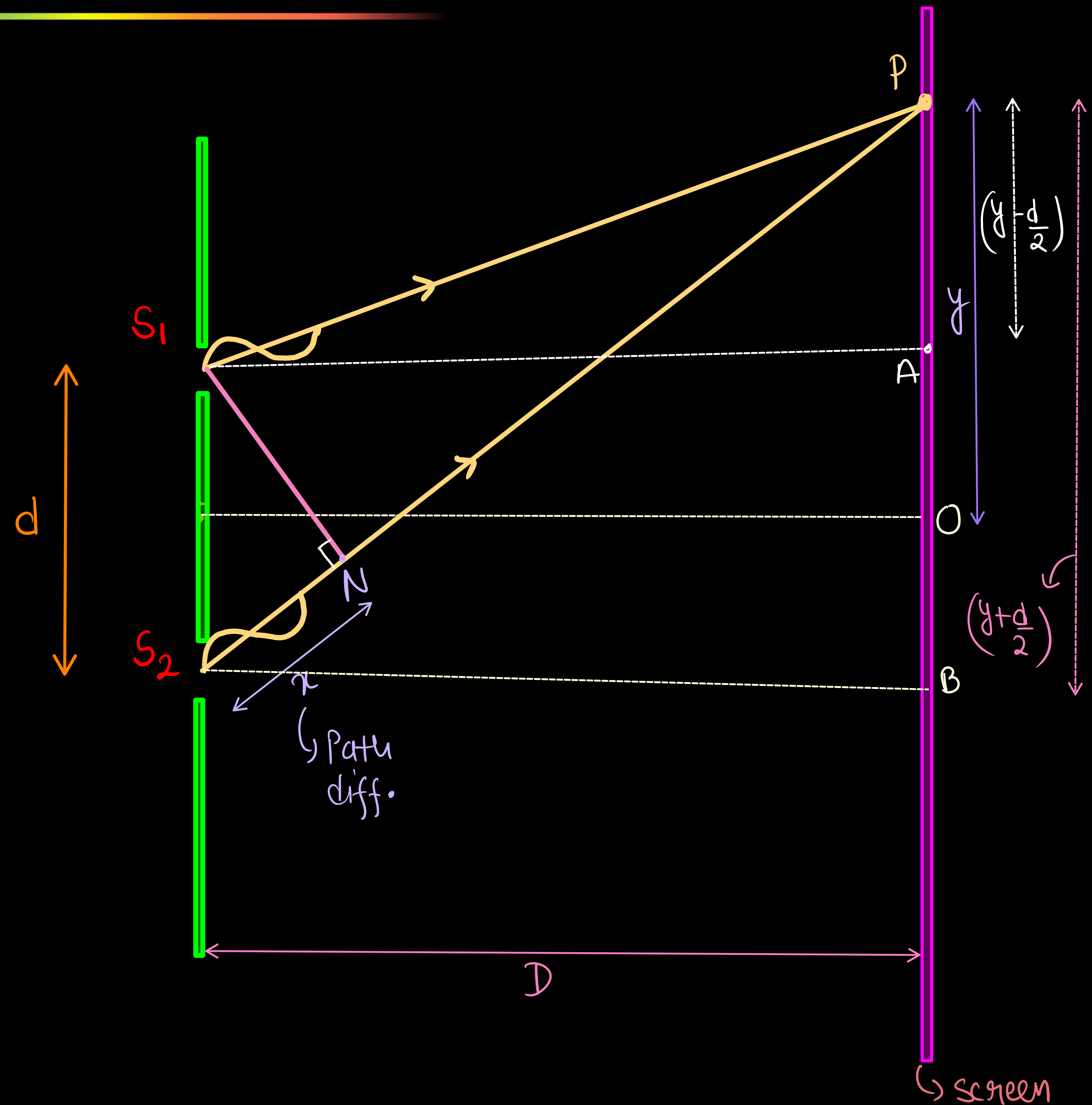
$$\theta = \frac{\beta}{D}$$

$$\boxed{\theta = \frac{\lambda}{d}} \quad \text{--- (7)}$$



Young's Double Slit Experiment

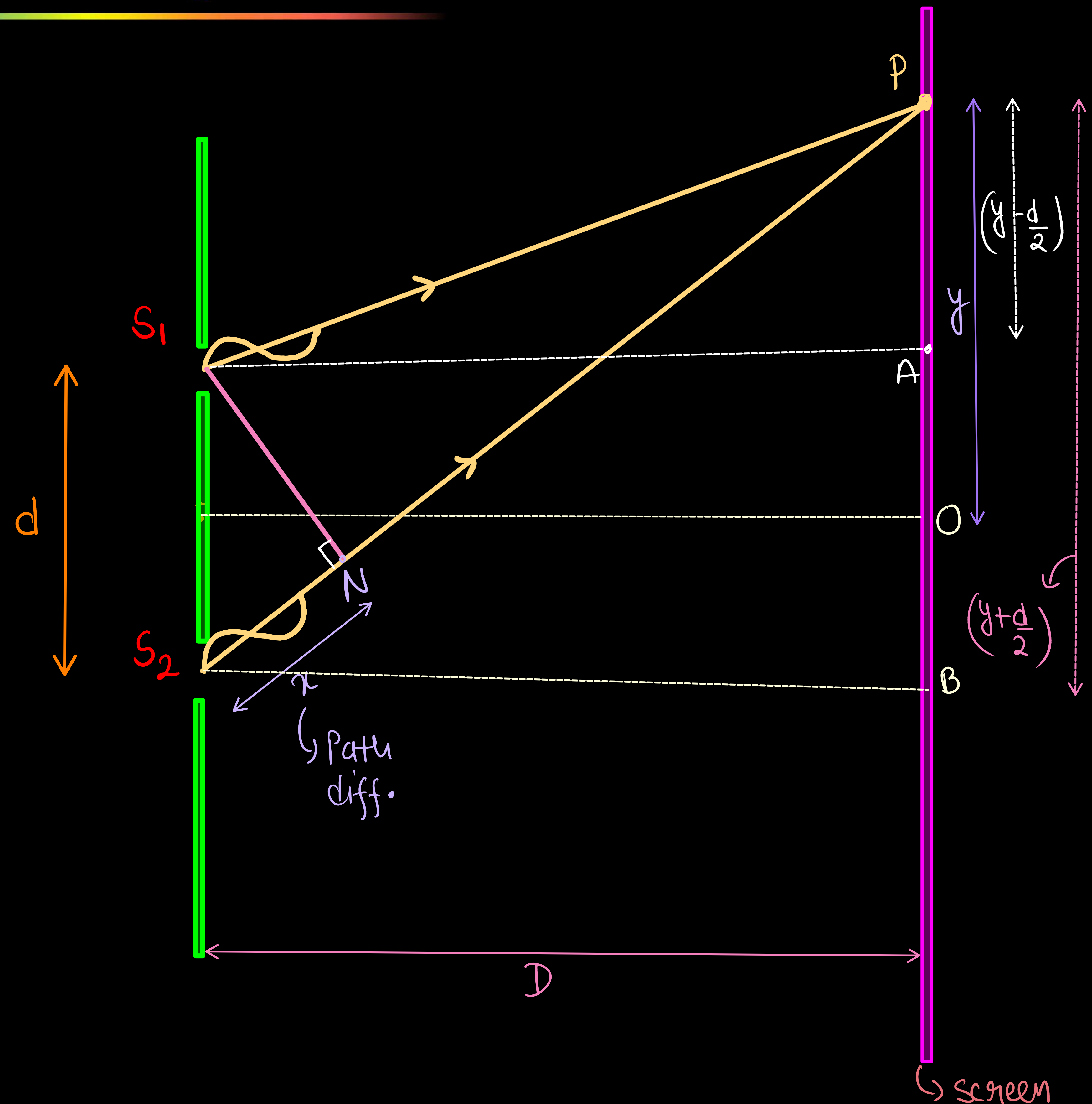
* If YDSE apparatus is immersed in a liquid of refractive index (μ)



Young's Double Slit Experiment

* If YDSE apparatus is immersed in a liquid of refractive index (μ)

$$\lambda' = \frac{\lambda}{\mu}$$



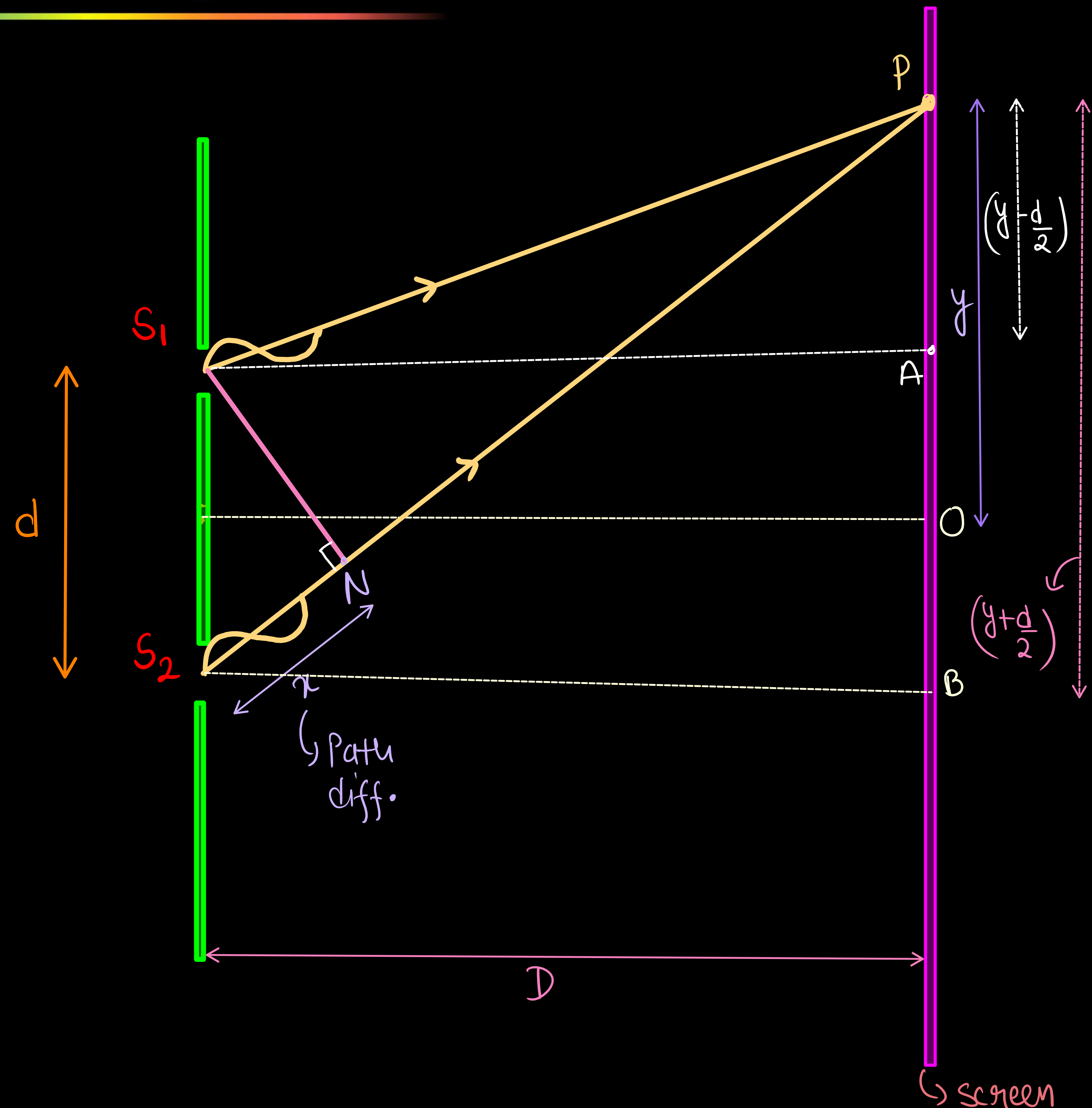
Young's Double Slit Experiment

* If YDSE apparatus is immersed in a liquid of refractive index (μ)

$$\lambda' = \frac{\lambda}{\mu}$$

$$\beta' = \frac{\lambda D}{\mu d} \quad (8)$$

↳ fringe width decreases



Young's Double Slit Experiment

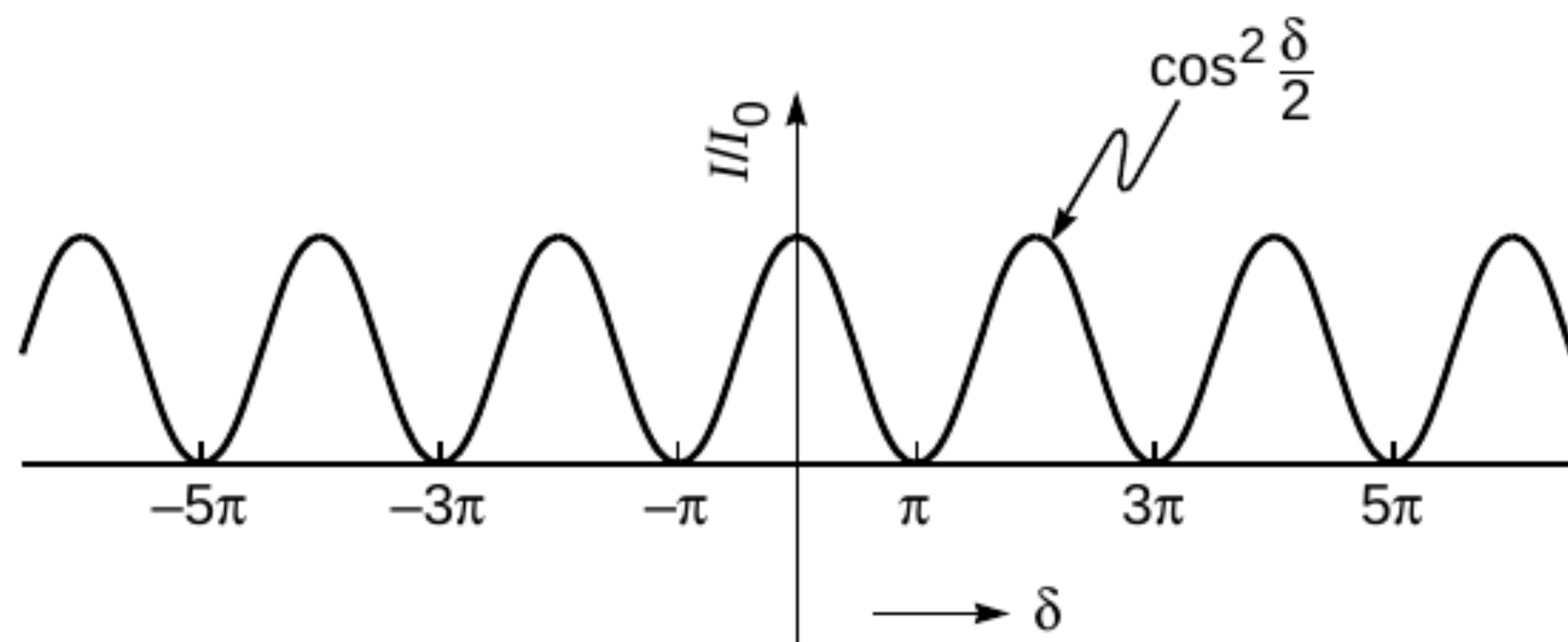
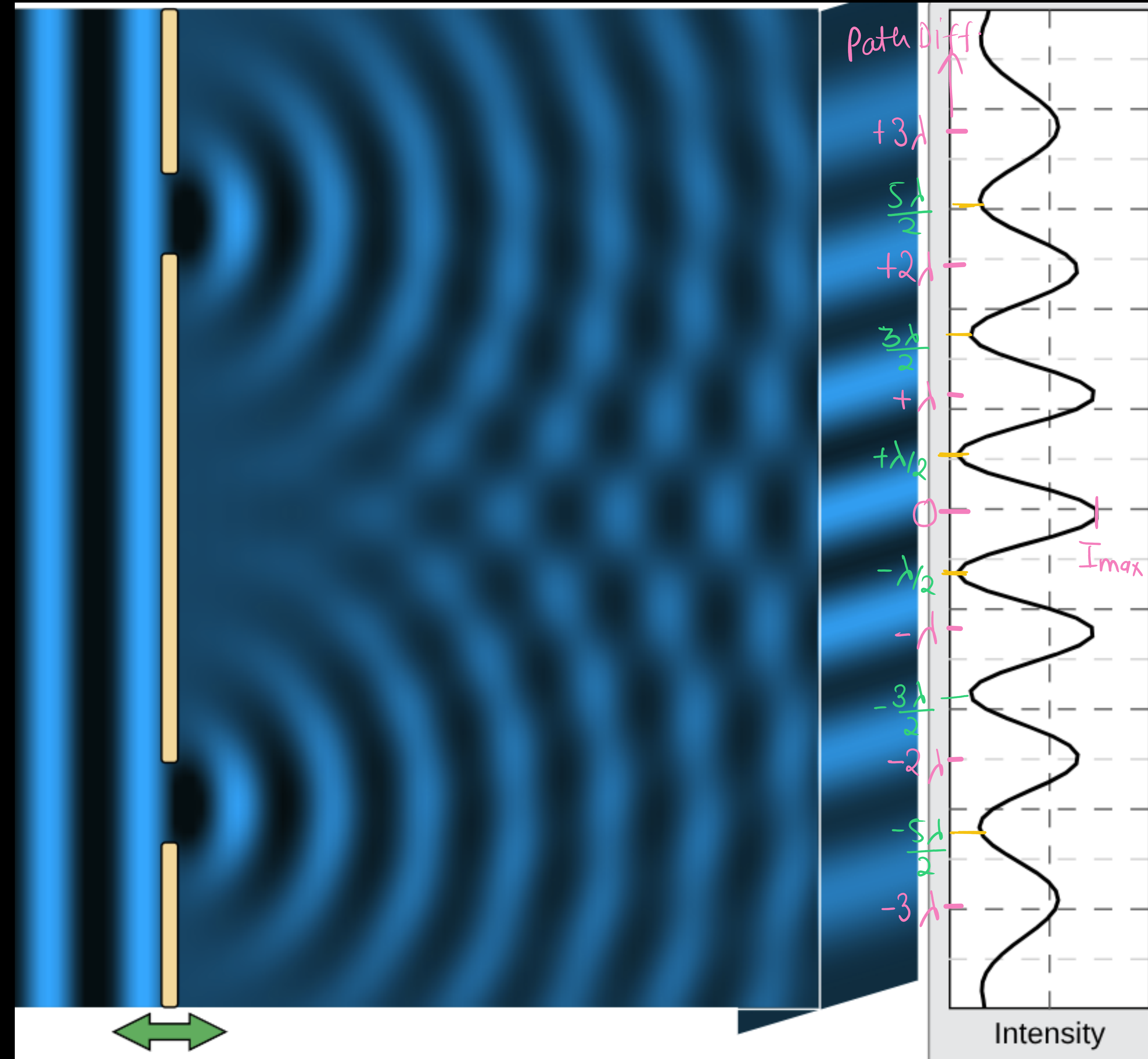
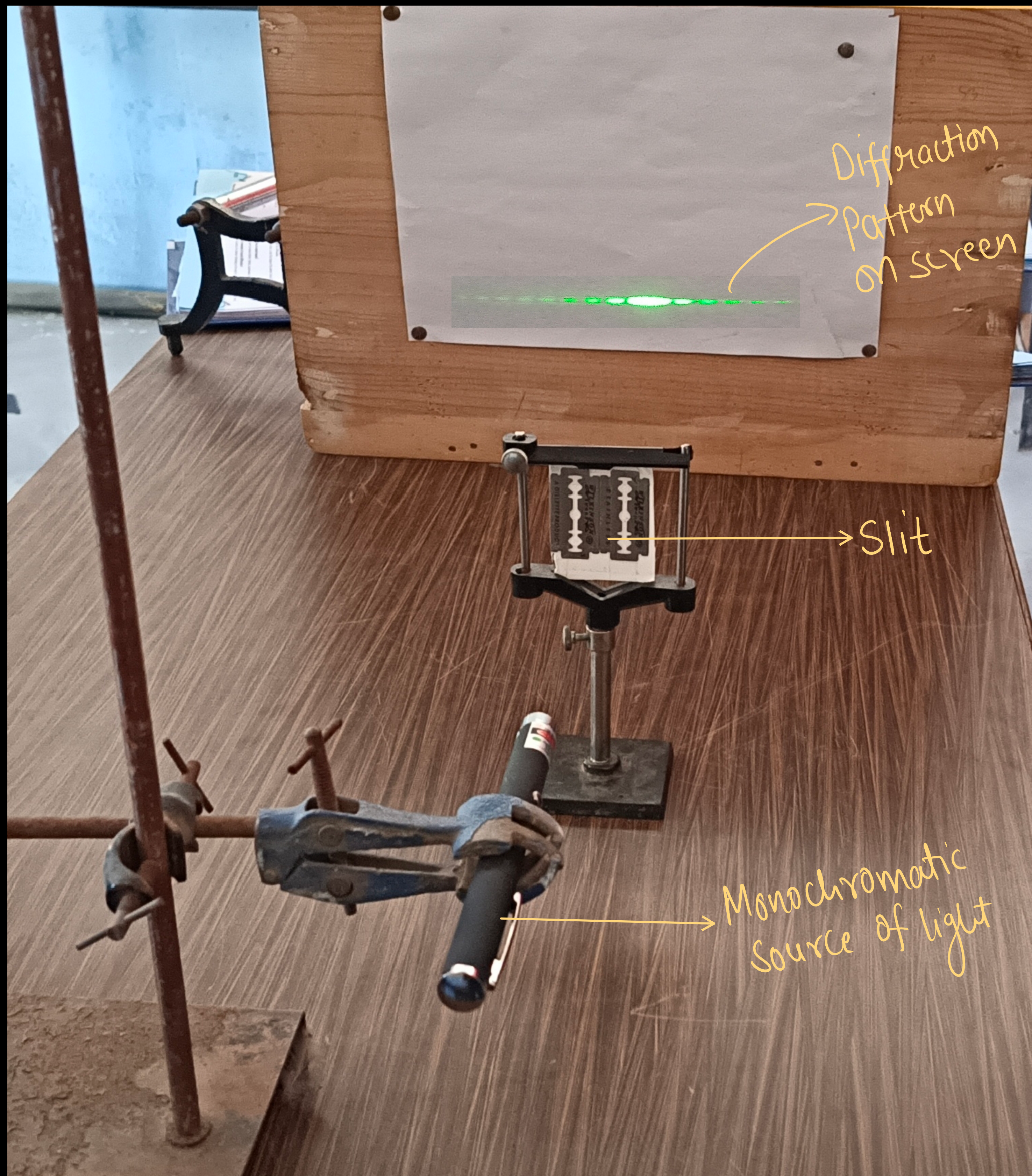


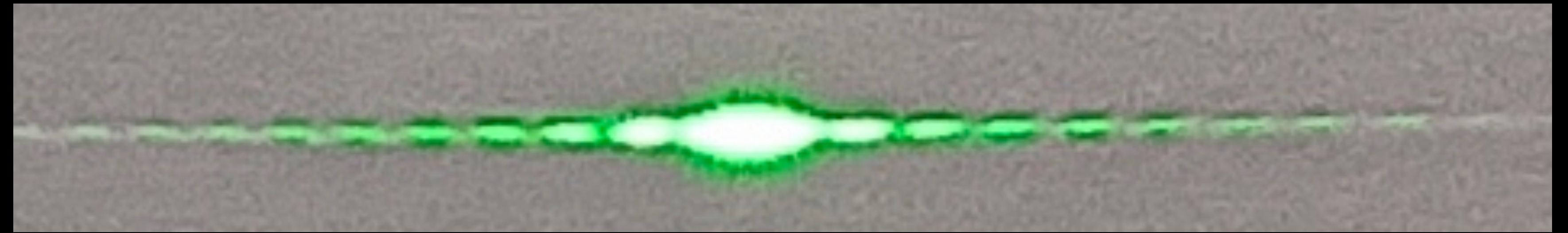
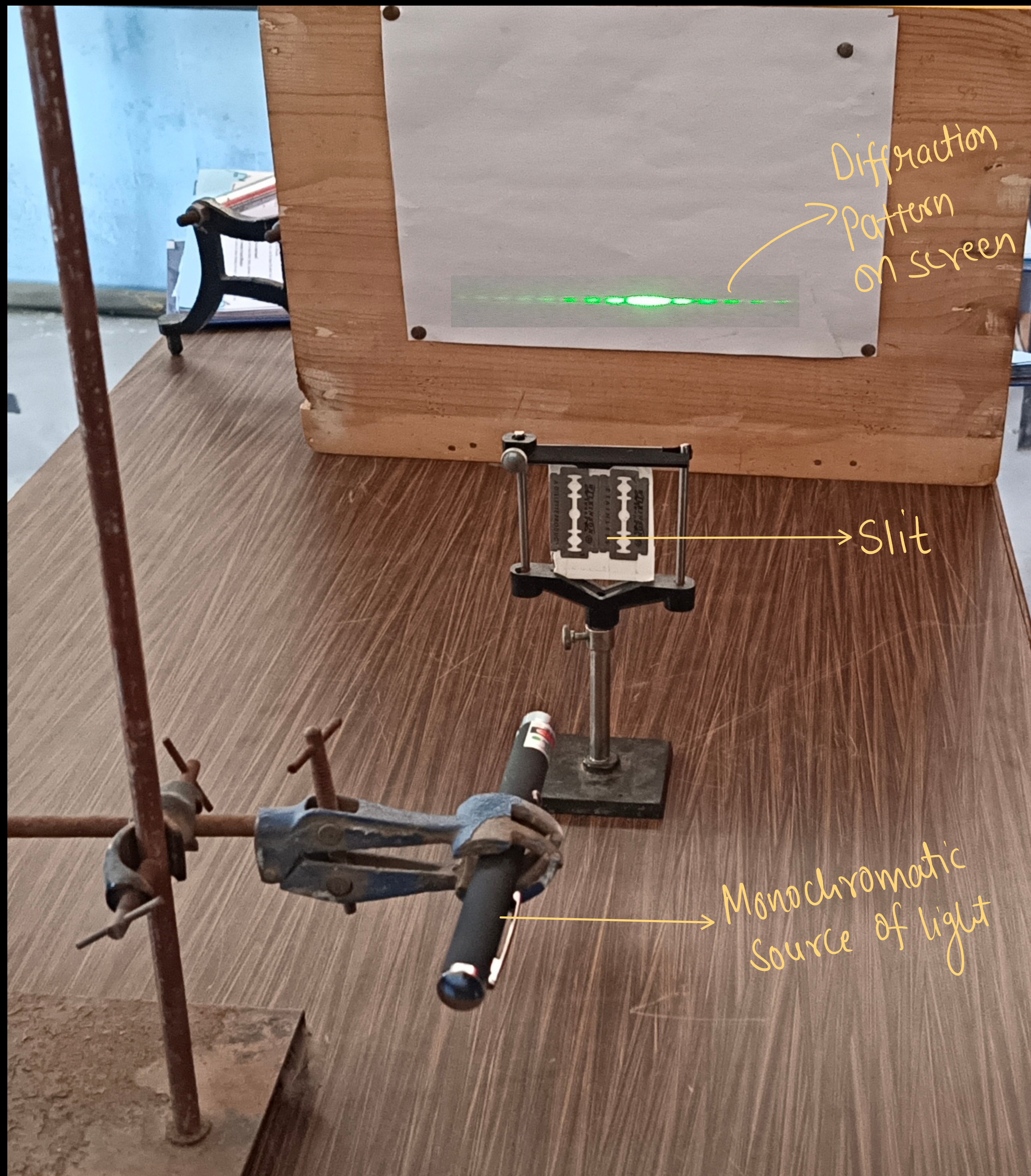
Fig. 14.9 The variation of intensity with δ .



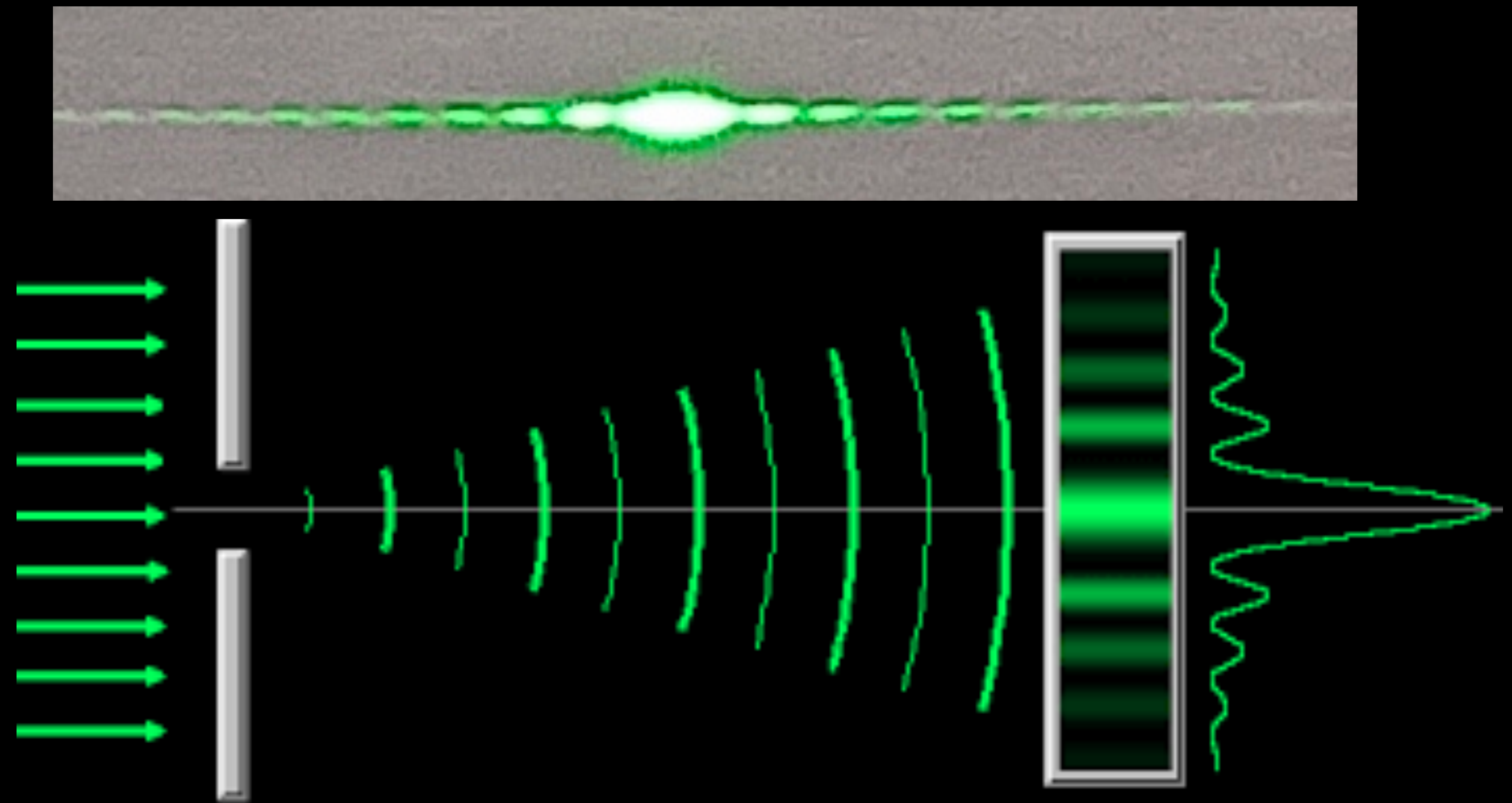
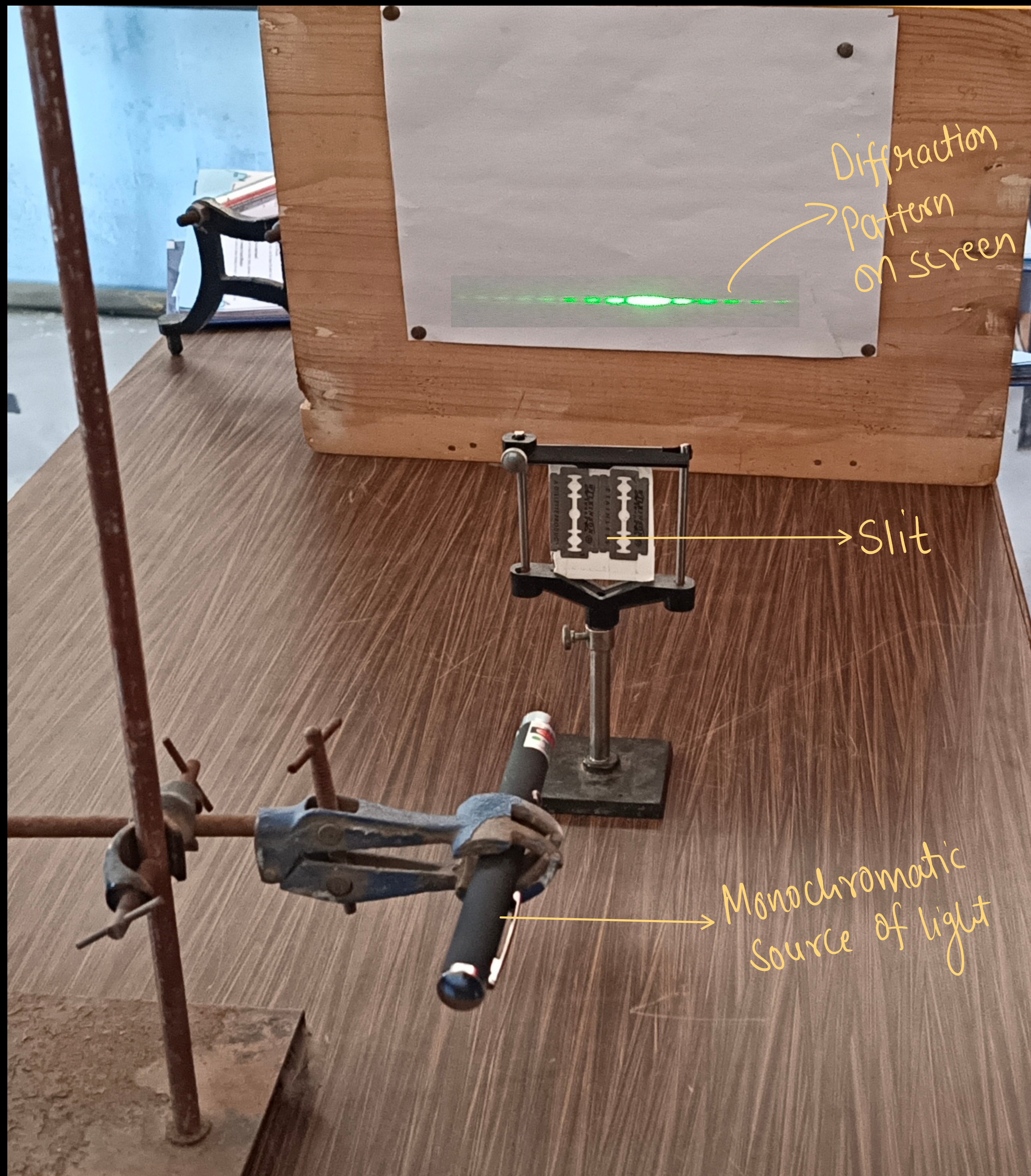
Diffraction due to a Single Slit



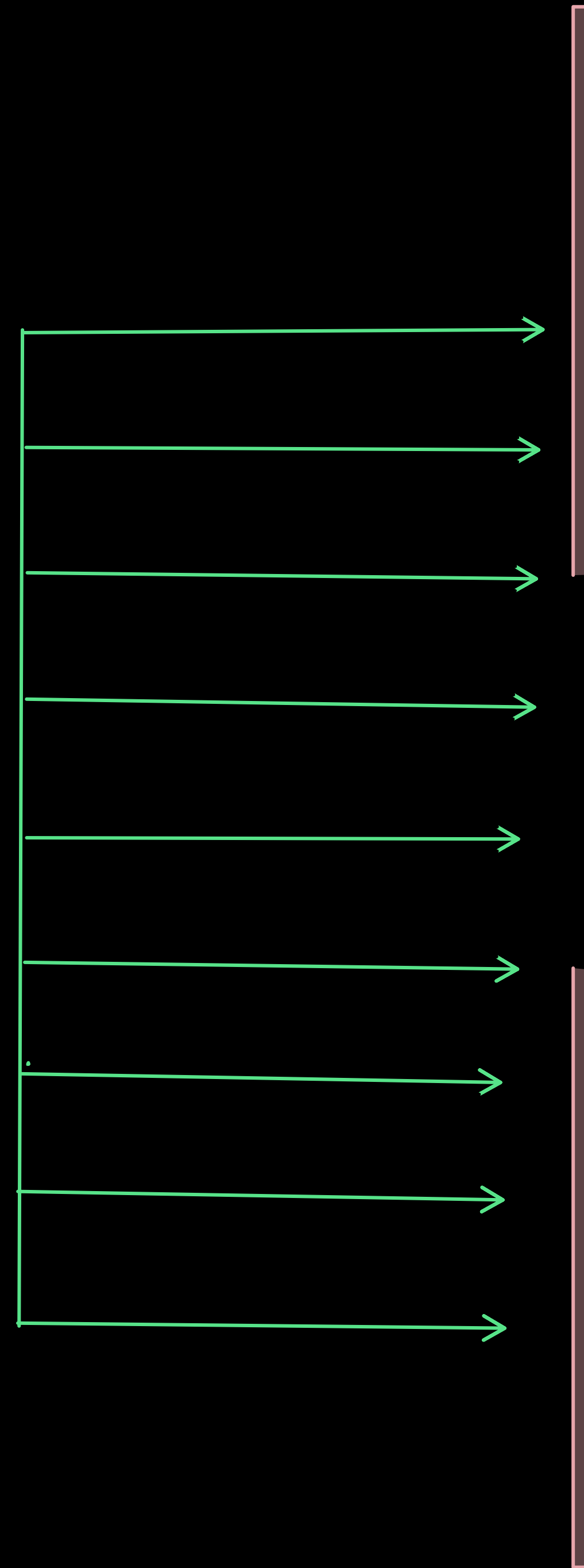
Diffraction due to a Single Slit



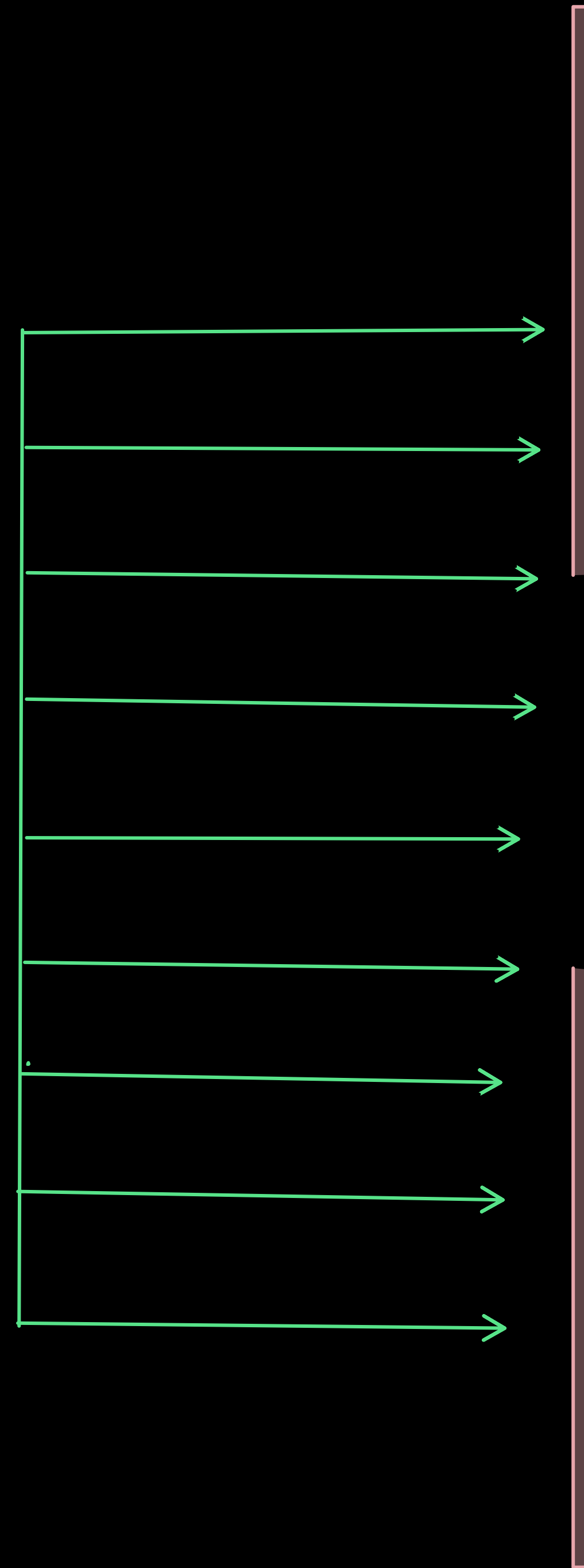
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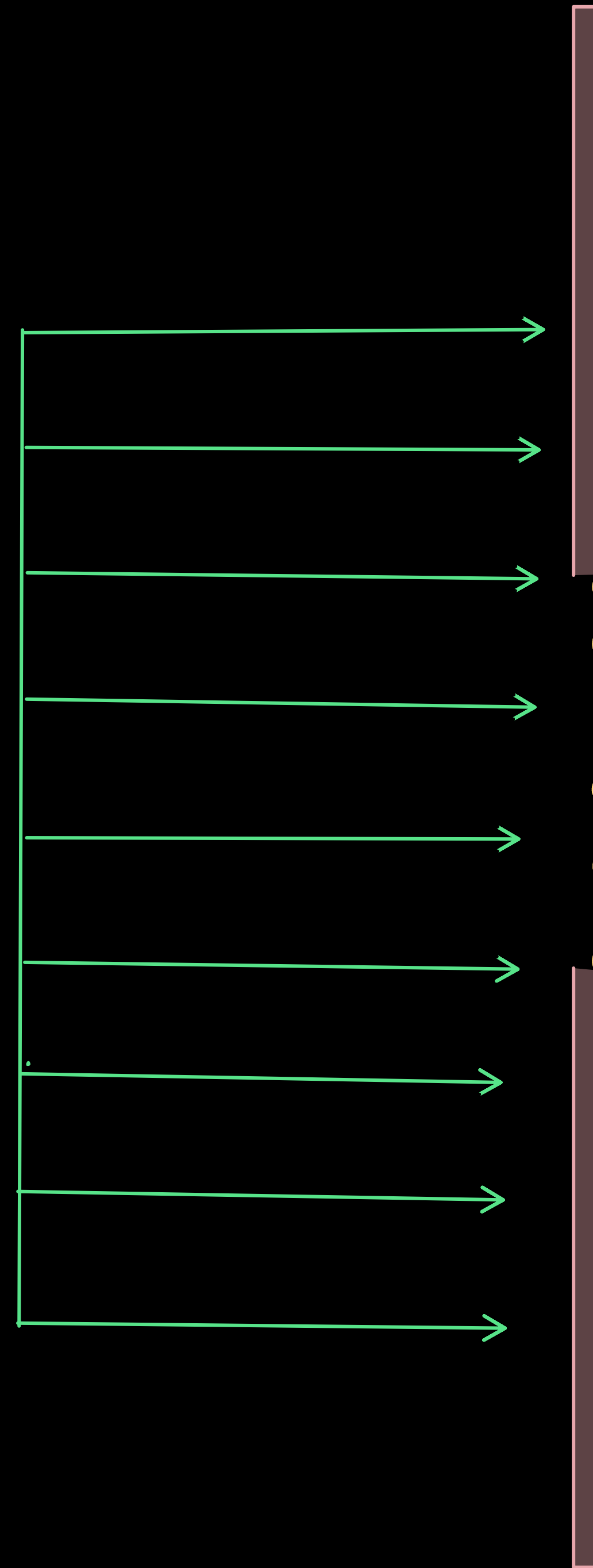
Diffraction due to a Single Slit



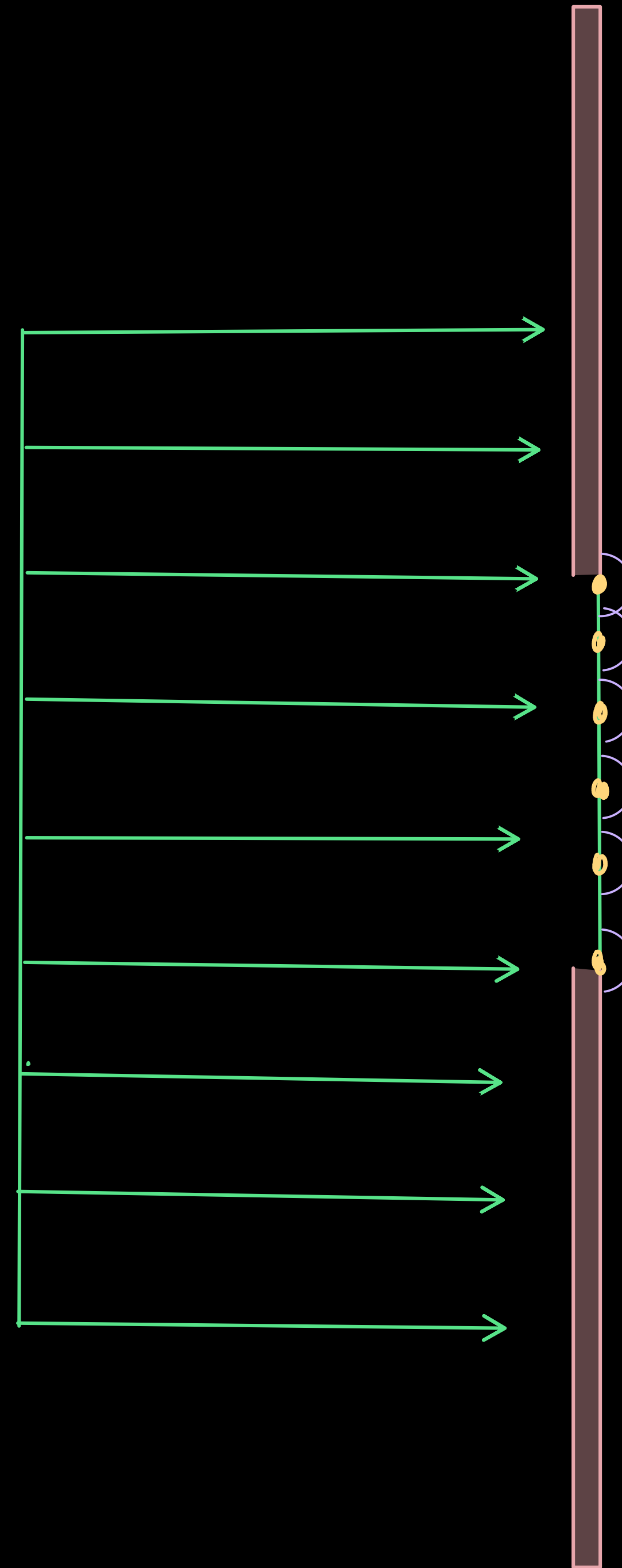
Diffraction due to a Single Slit



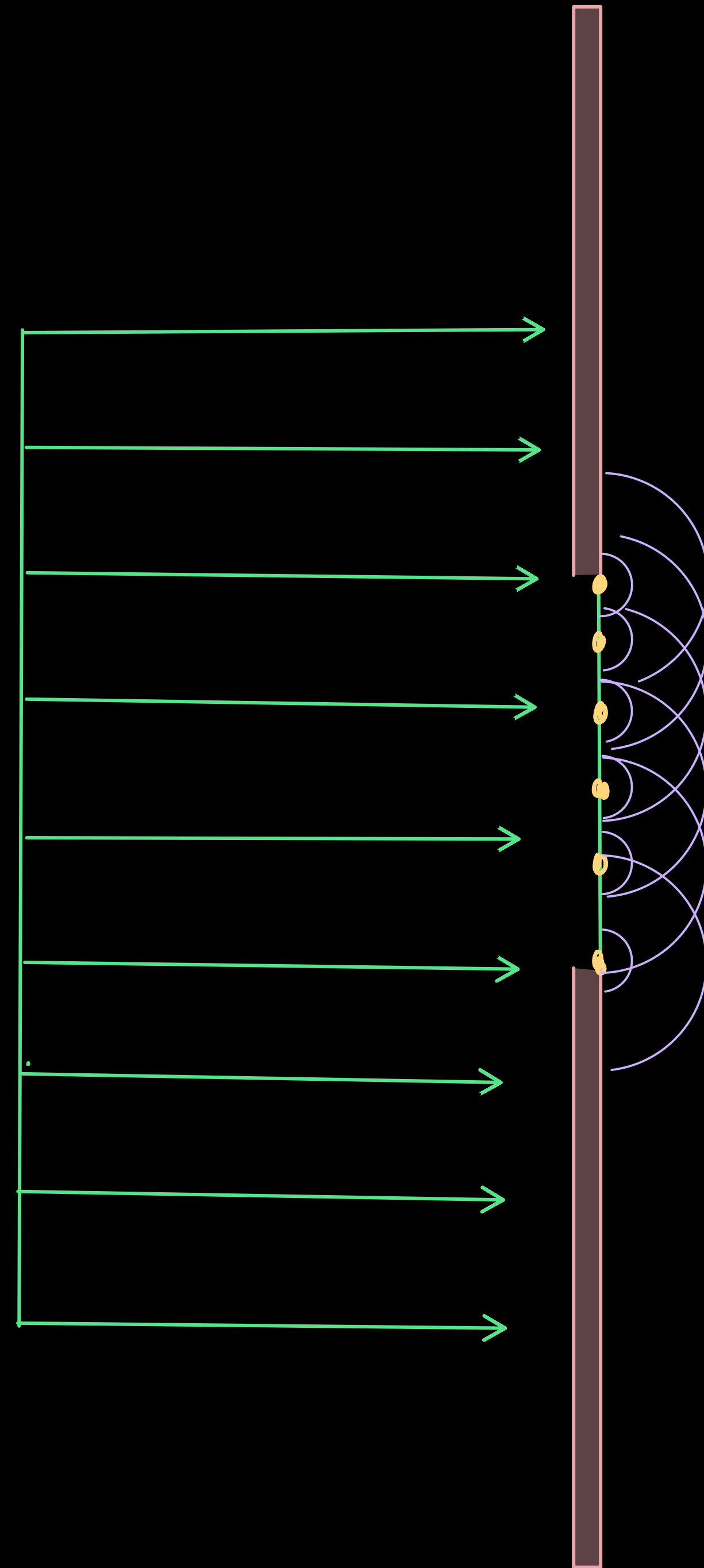
Diffraction due to a Single Slit



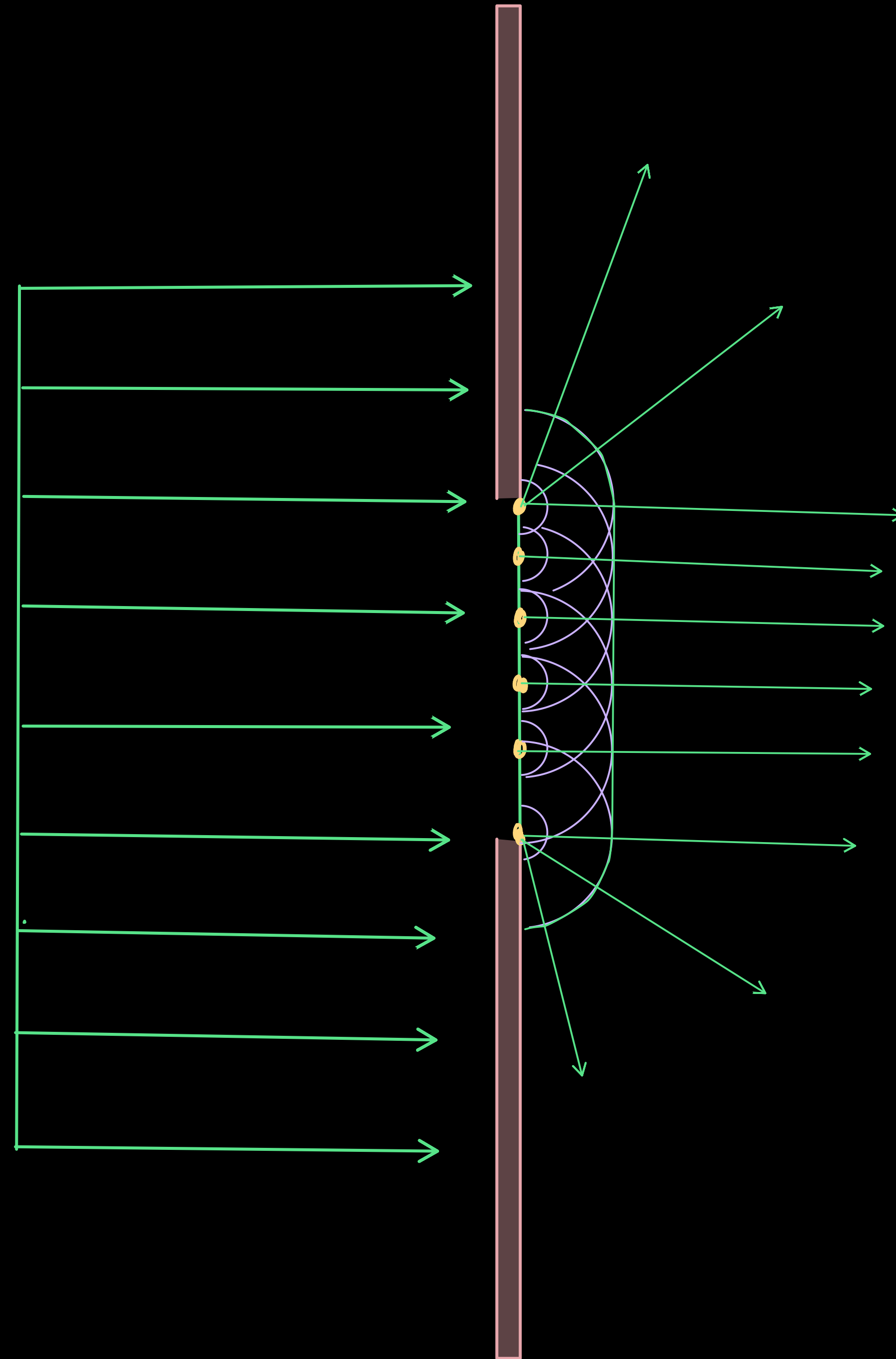
Diffraction due to a Single Slit



Diffraction due to a Single Slit

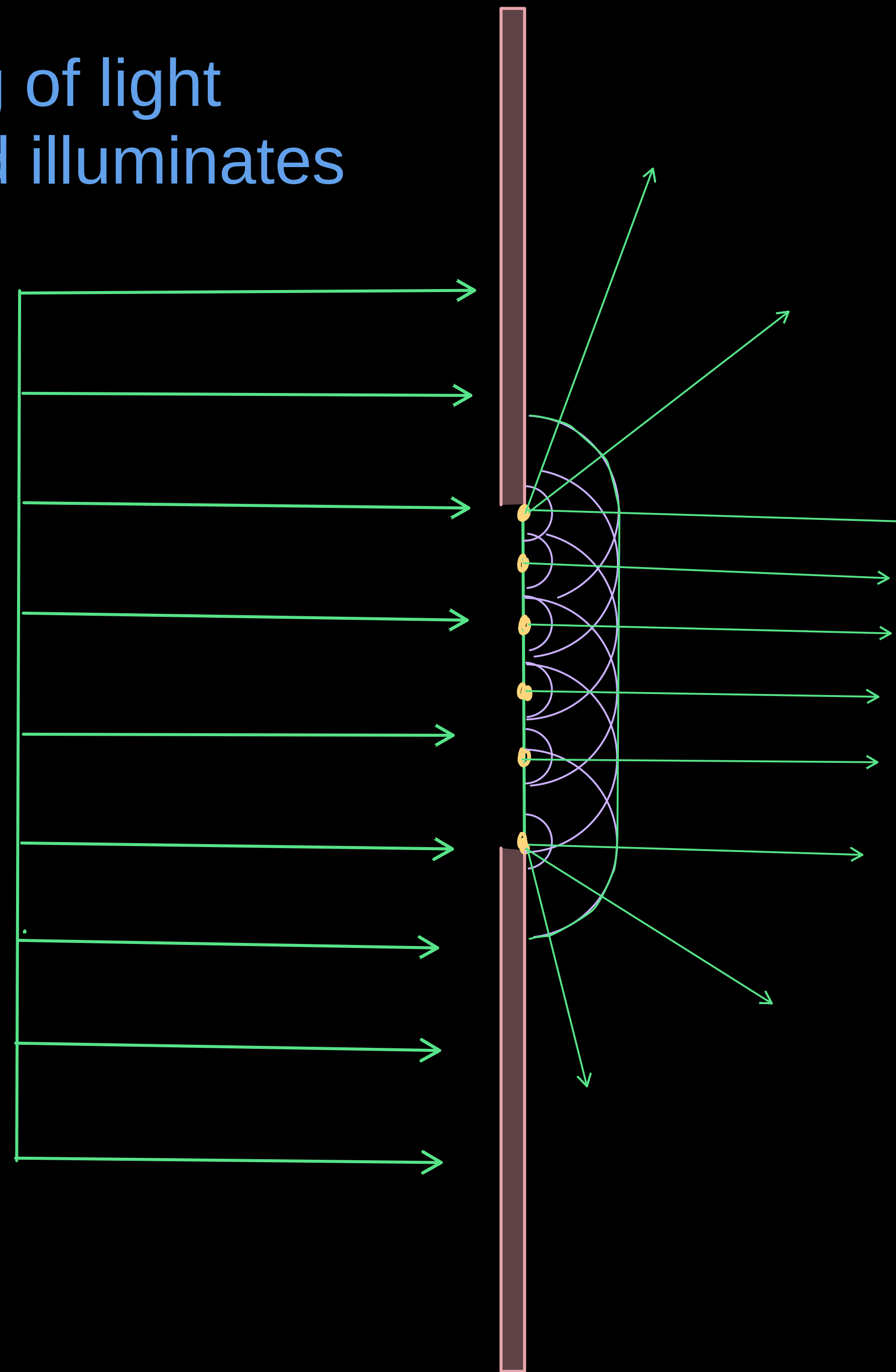


Diffraction due to a Single Slit



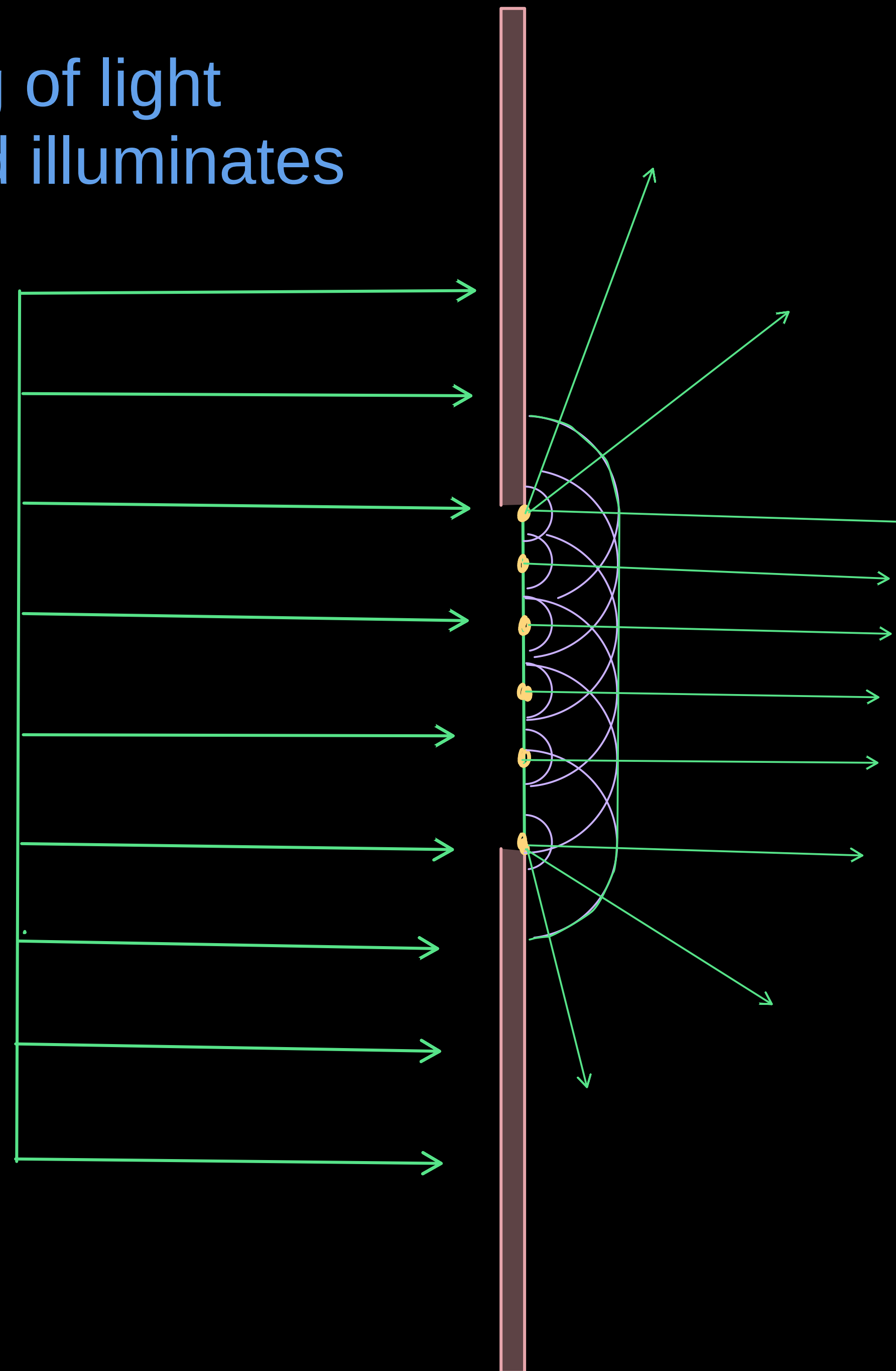
Diffraction due to a Single Slit

- * Diffraction of light is defined as the bending of light around corners such that it spreads out and illuminates areas where a shadow is expected.



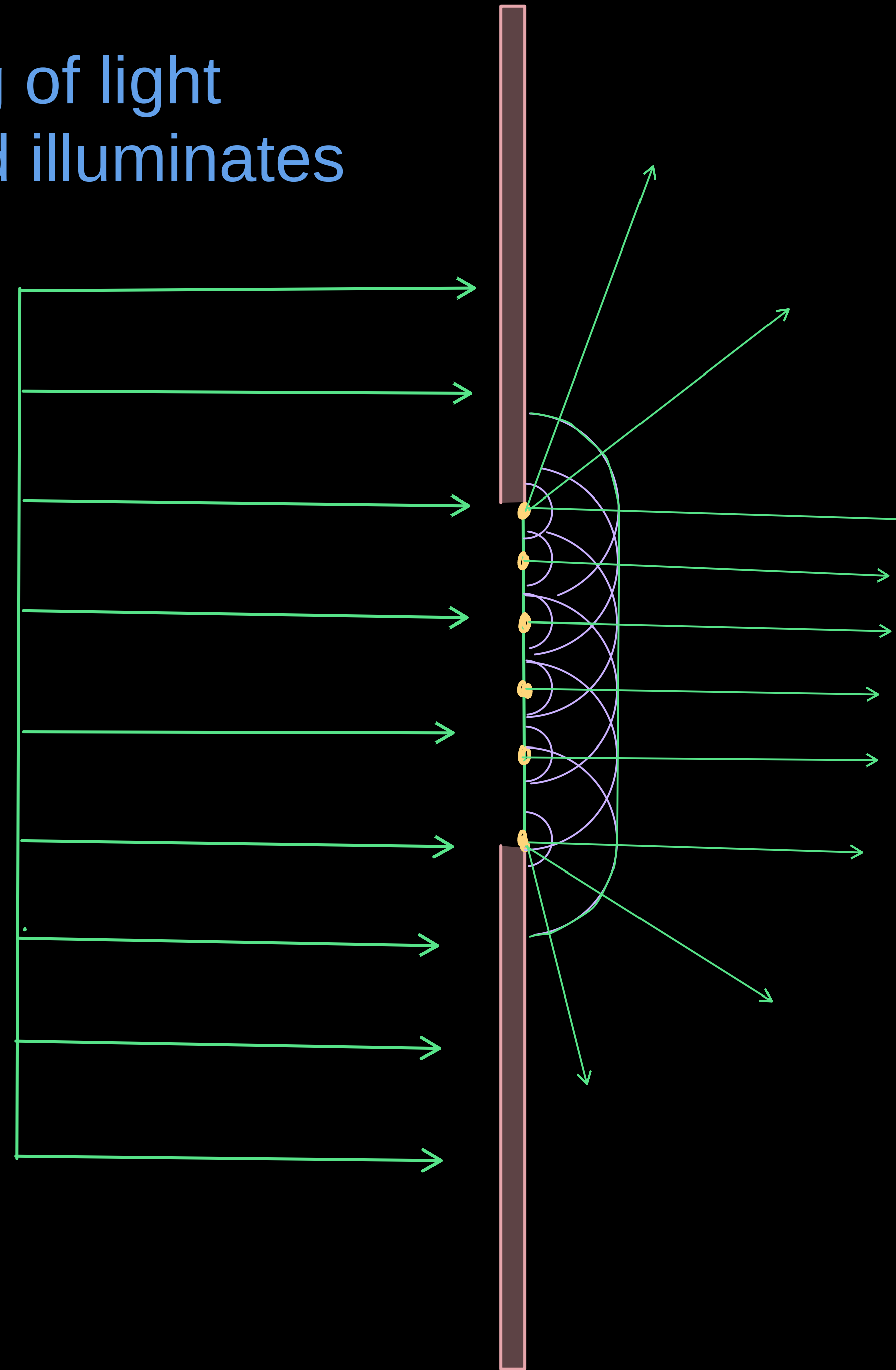
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- * Diffraction of light is defined as the bending of light around corners such that it spreads out and illuminates areas where a shadow is expected.
- * Diffraction is a general characteristic exhibited by all types of waves, be it sound waves, light waves, water waves or matter waves.



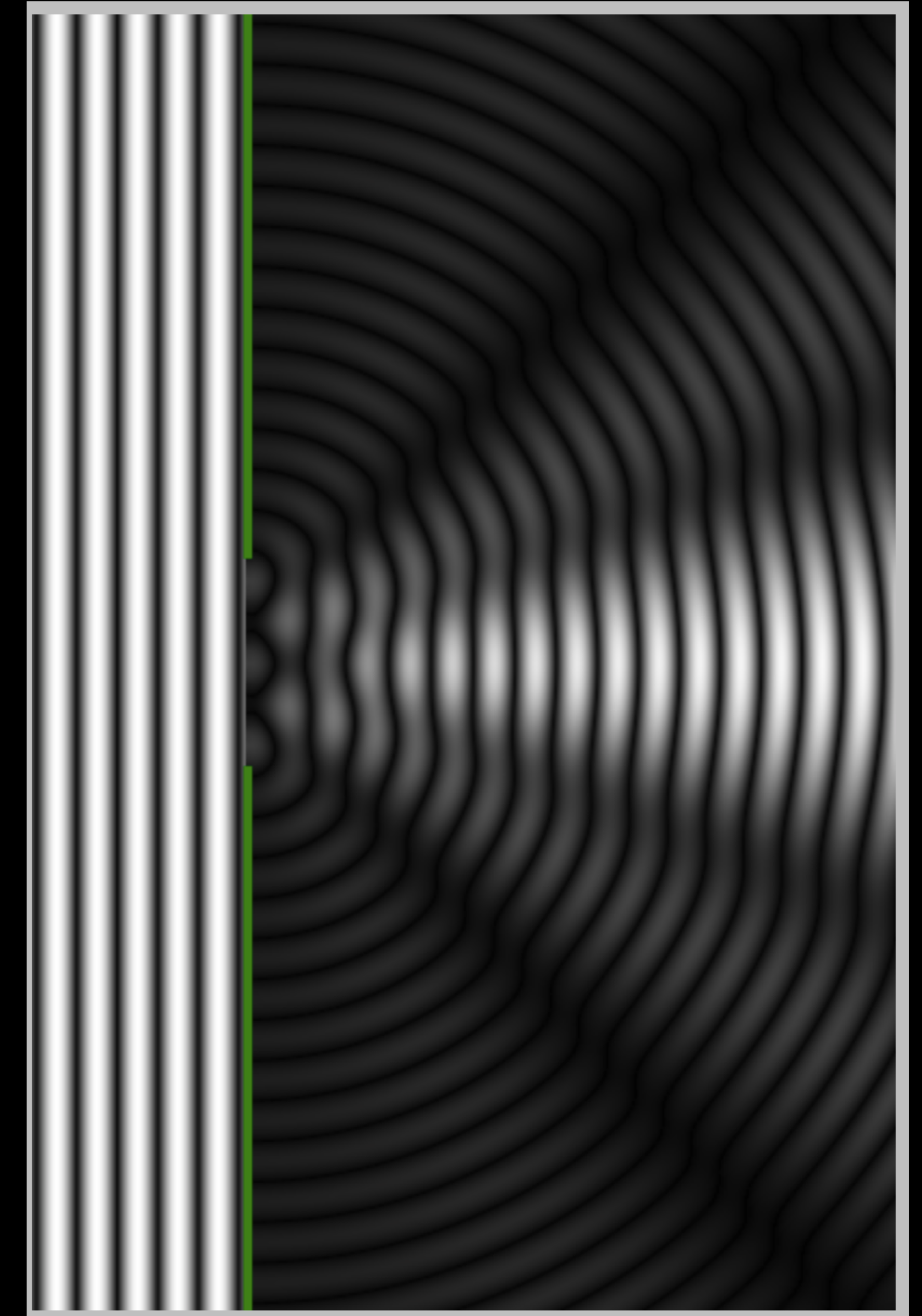
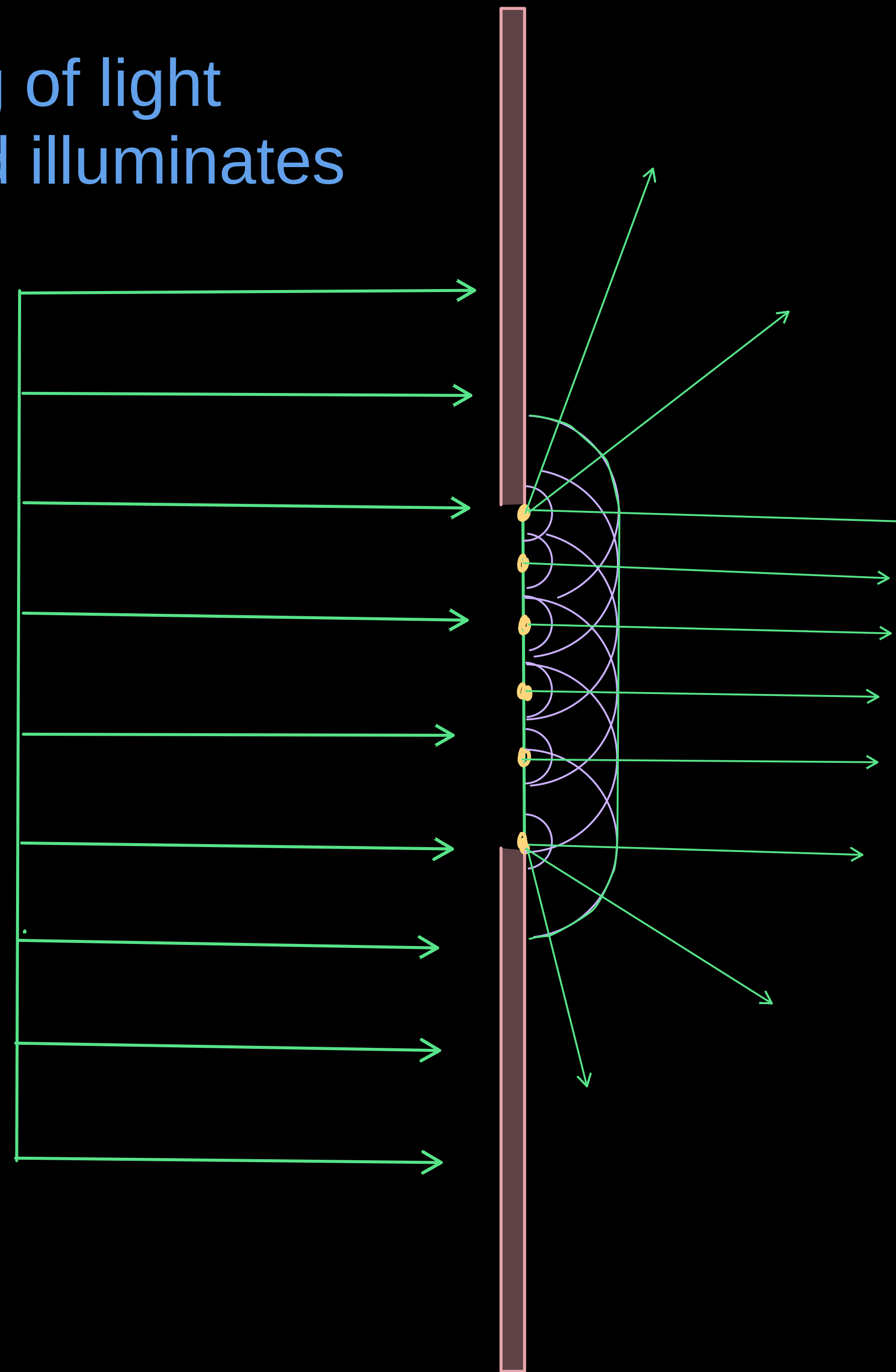
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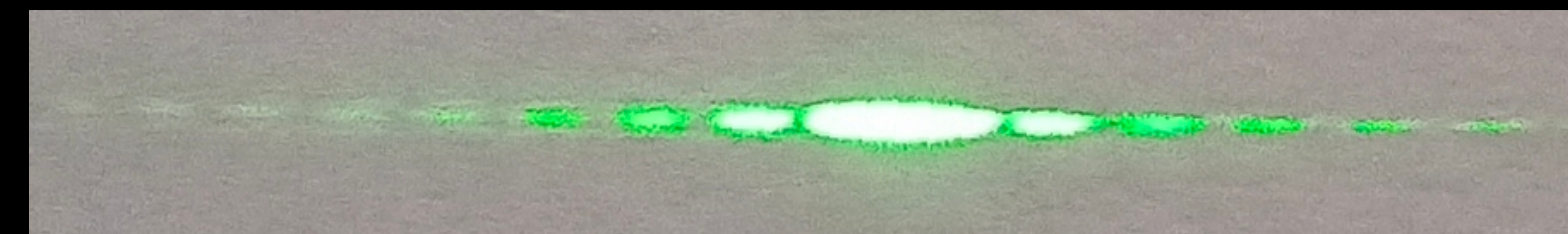
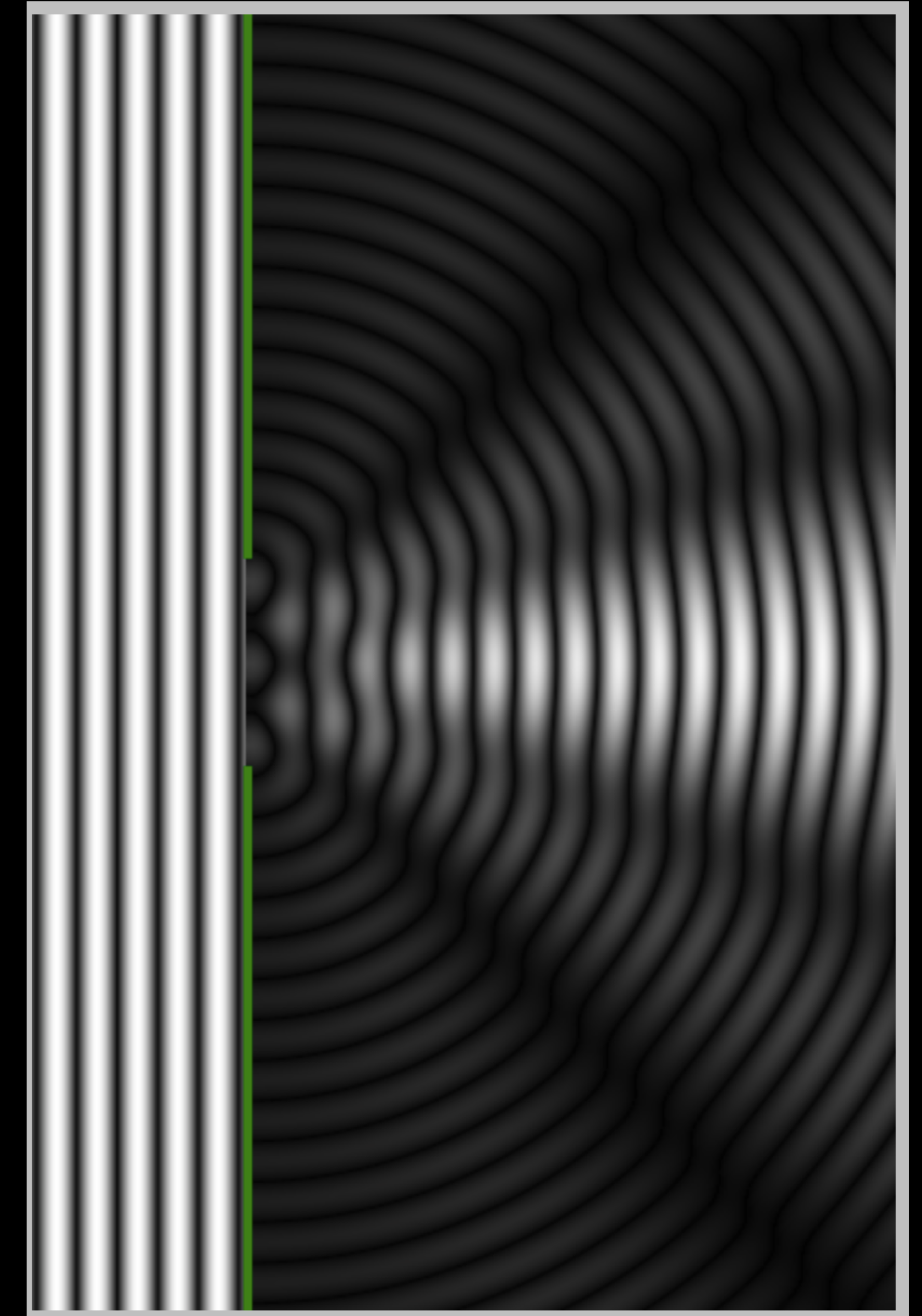
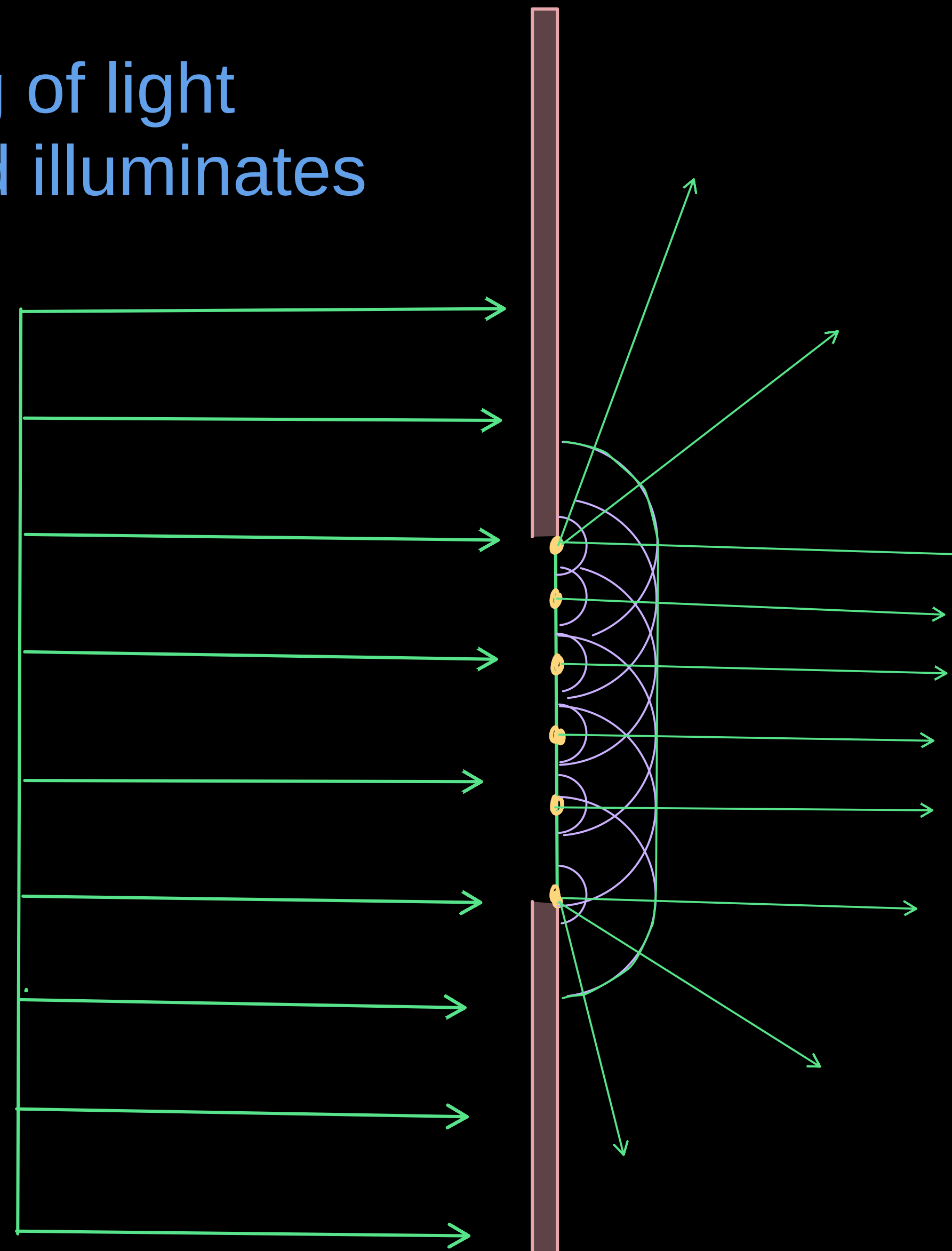
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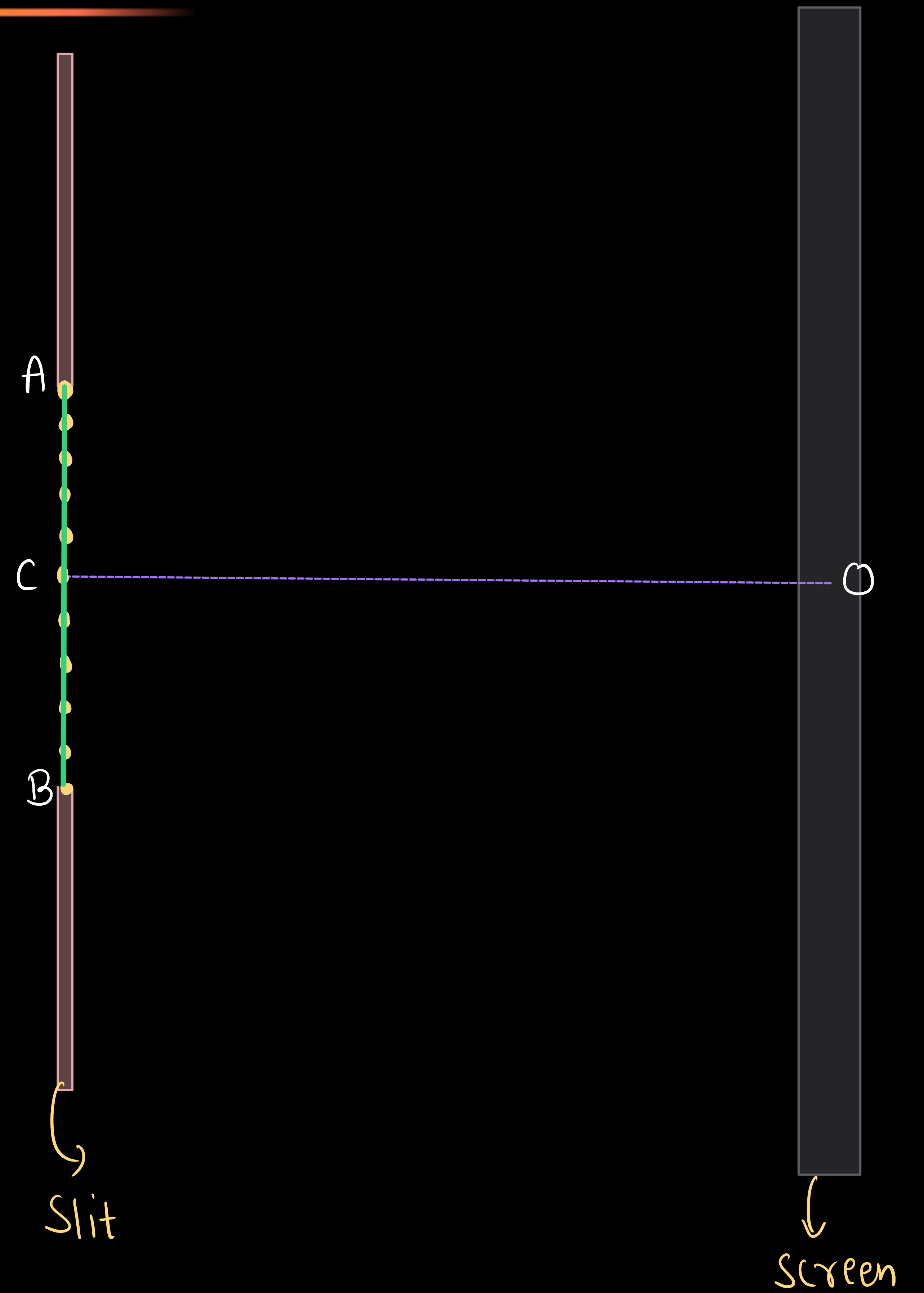


Diffraction due to a Single Slit

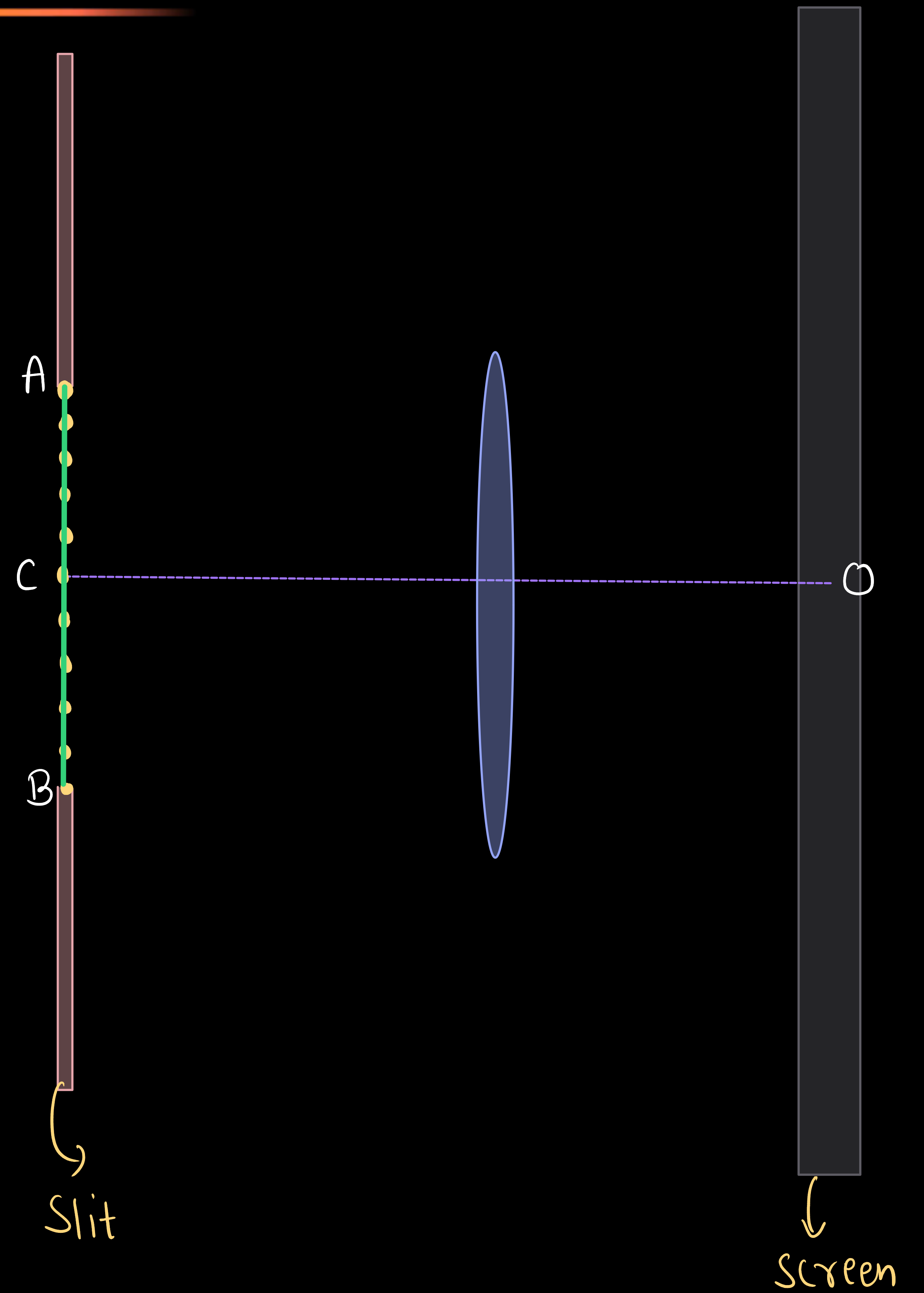
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- * Since the wavelength of light is much smaller than the dimensions of most obstacles; we do not encounter diffraction effects of light in everyday observations.
- * When the double slit in Young's experiment is replaced by a single narrow slit (illuminated by a monochromatic source), a broad pattern with a central bright region is seen. On both sides, there are alternate dark and bright regions, the intensity becoming weaker away from the centre.



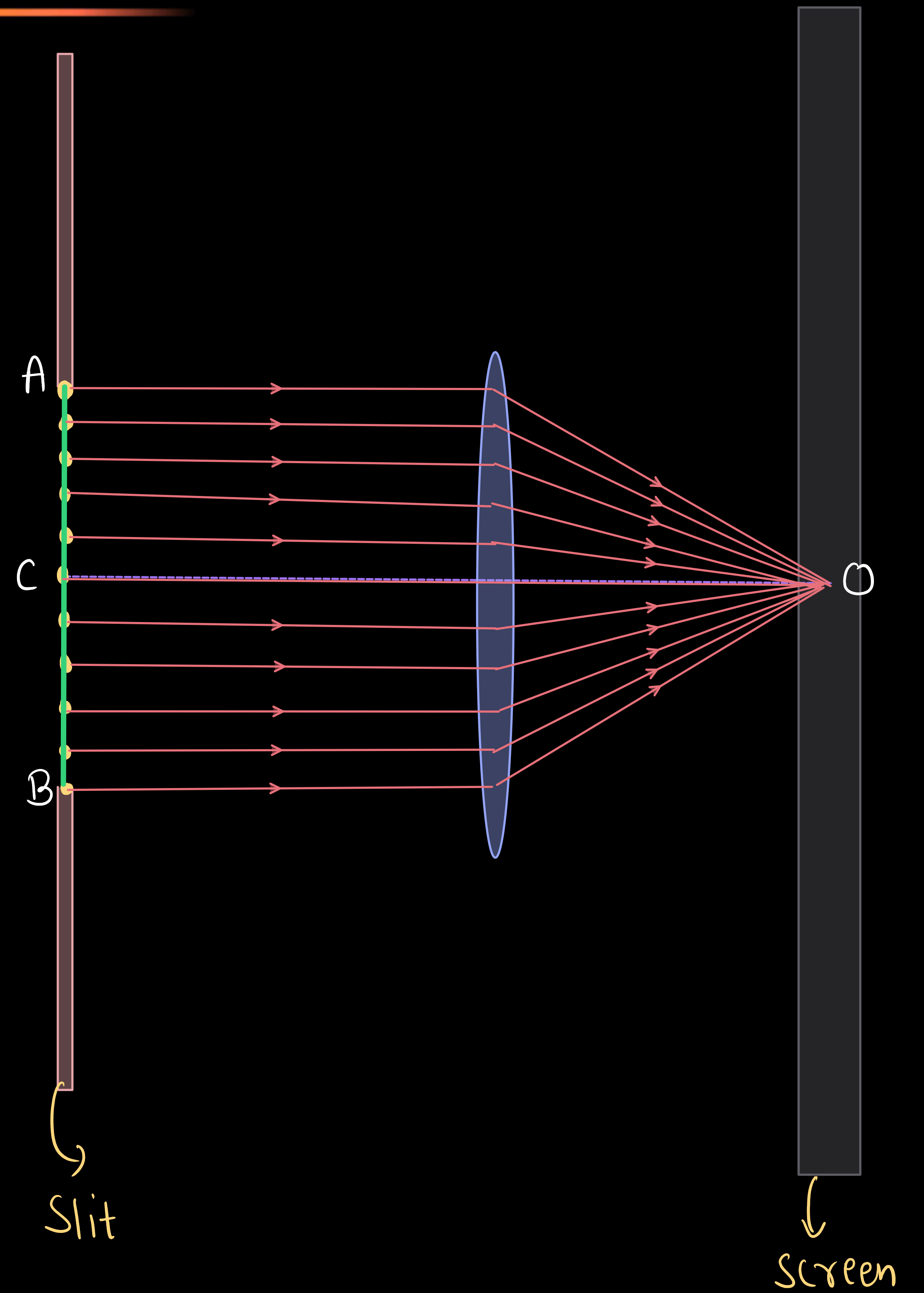
Diffraction due to a Single Slit



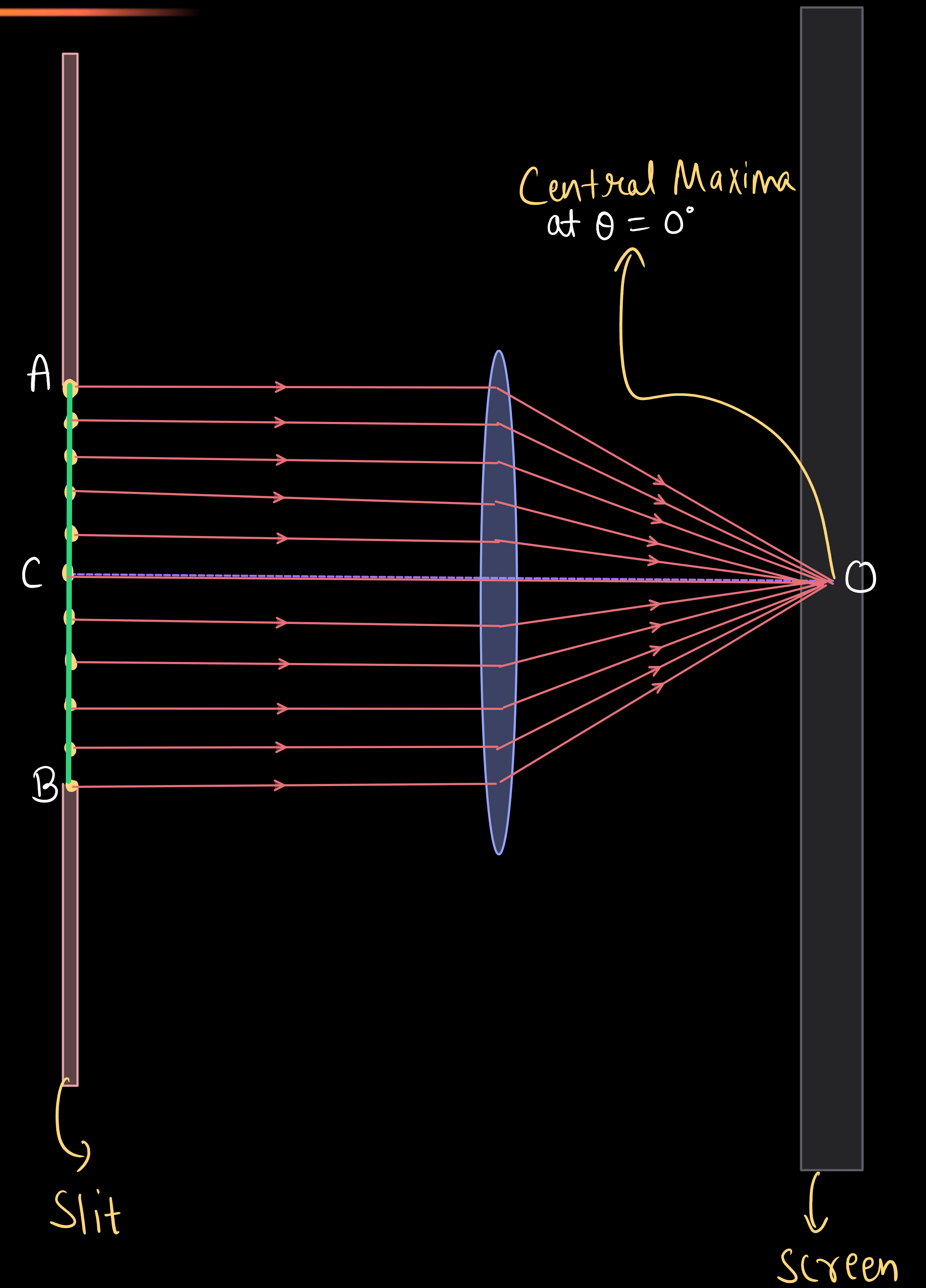
Diffraction due to a Single Slit



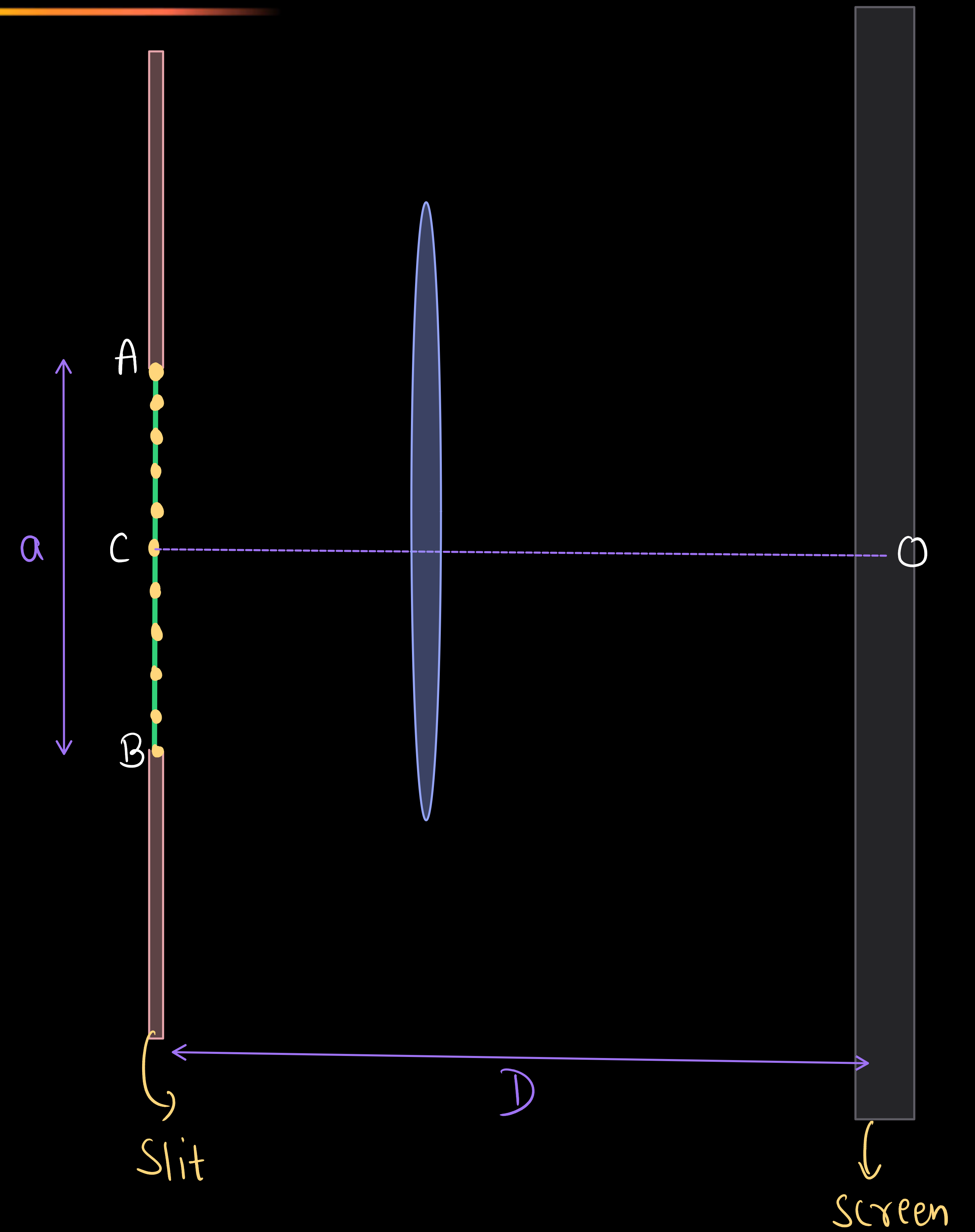
Diffraction due to a Single Slit



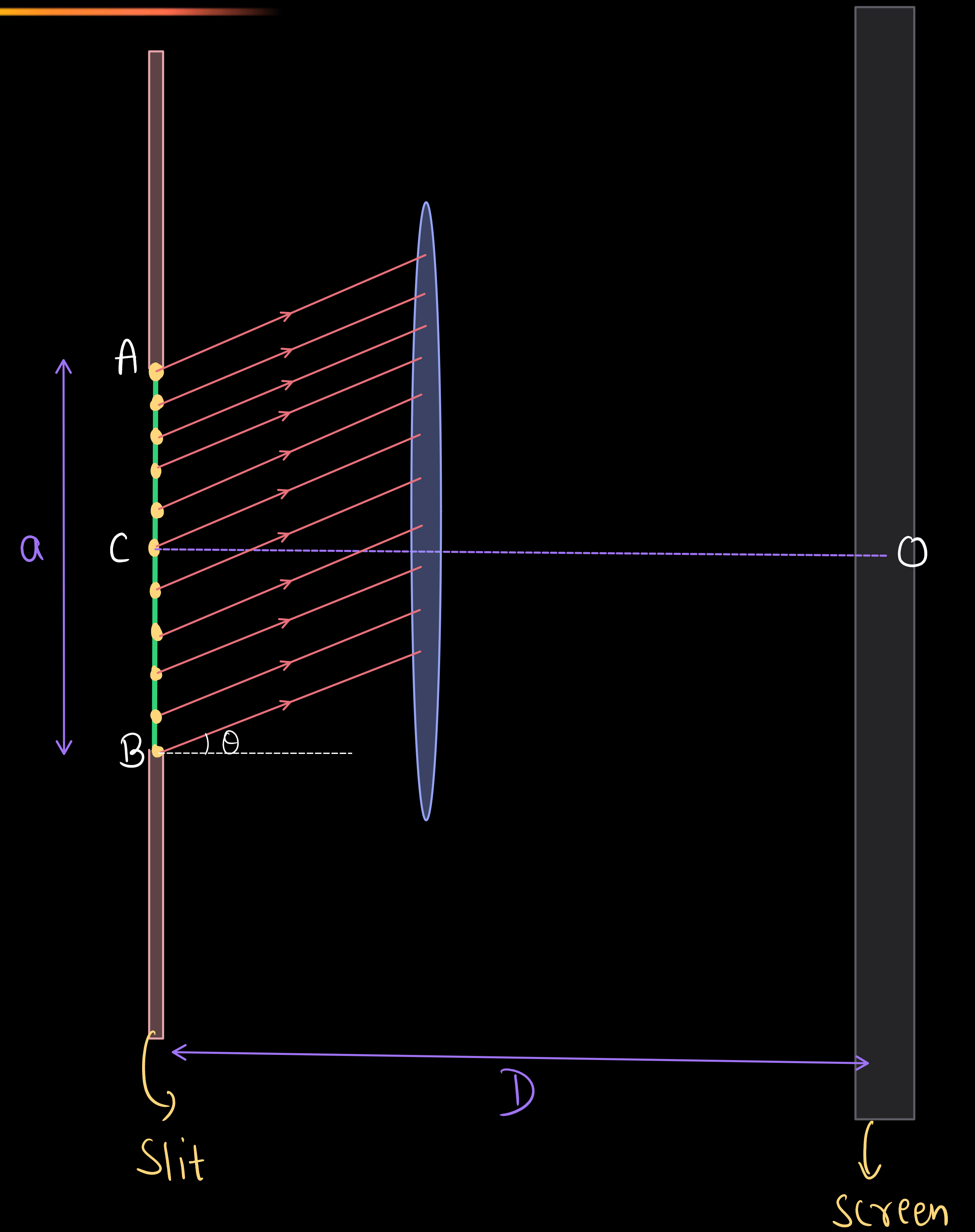
Diffraction due to a Single Slit



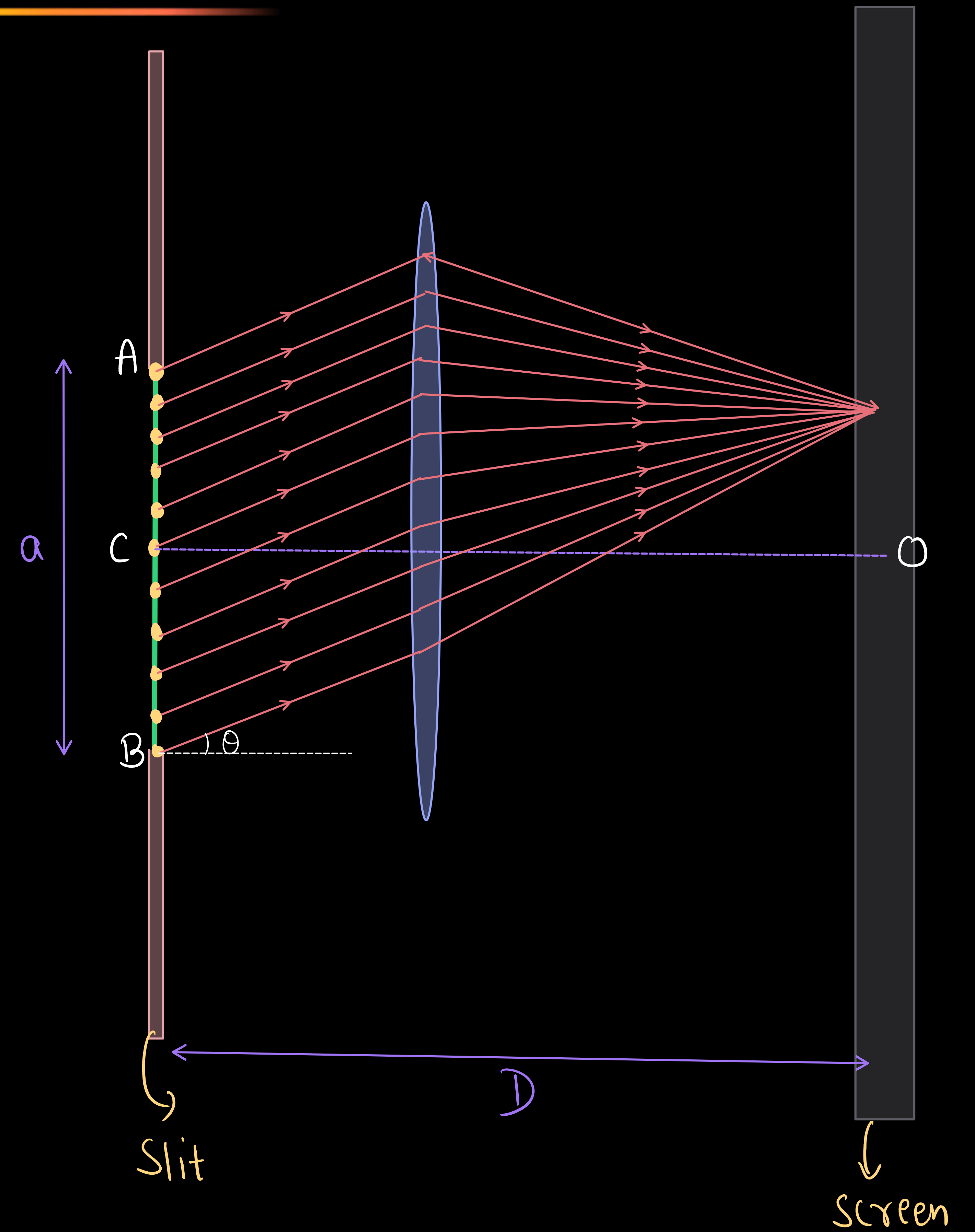
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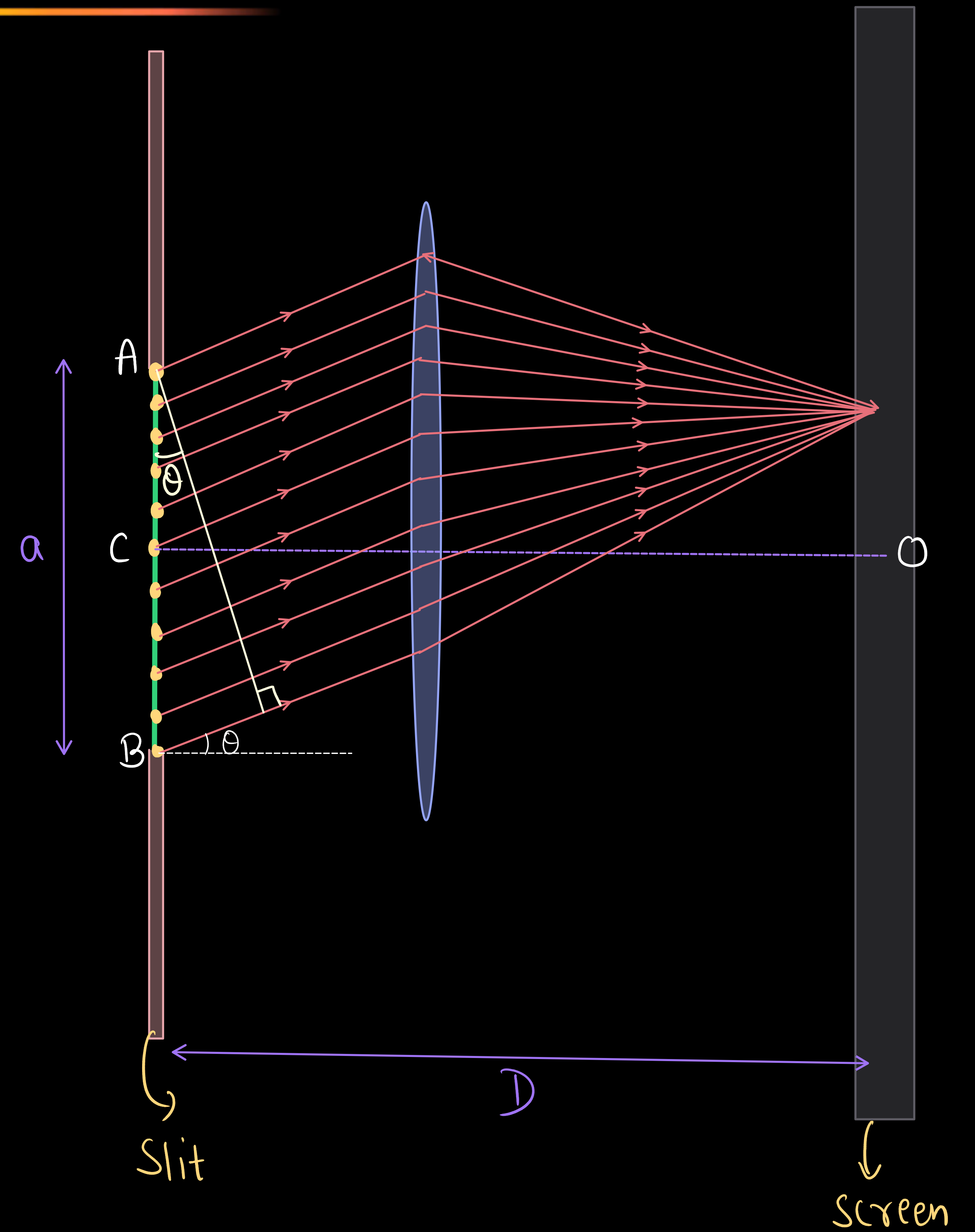
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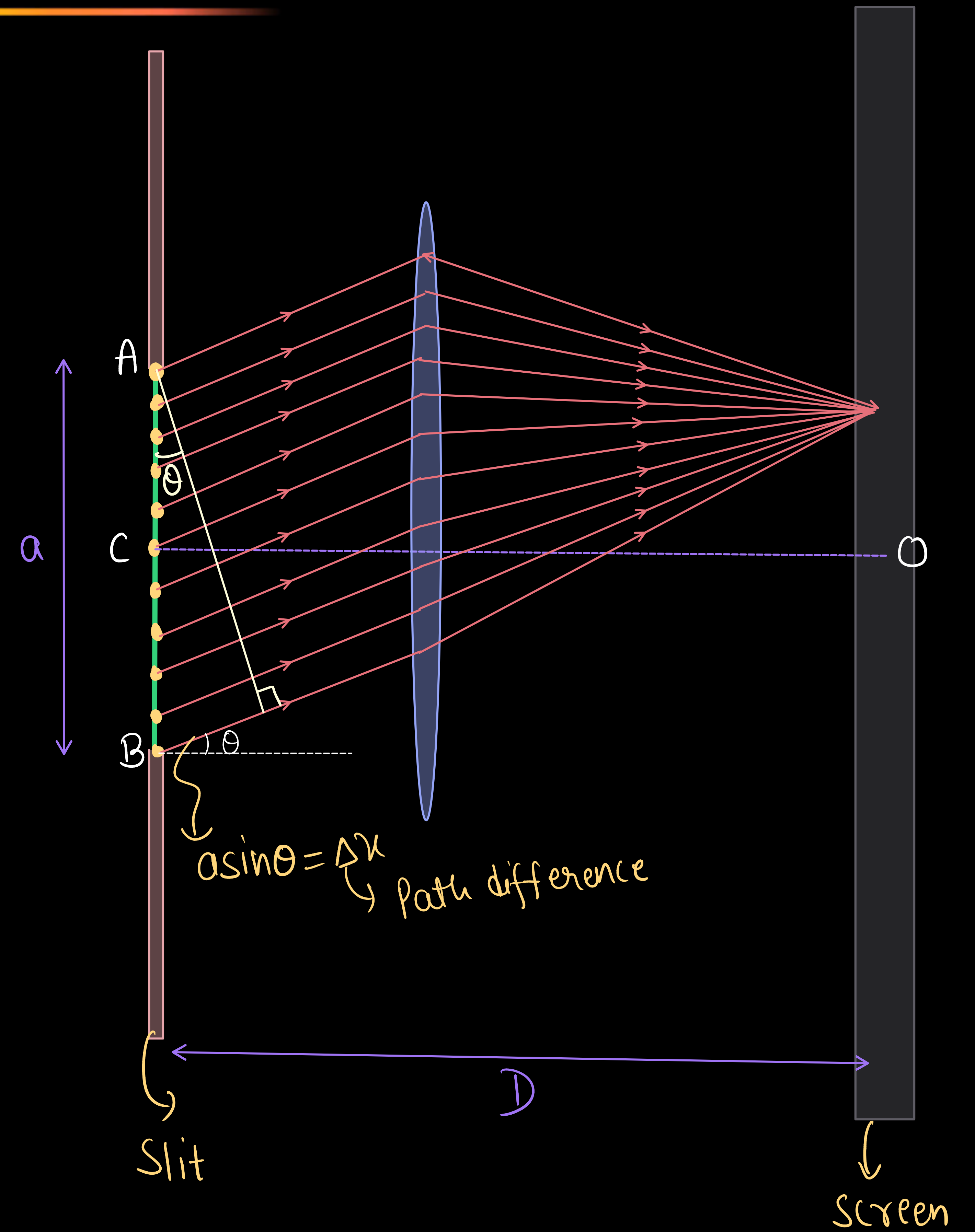
Diffraction due to a Single Slit



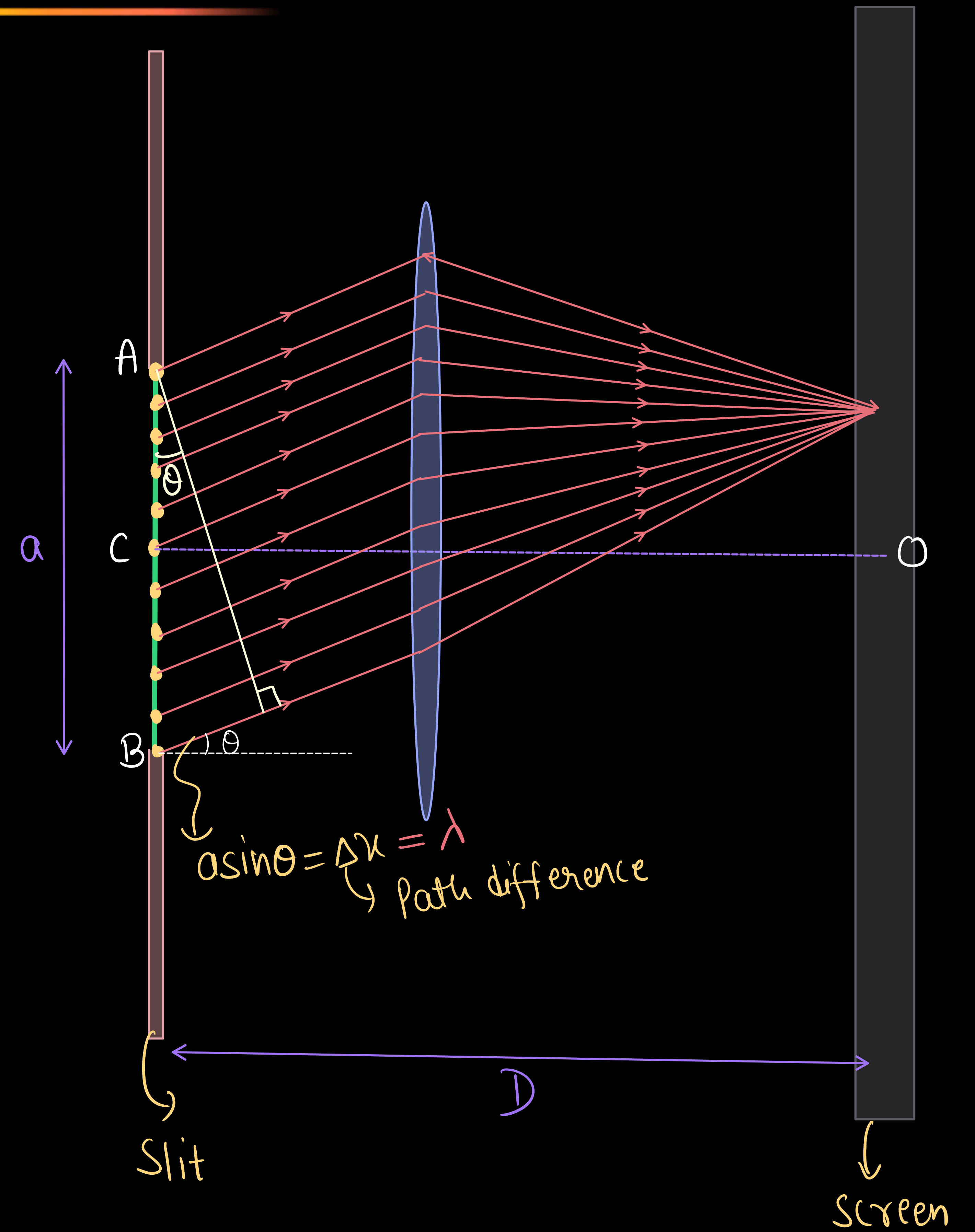
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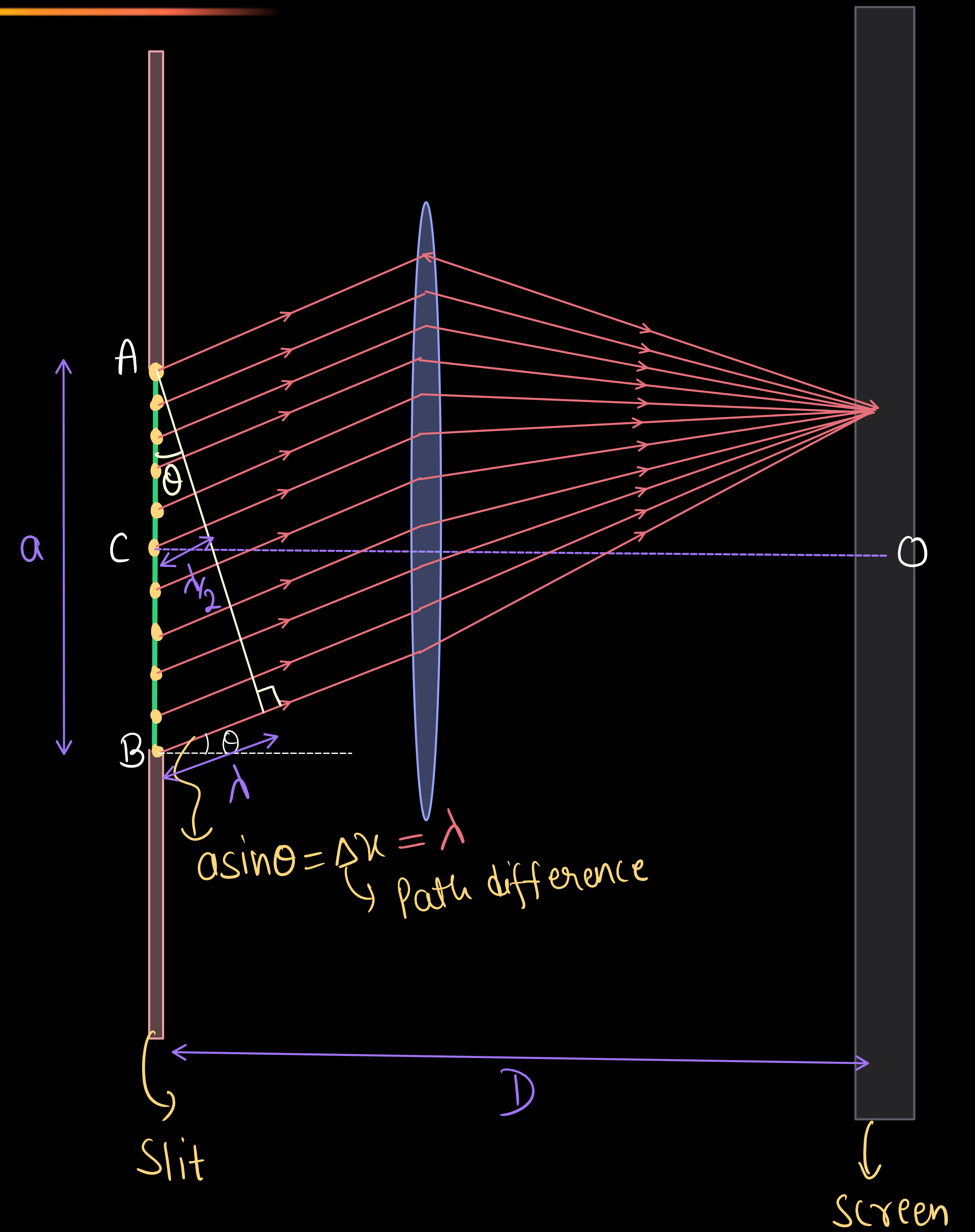
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Diffraction due to a Single Slit



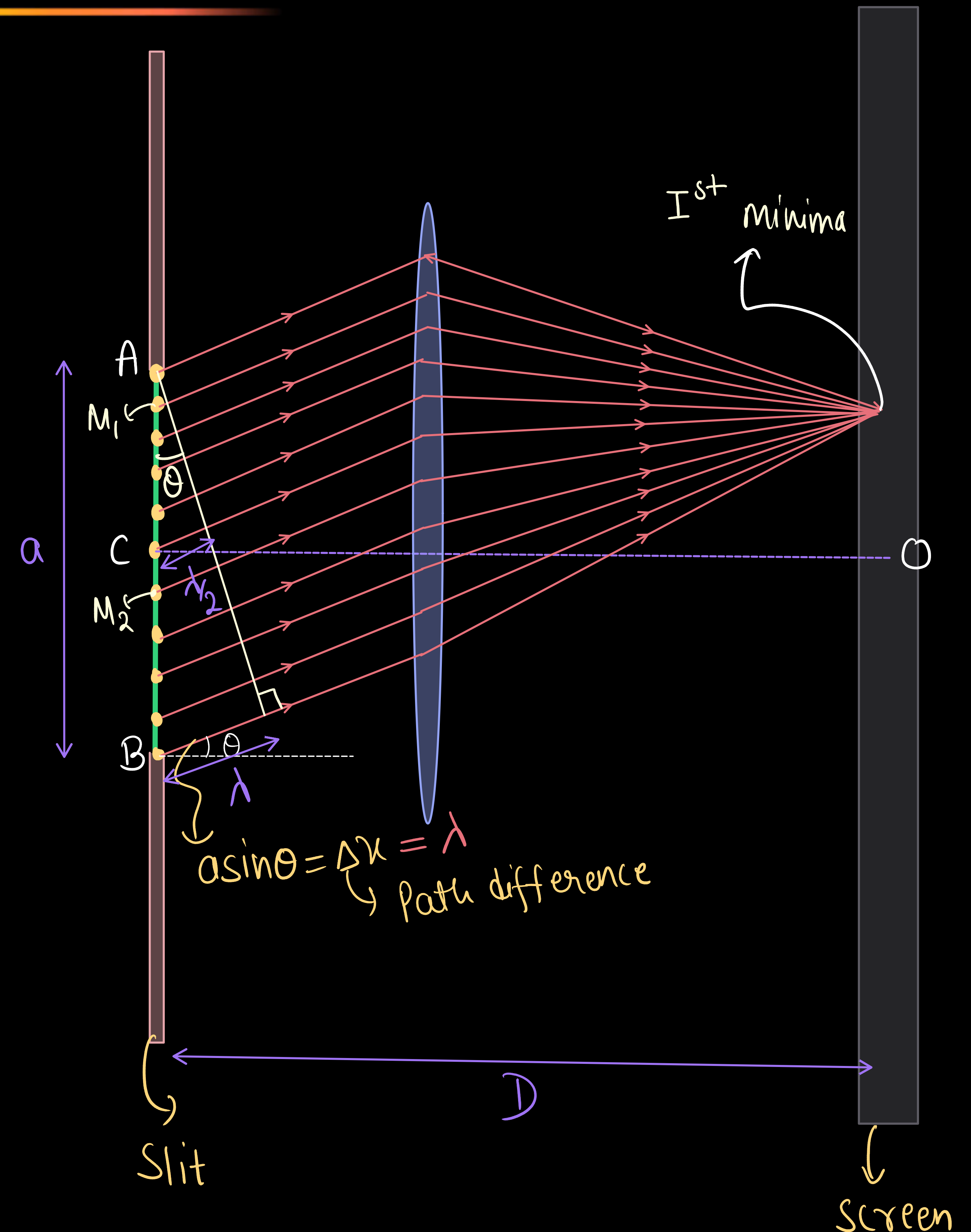
Diffraction due to a Single Slit



Diffraction due to a Single Slit

* the contributions from M1 and M2 are 180° out of phase and cancel in the direction $\theta = \lambda/a$. Contributions from the two halves of the slit AC and CB, therefore, cancel each other.

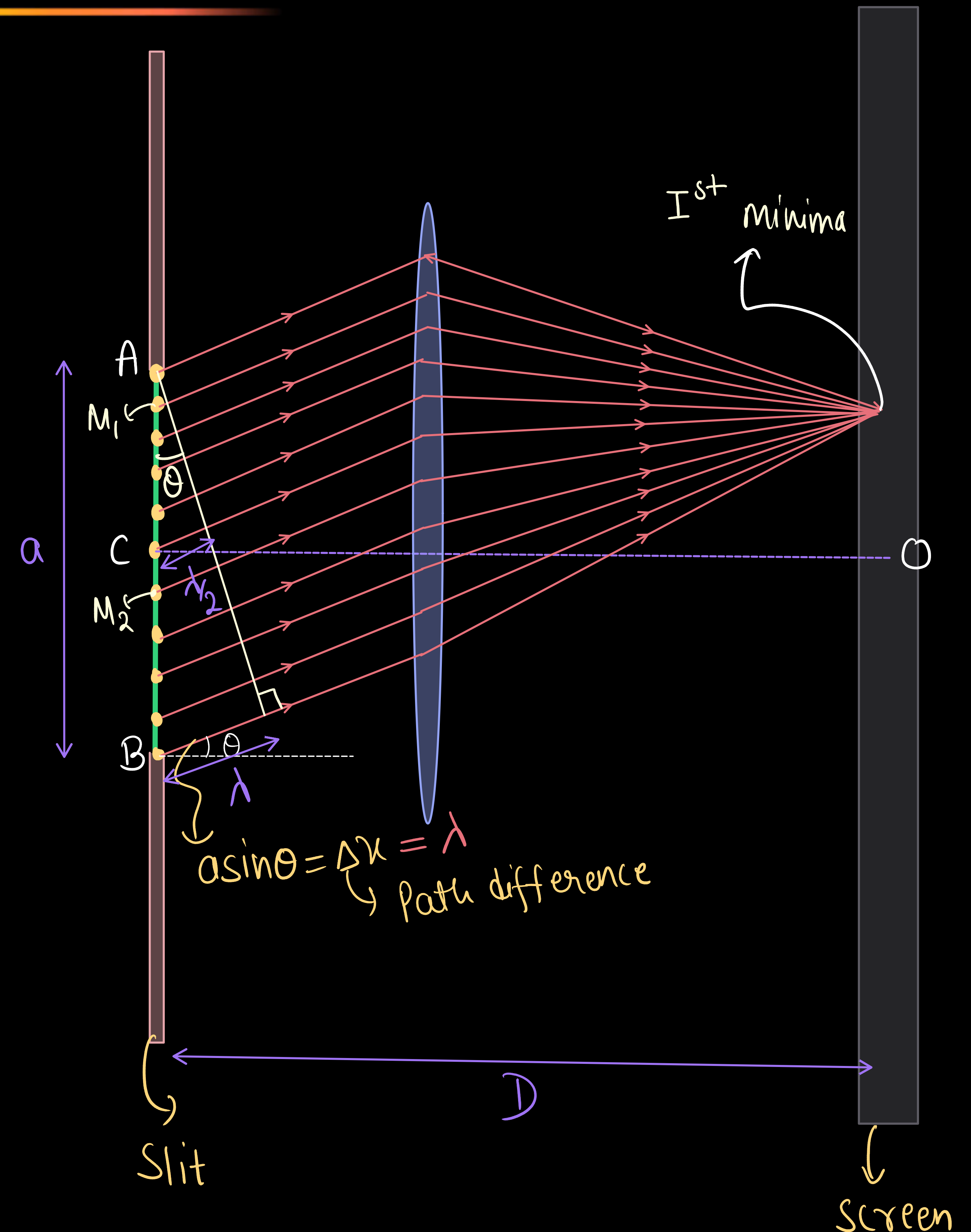
I^{st} minima - $a \sin \theta = \lambda$
 $\sin \theta = \frac{\lambda}{a} \approx \theta$



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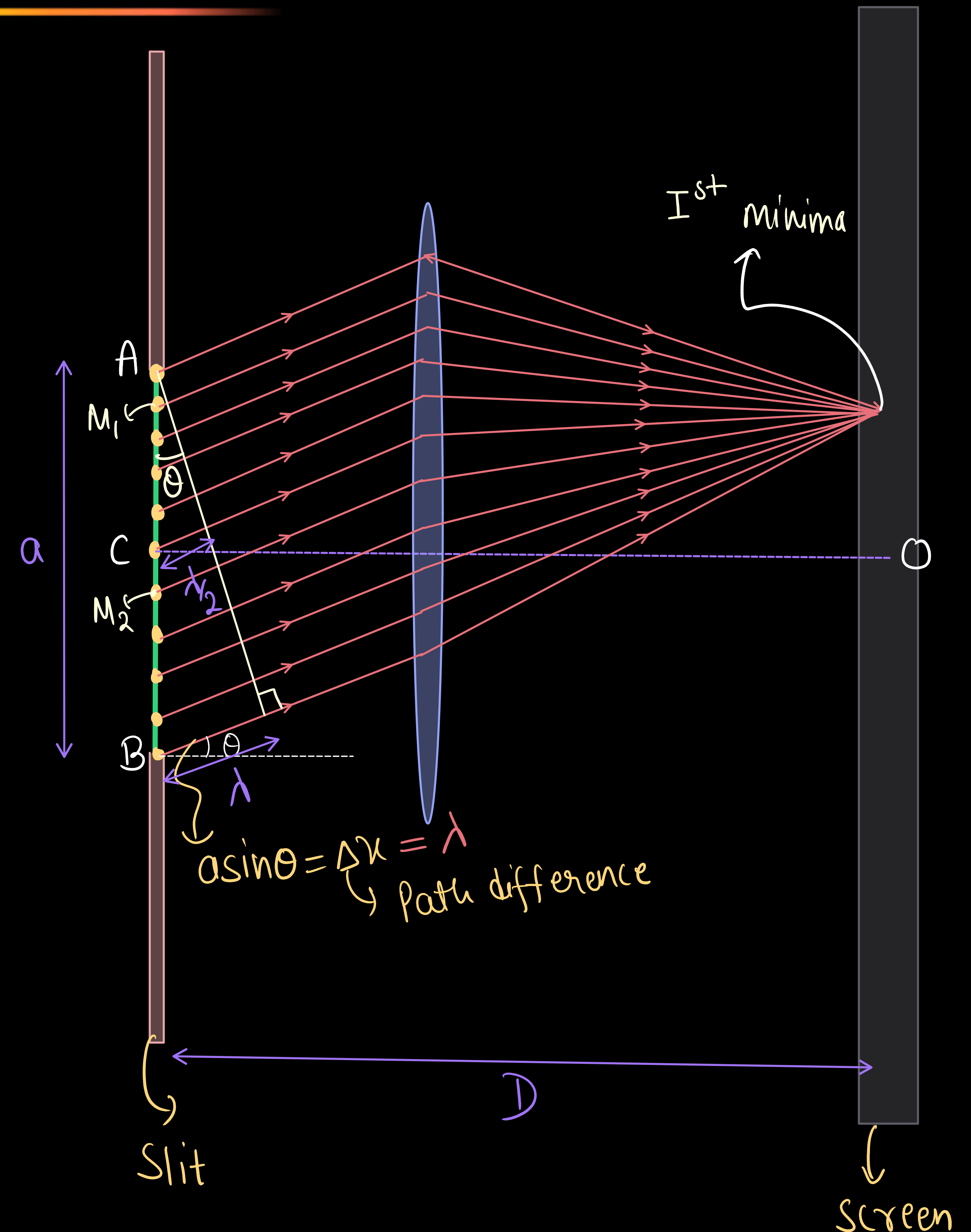


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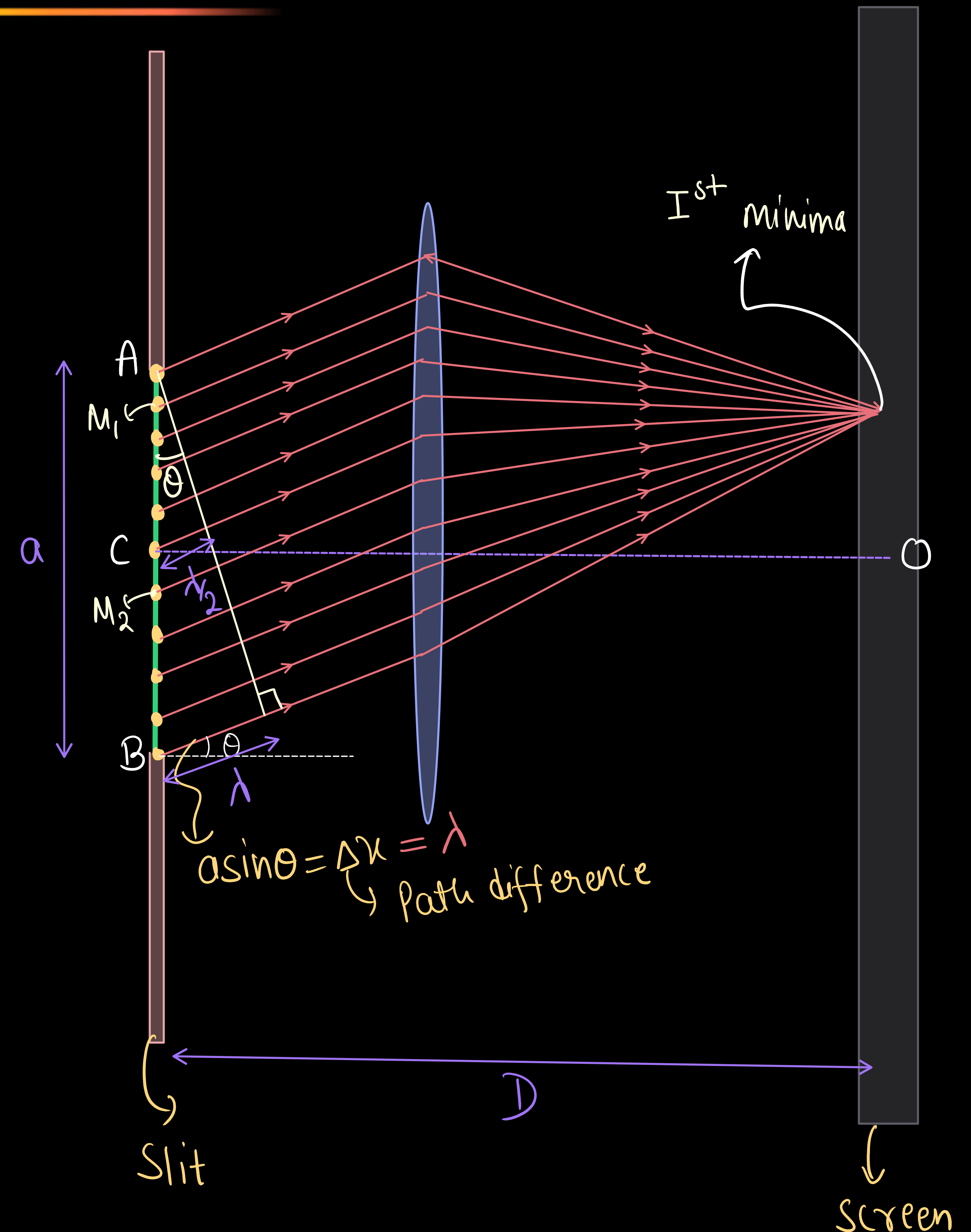


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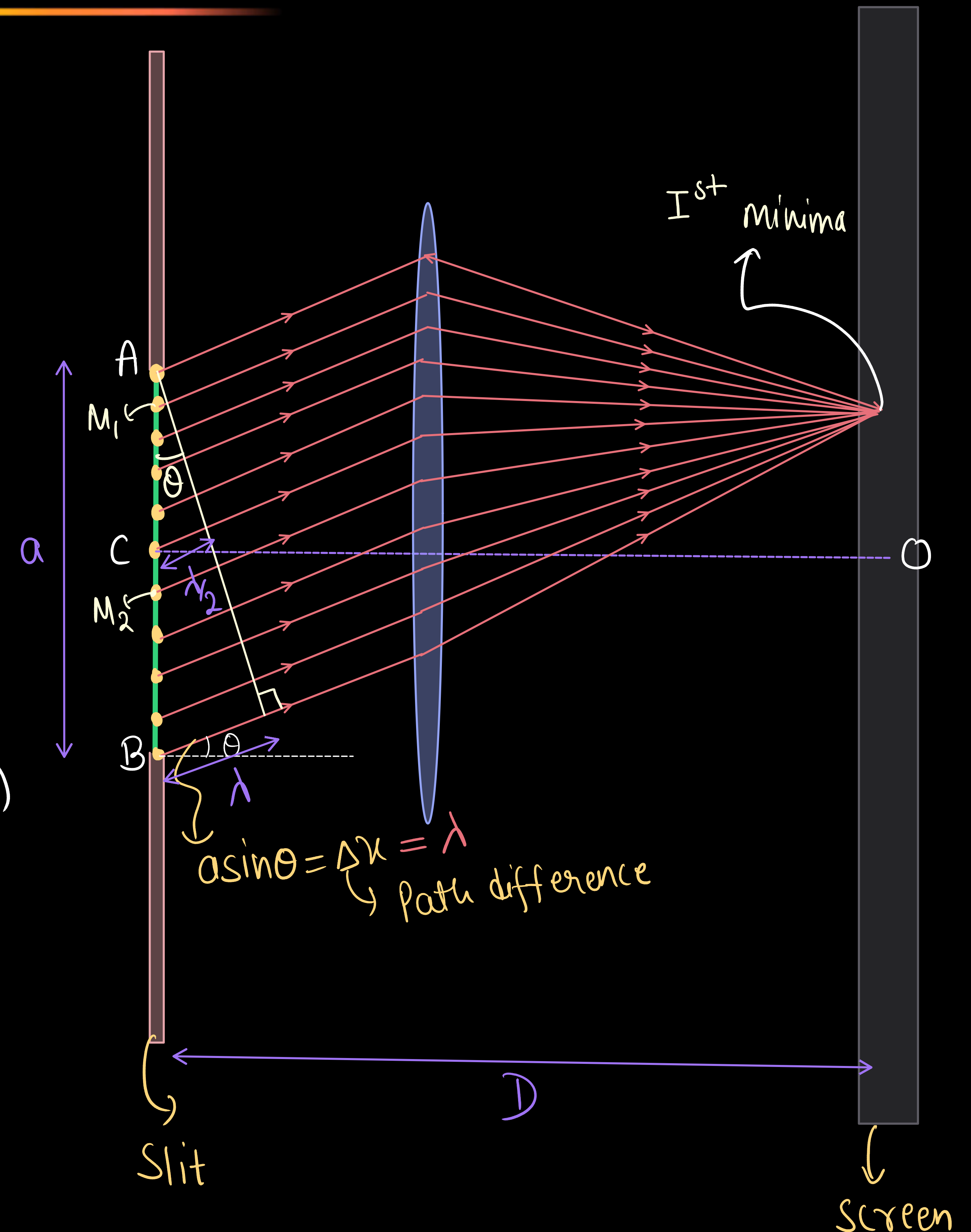
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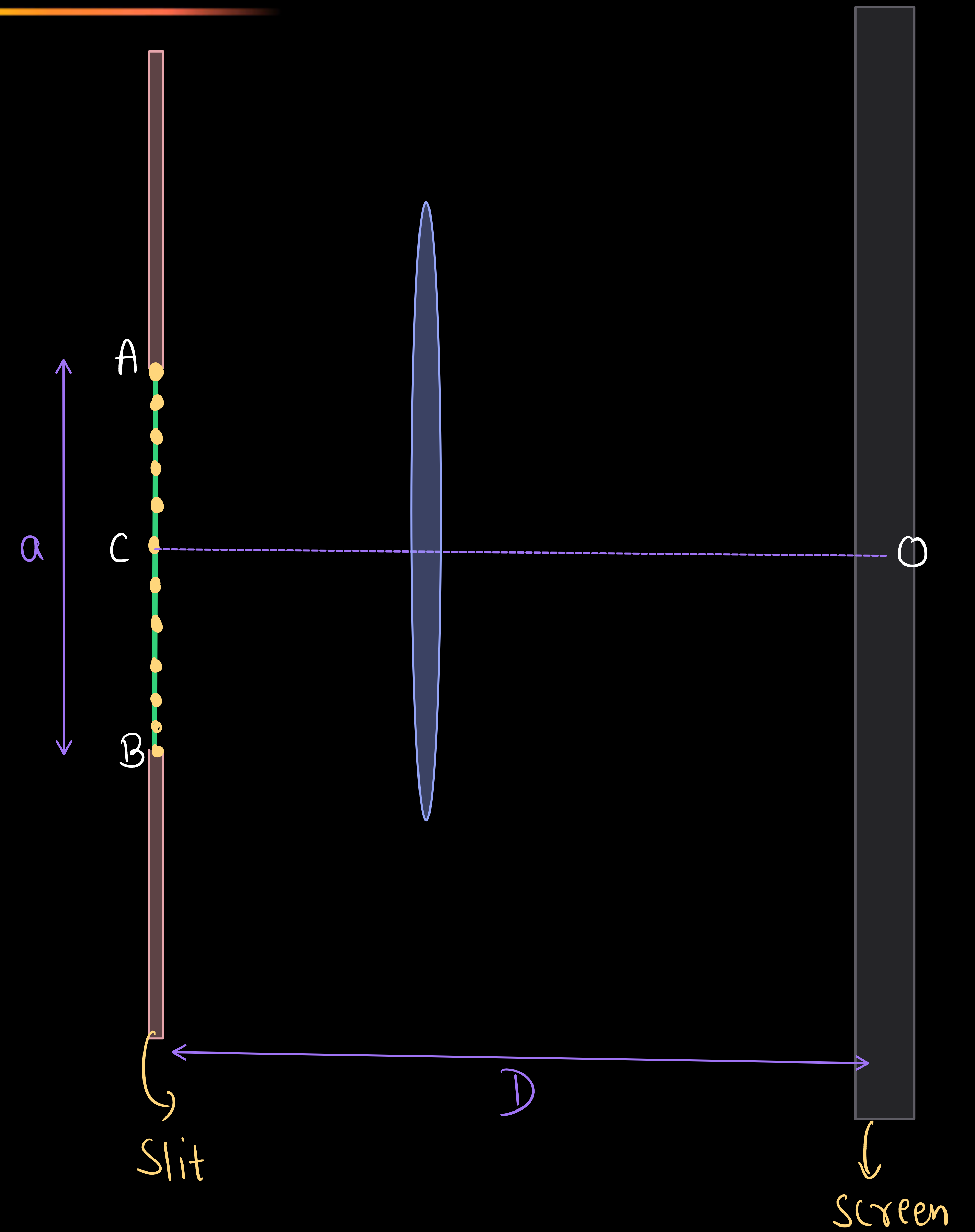
In general $\rightarrow$
 for n^{th} minima $\rightarrow$

$a \sin \theta_n = n\lambda$ ($n = \pm 1, \pm 2, \pm 3, \dots$)

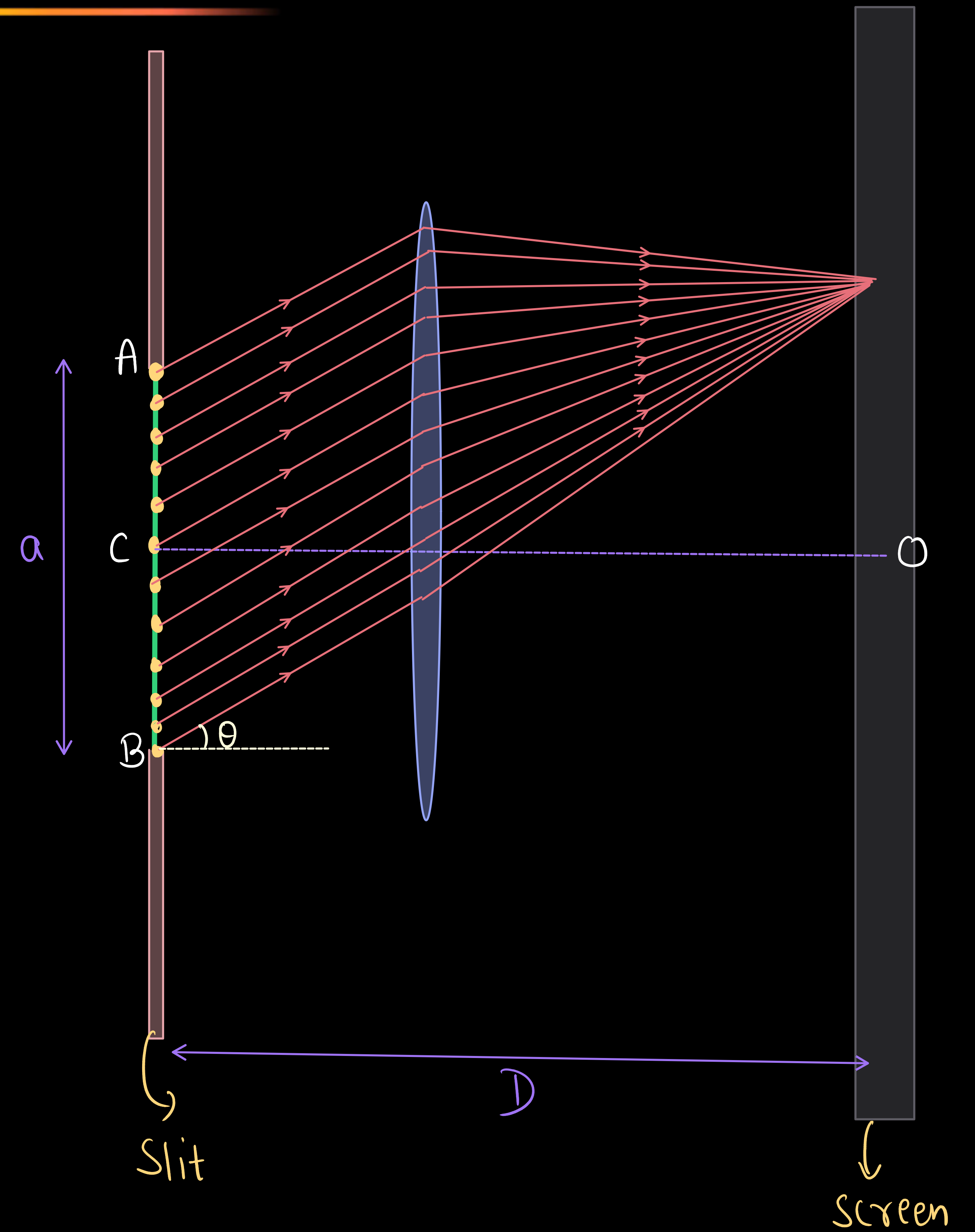
$\sin \theta_n \approx \theta_n = \frac{n\lambda}{a}$ $\rightarrow$ Angular position of n^{th} minima

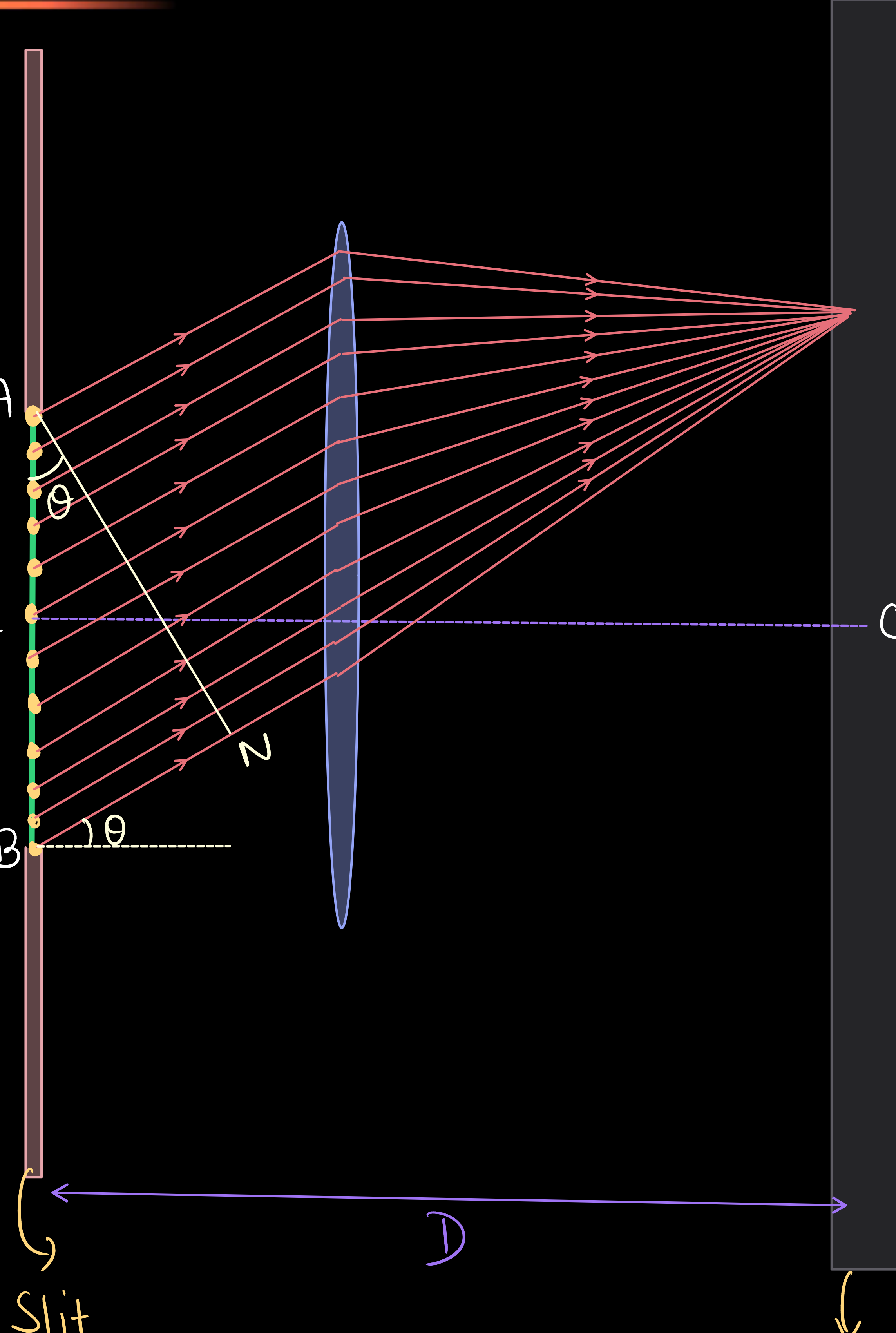


Diffraction due to a Single Slit

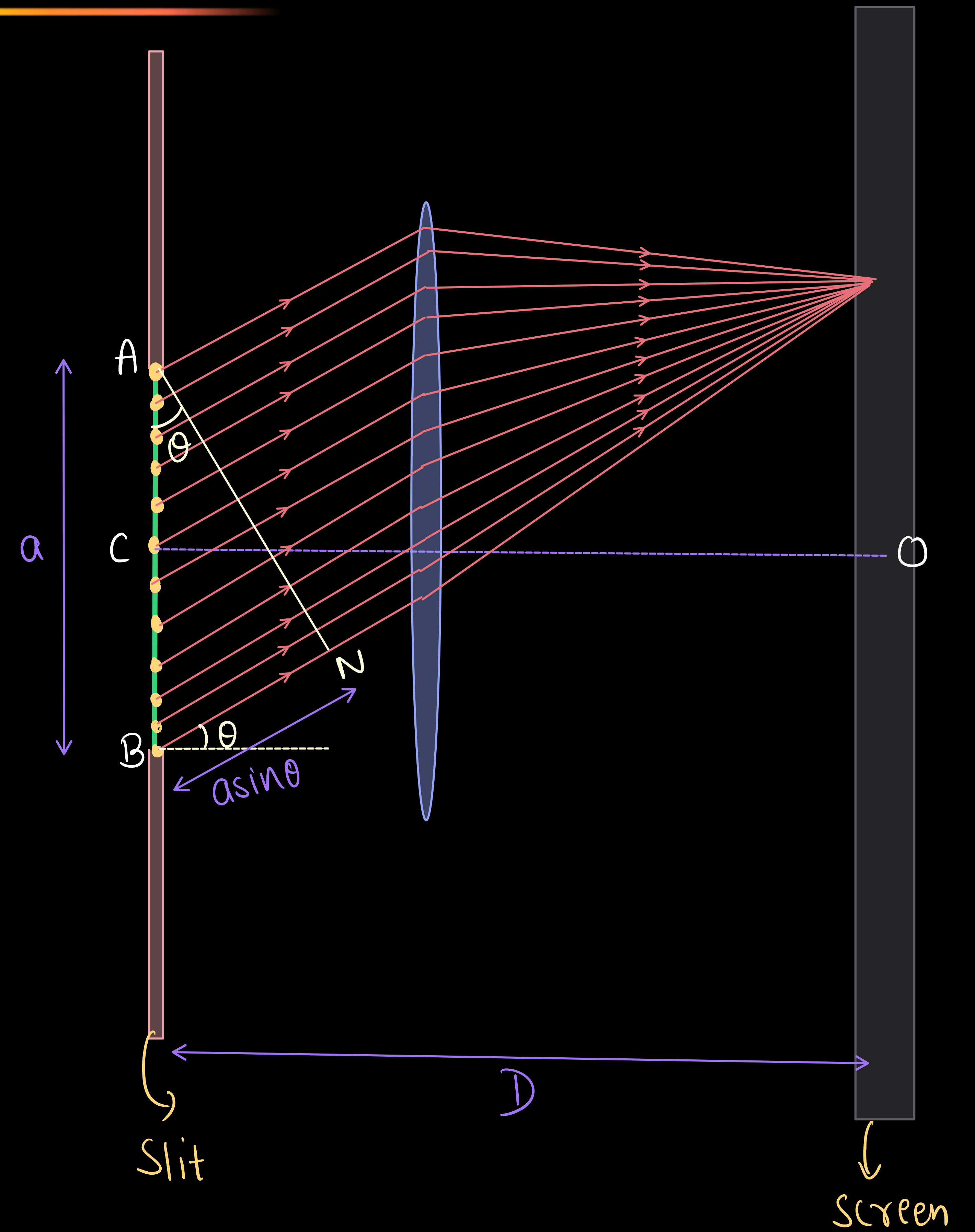


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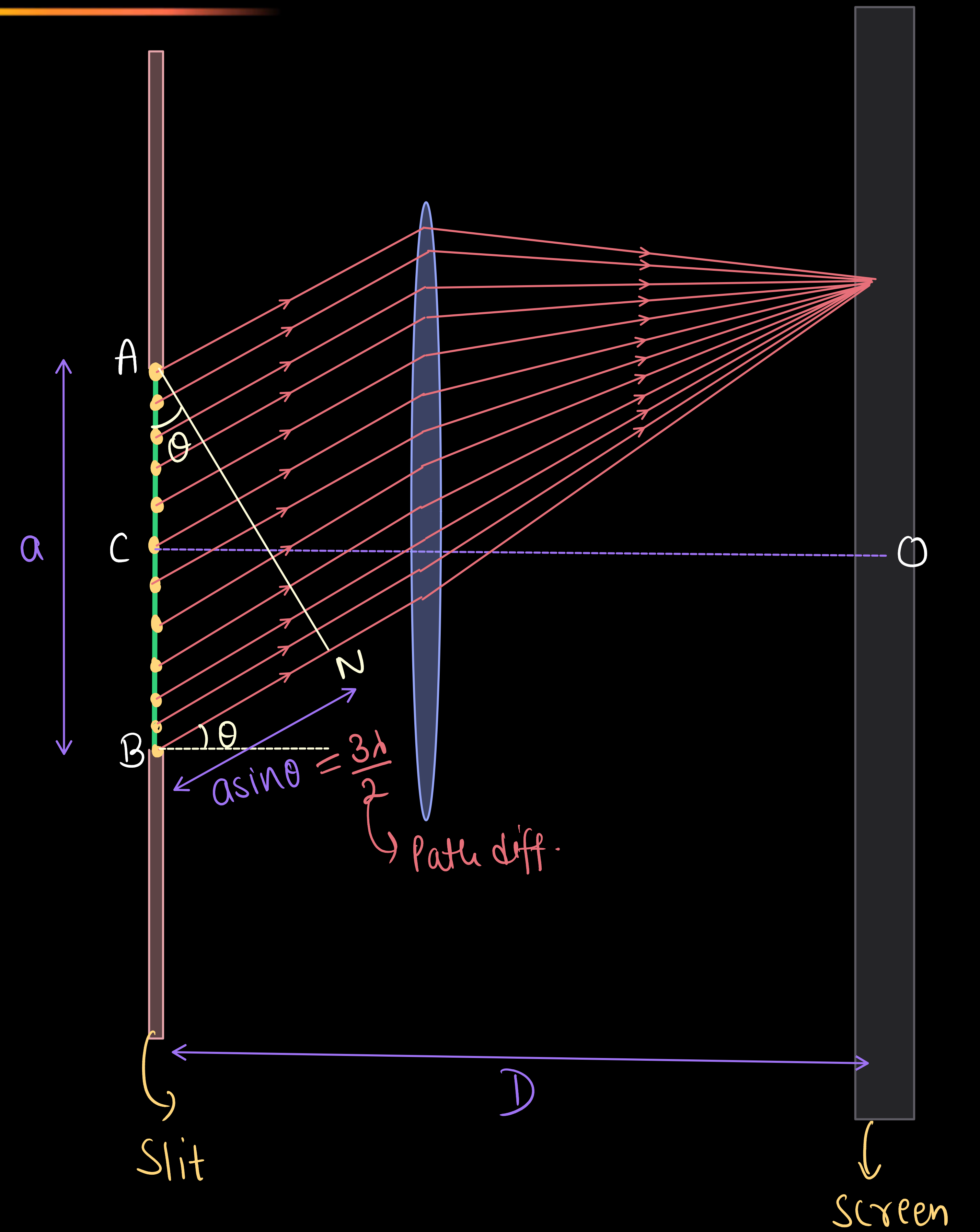




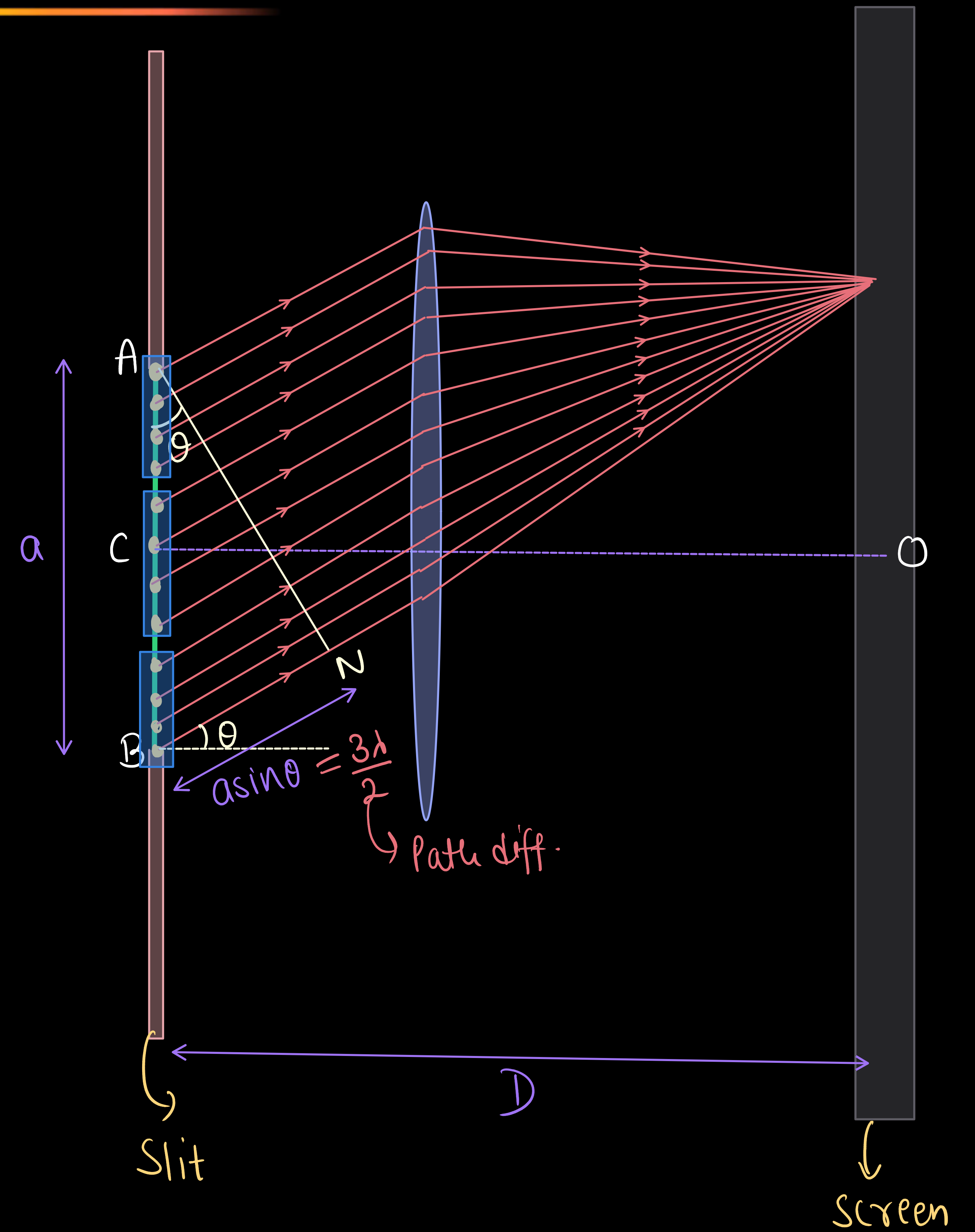
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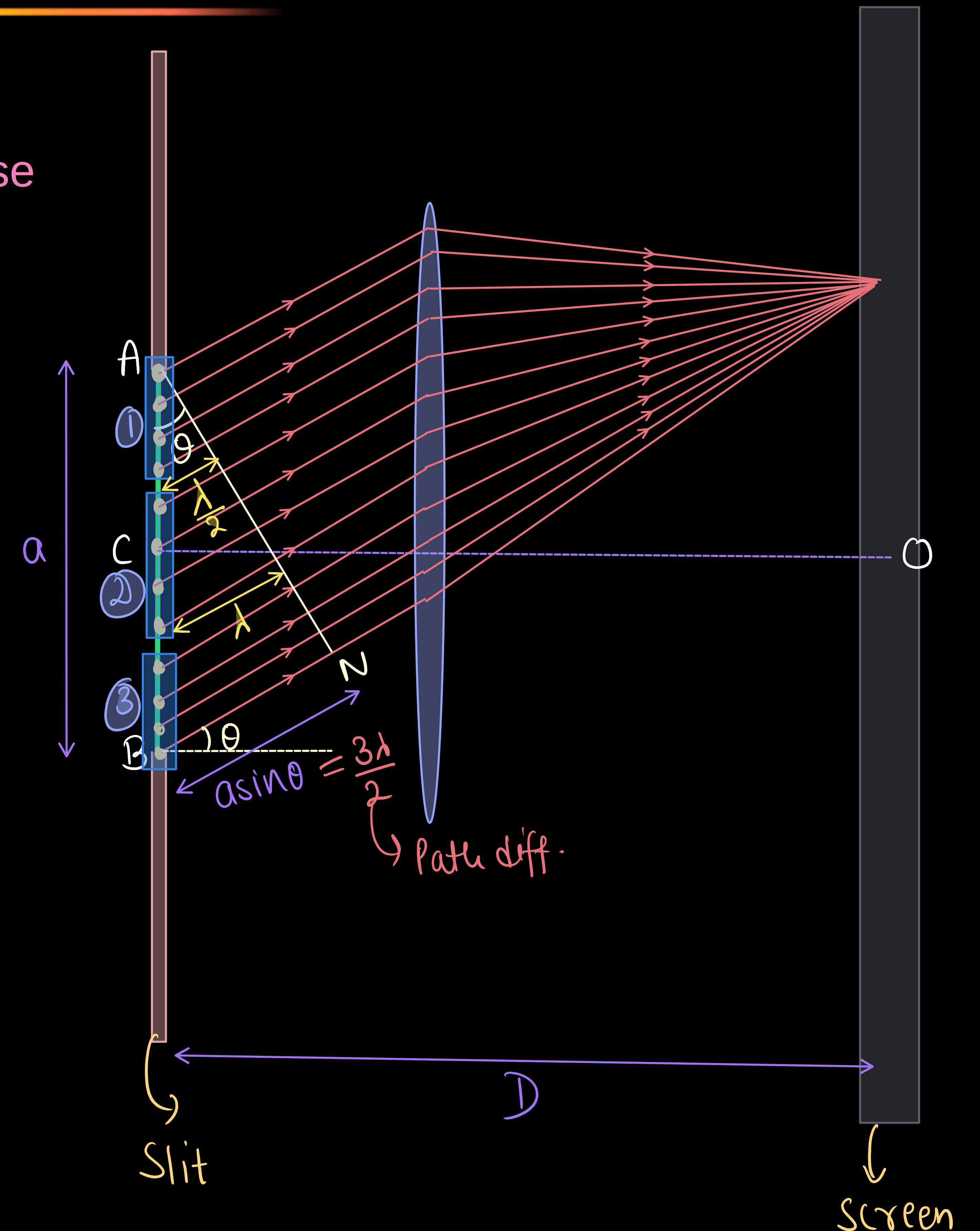


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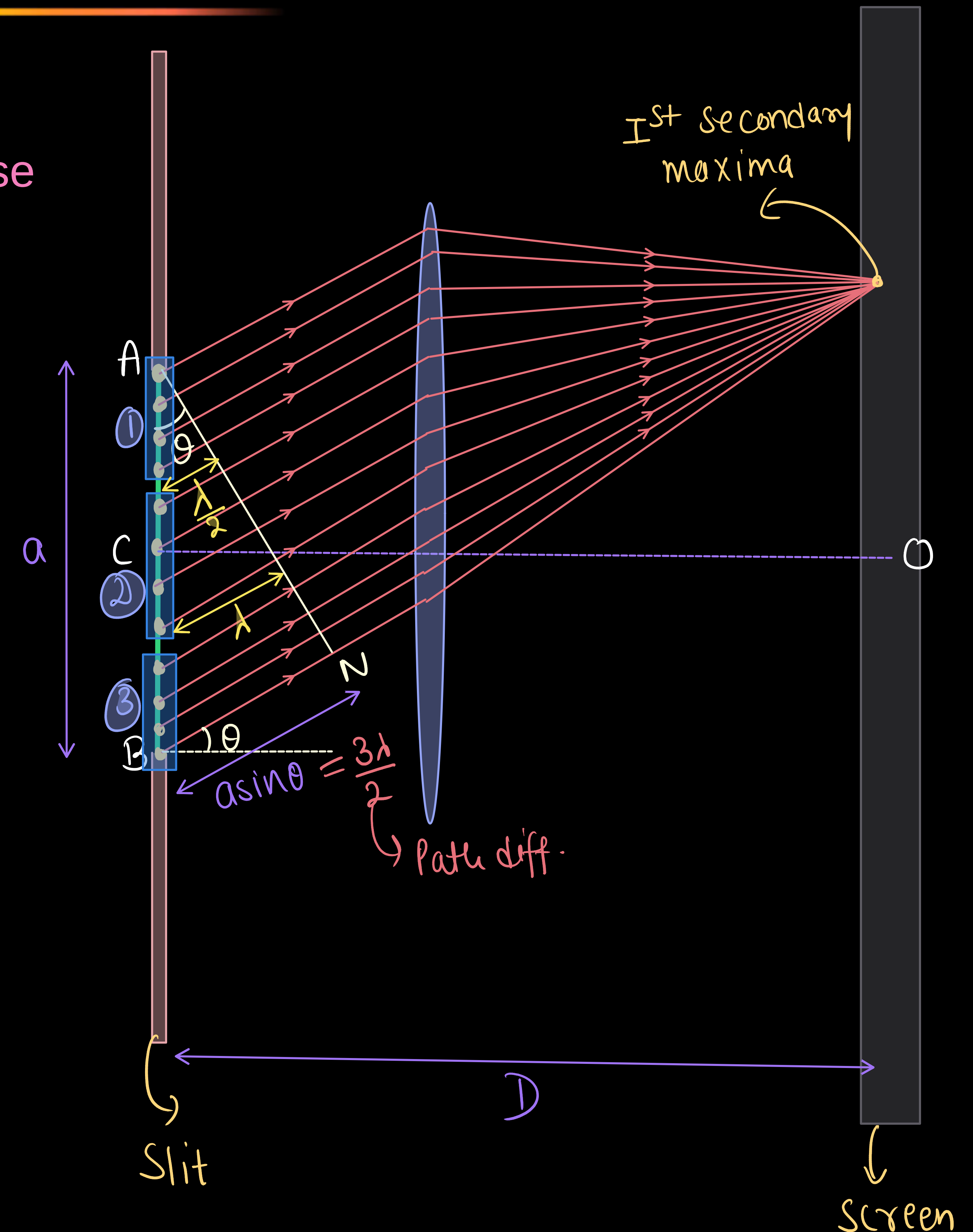
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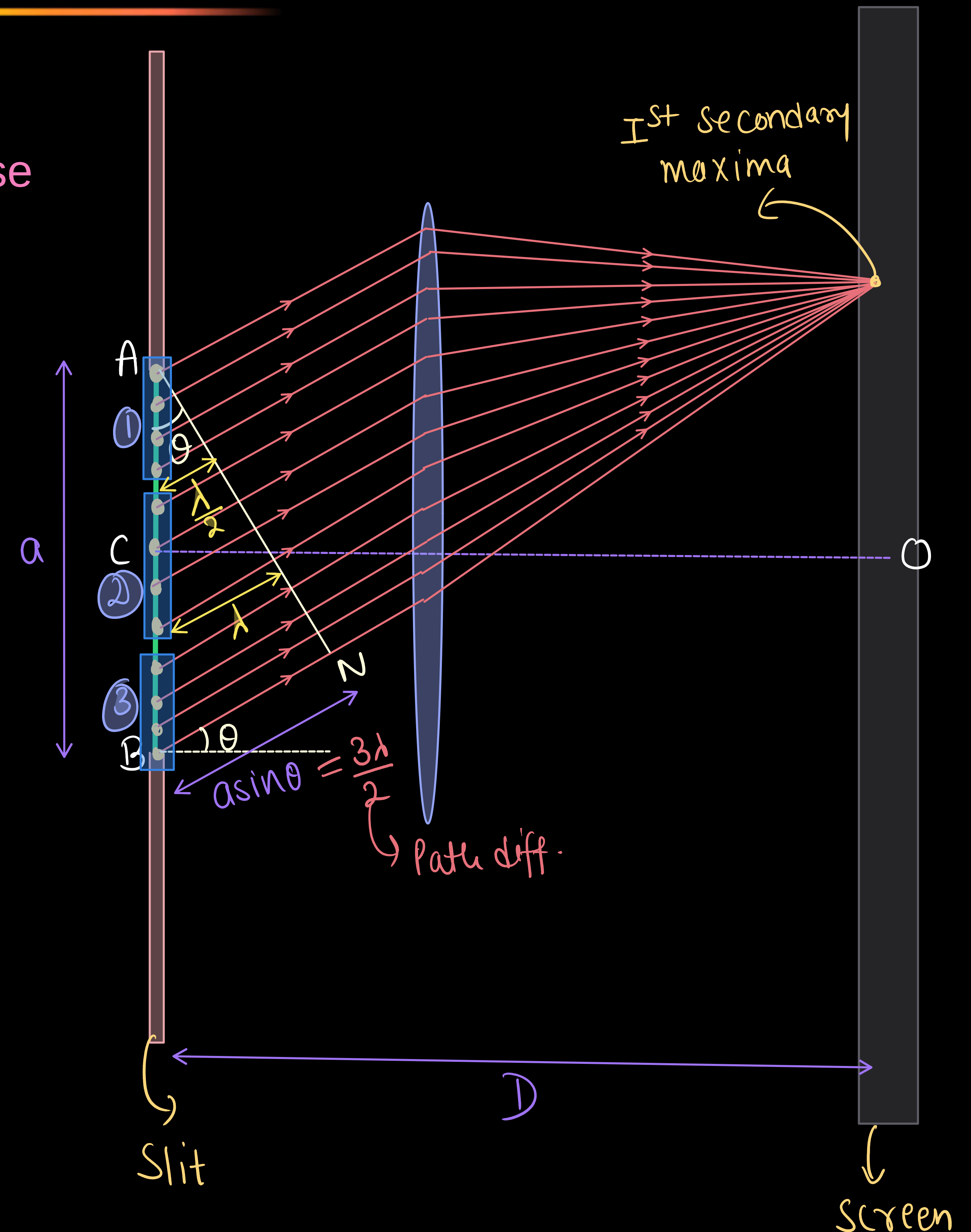


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Angular position of 1st secondary maxima $\rightarrow$

$$a \sin \theta_1 = \frac{3\lambda}{2}$$

$$\sin \theta_1 \approx \theta_1 = \frac{3\lambda}{2a}$$


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Angular position of 1st secondary maxima →

$$a \sin \theta_1 = \frac{3\lambda}{2}$$

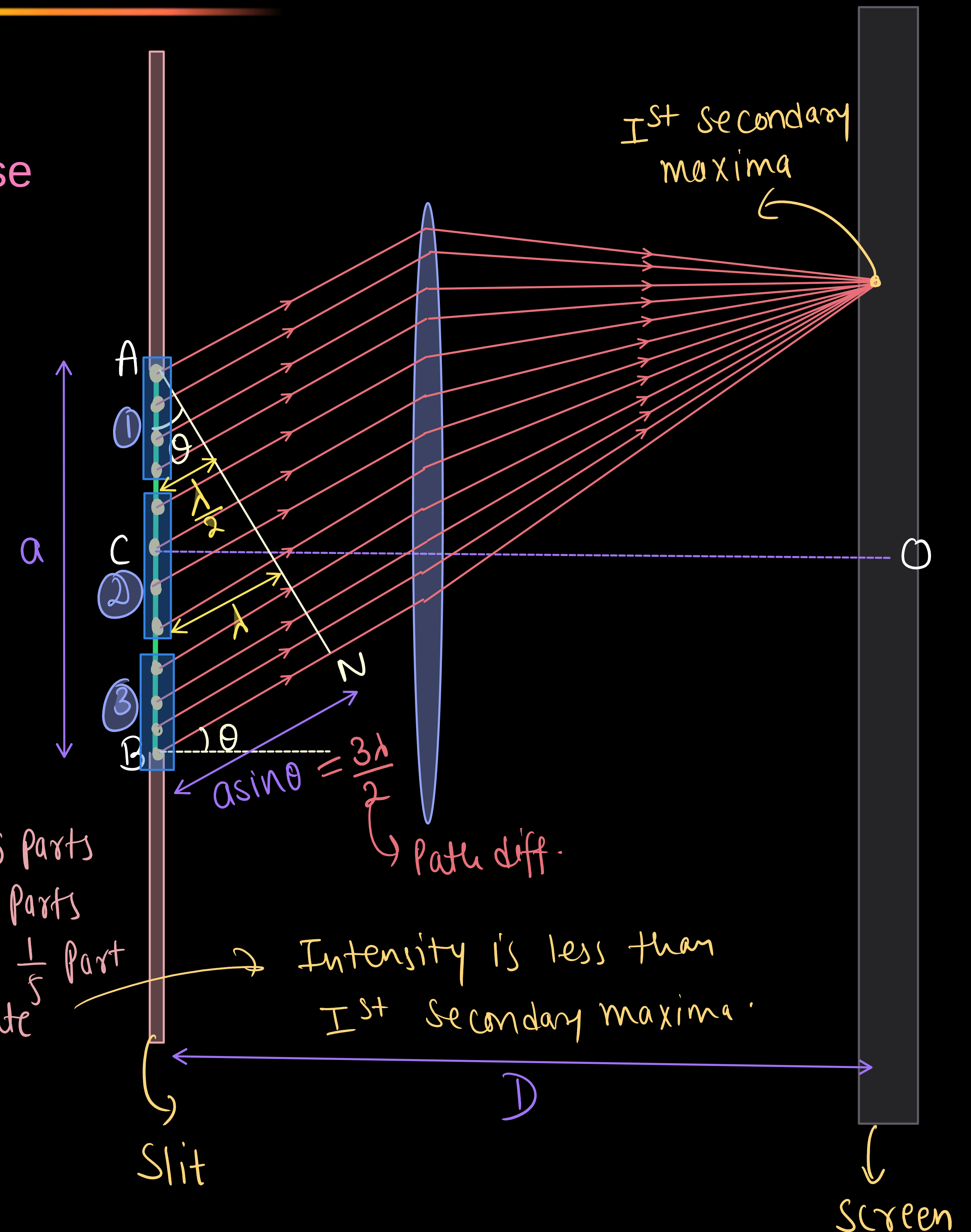
$$\sin \theta_1 \approx \theta_1 = \frac{3\lambda}{2a}$$

Similarly Angular position of 2nd secondary maxima →

$$a \sin \theta_2 = \frac{5\lambda}{2} \Rightarrow \theta_2 = \frac{5\lambda}{2a}$$

Divide slit into 5 parts
Contribution from 4 parts
Cancel only $\frac{1}{5}$ part
Contribute

Intensity is less than
1st Secondary maxima.



Diffraction due to a Single Slit

- * Divide the slit into three equal parts.
- * The first two parts have a $\lambda/2$ path difference. The contributions of these two halves cancel in the same manner as described earlier.
- * Only the remaining one-third of the slit contributes to the intensity at a point between the two minima.
- * Clearly, this will be much weaker than the central maximum (where the entire slit contributes in phase).

Angular position of 1st secondary maxima $\rightarrow$

$$a \sin \theta_1 = \frac{3\lambda}{2}$$

$$\sin \theta_1 \approx \theta_1 = \frac{3\lambda}{2a}$$

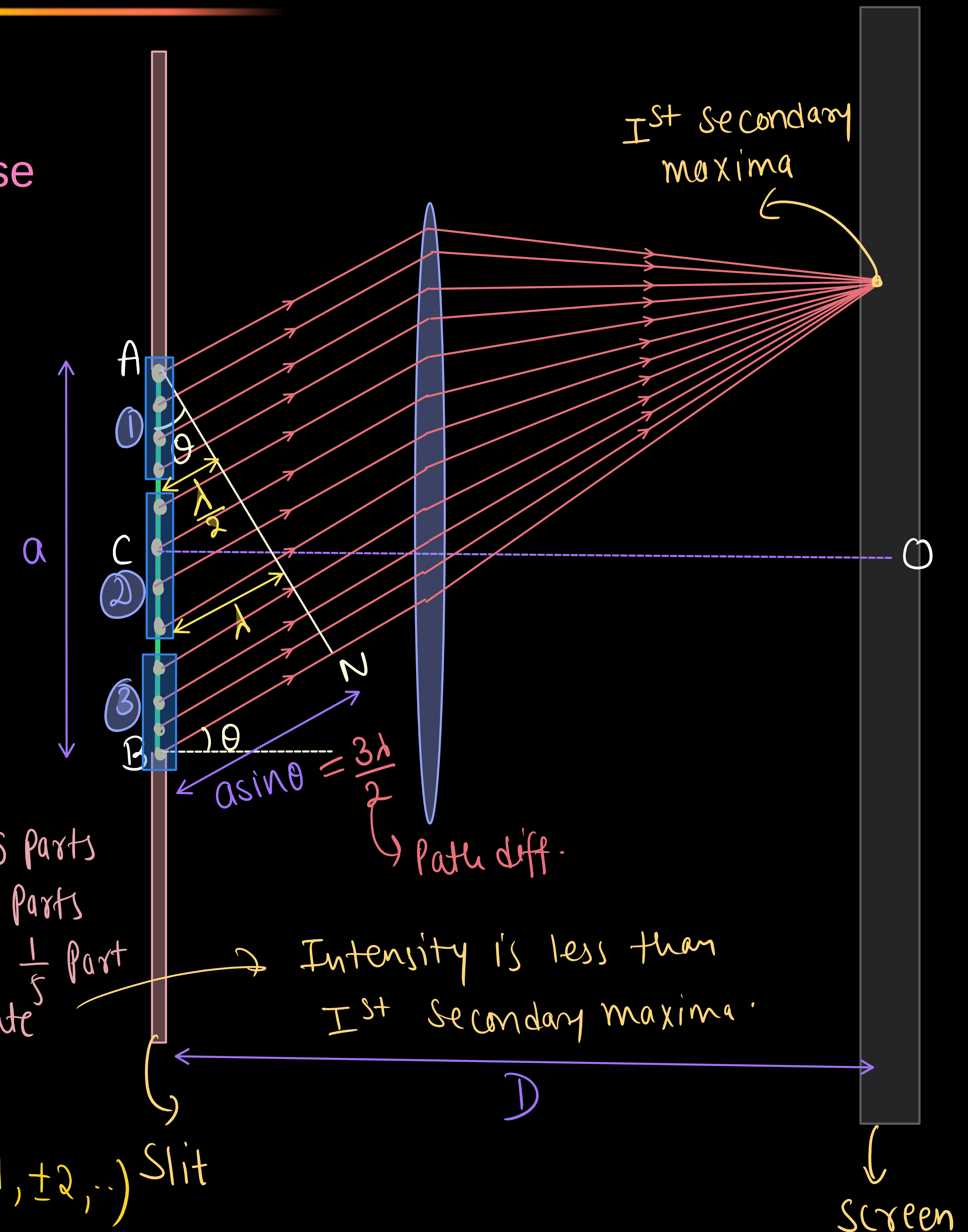
Similarly Angular position of 2nd secondary maxima $\rightarrow$

$$a \sin \theta_2 = \frac{5\lambda}{2} \Rightarrow \theta_2 = \frac{5\lambda}{2a}$$

Divide slit into 5 parts
Contribution from 4 parts
Cancel only $\frac{1}{5}$ part
Contribute

* Angular position of n^{th} secondary maxima $\rightarrow$

$$\theta_n^{\text{max}} = (2n+1) \frac{\lambda}{2} \quad (n = \pm 1, \pm 2, \dots) \text{ Slit}$$

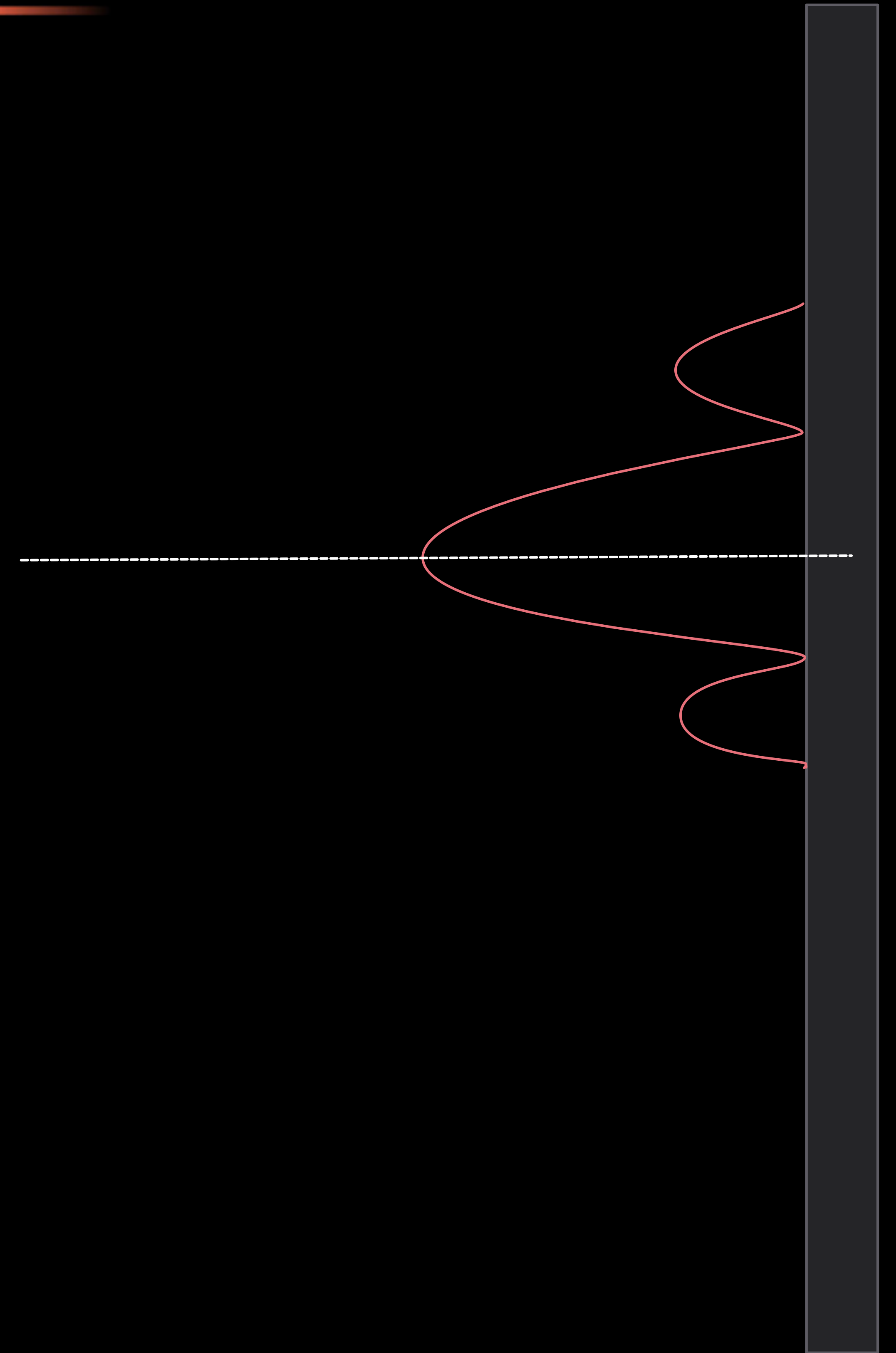


Diffraction due to a Single Slit

* Angular Width of Central Maxima (β_0)

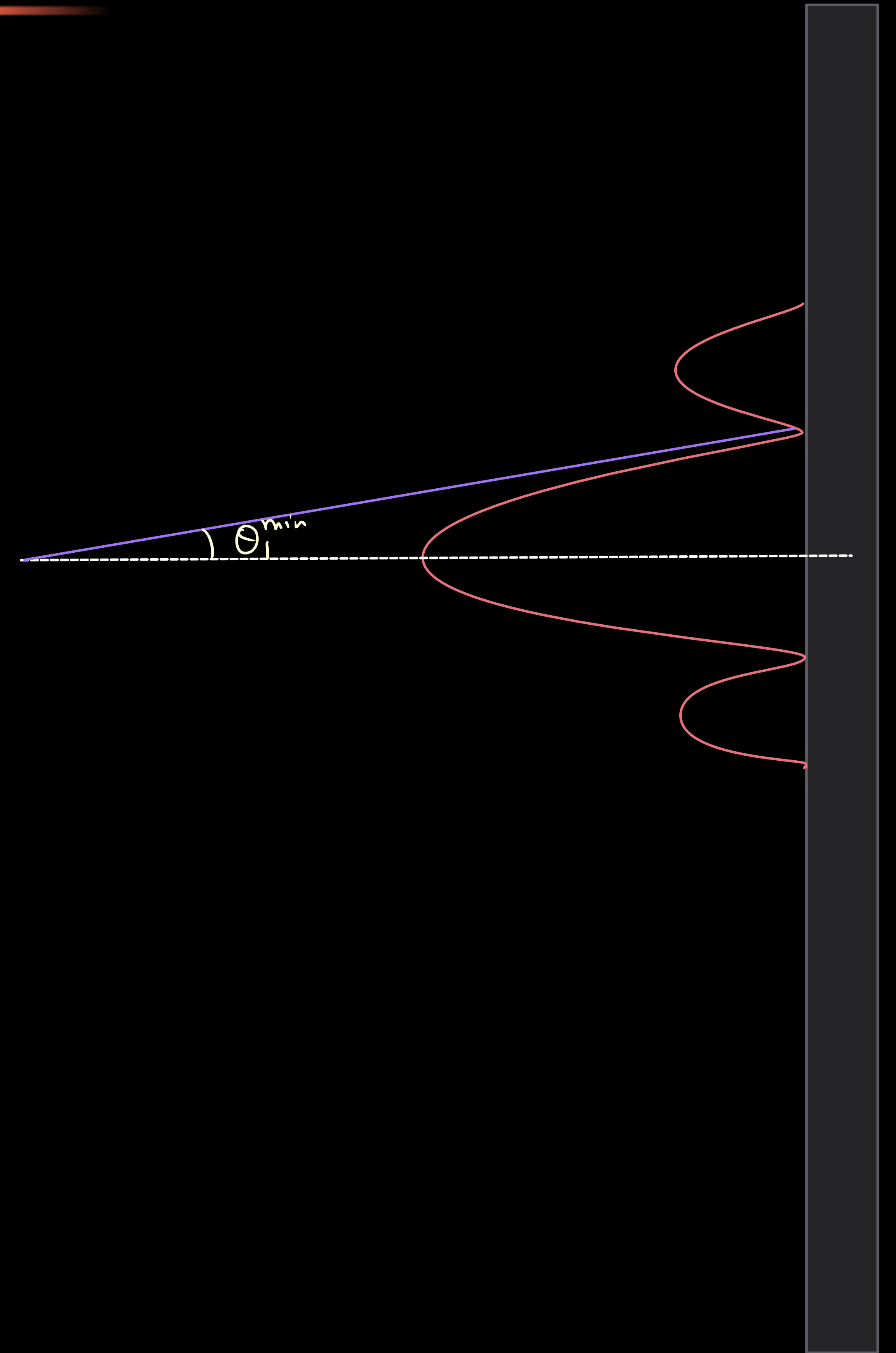
Diffraction due to a Single Slit

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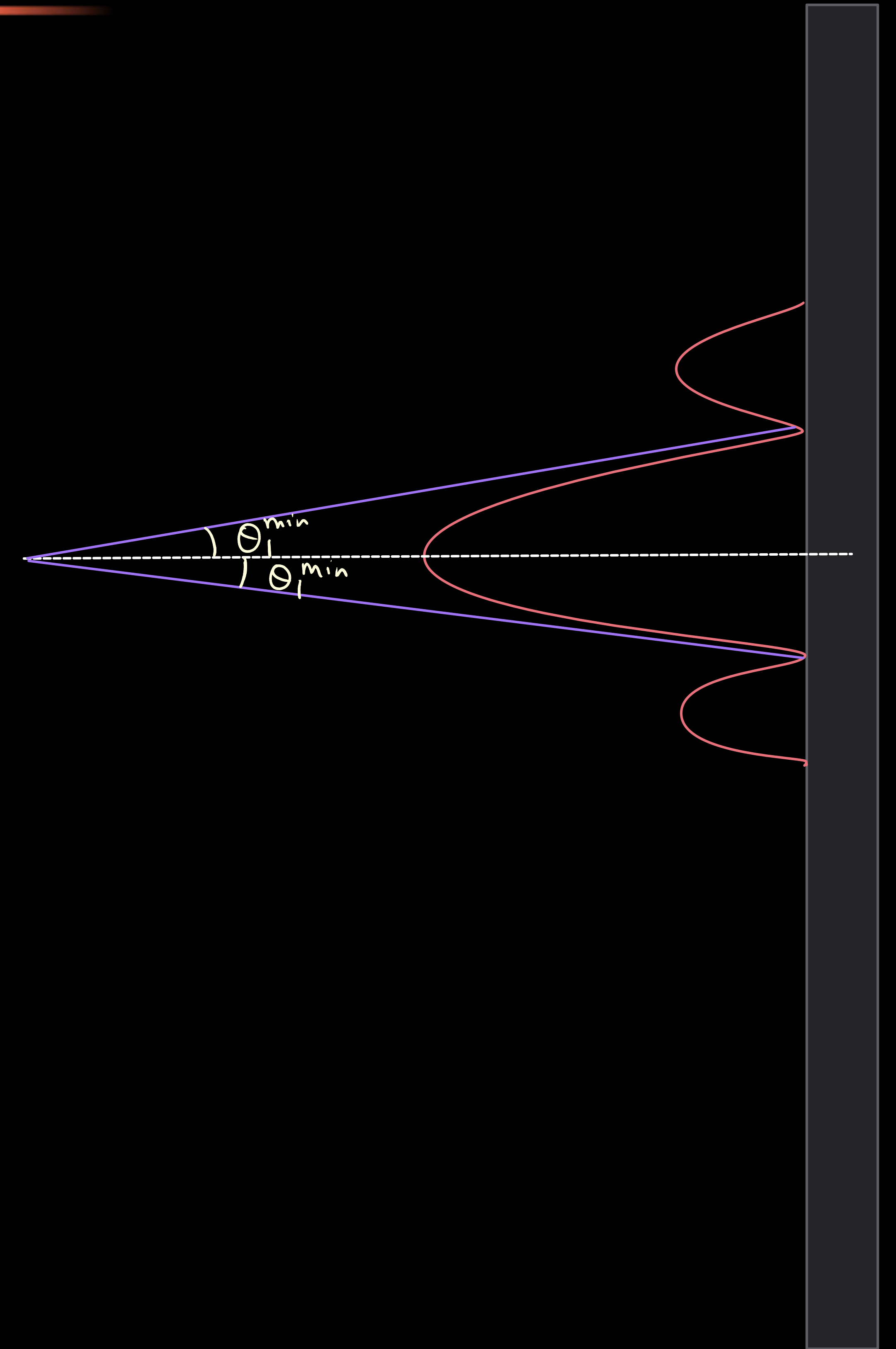
Diffraction due to a Single Slit

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Diffraction due to a Single Slit

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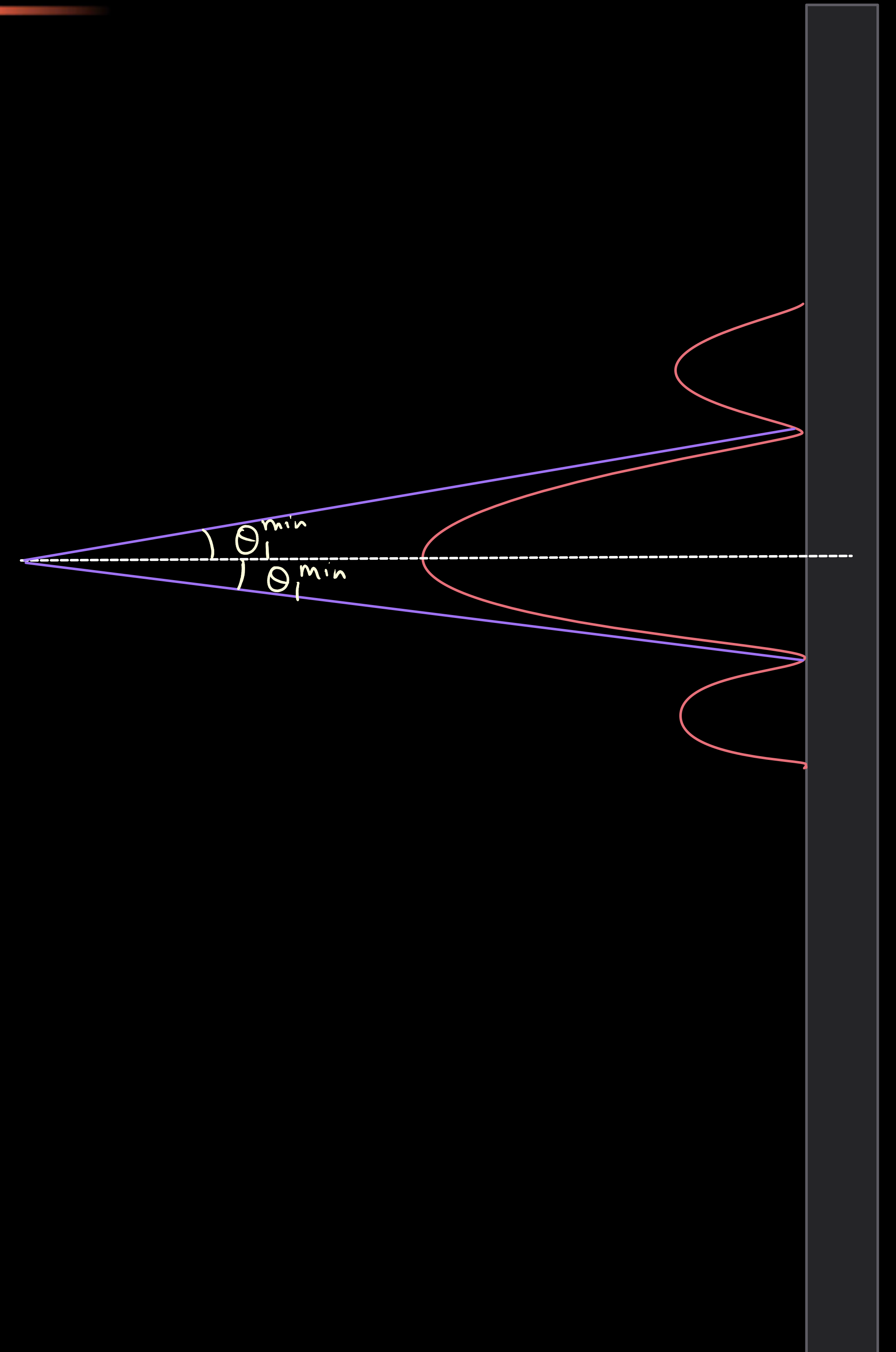


Diffraction due to a Single Slit

* Angular Width of Central Maxima (β_0)

$$\beta_0 = 2 \times \theta_1^{\min}$$

$$\beta_0 = 2 \times \frac{\lambda}{a}$$



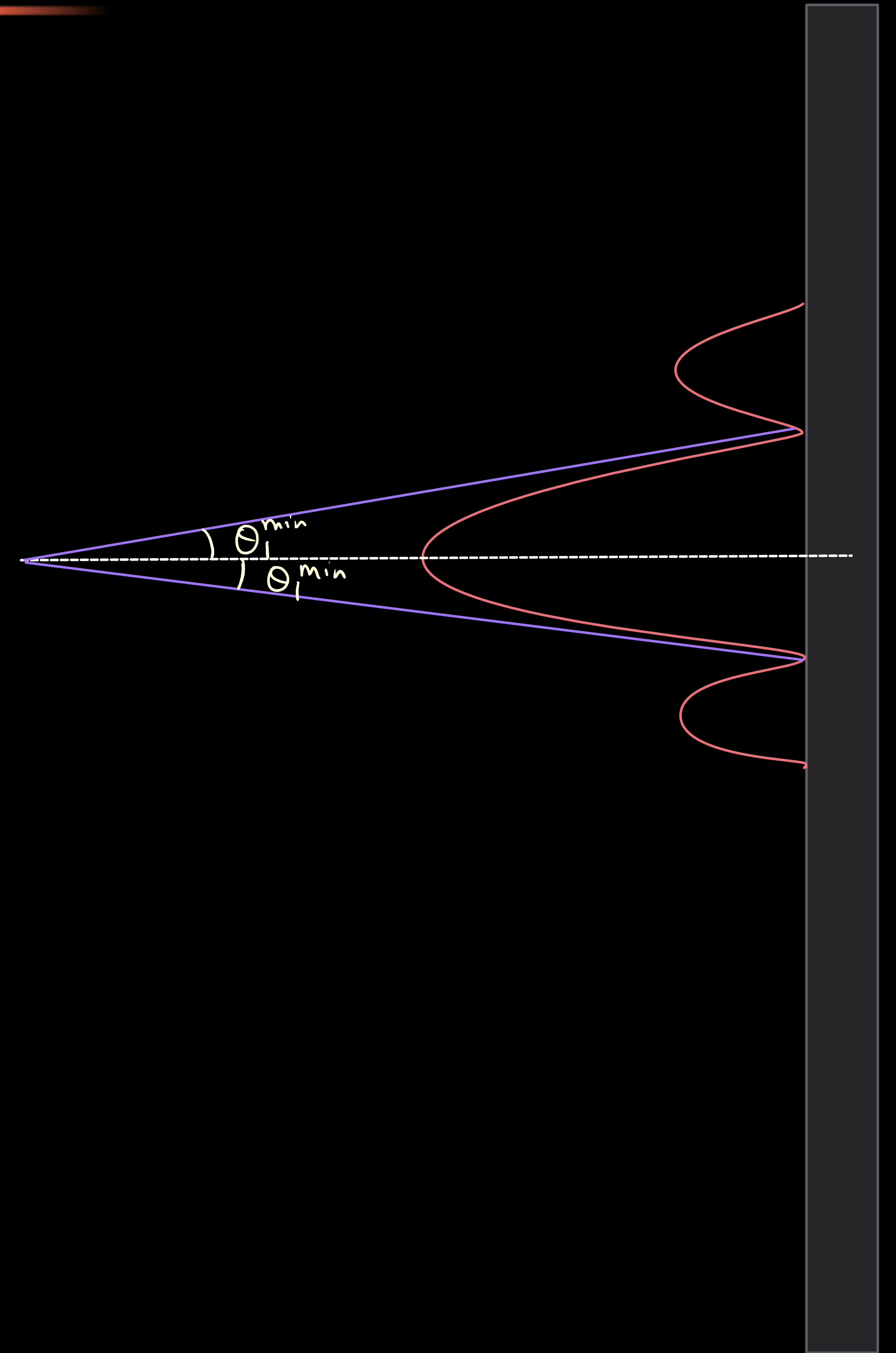
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Twice the angular width of secondary maxima.



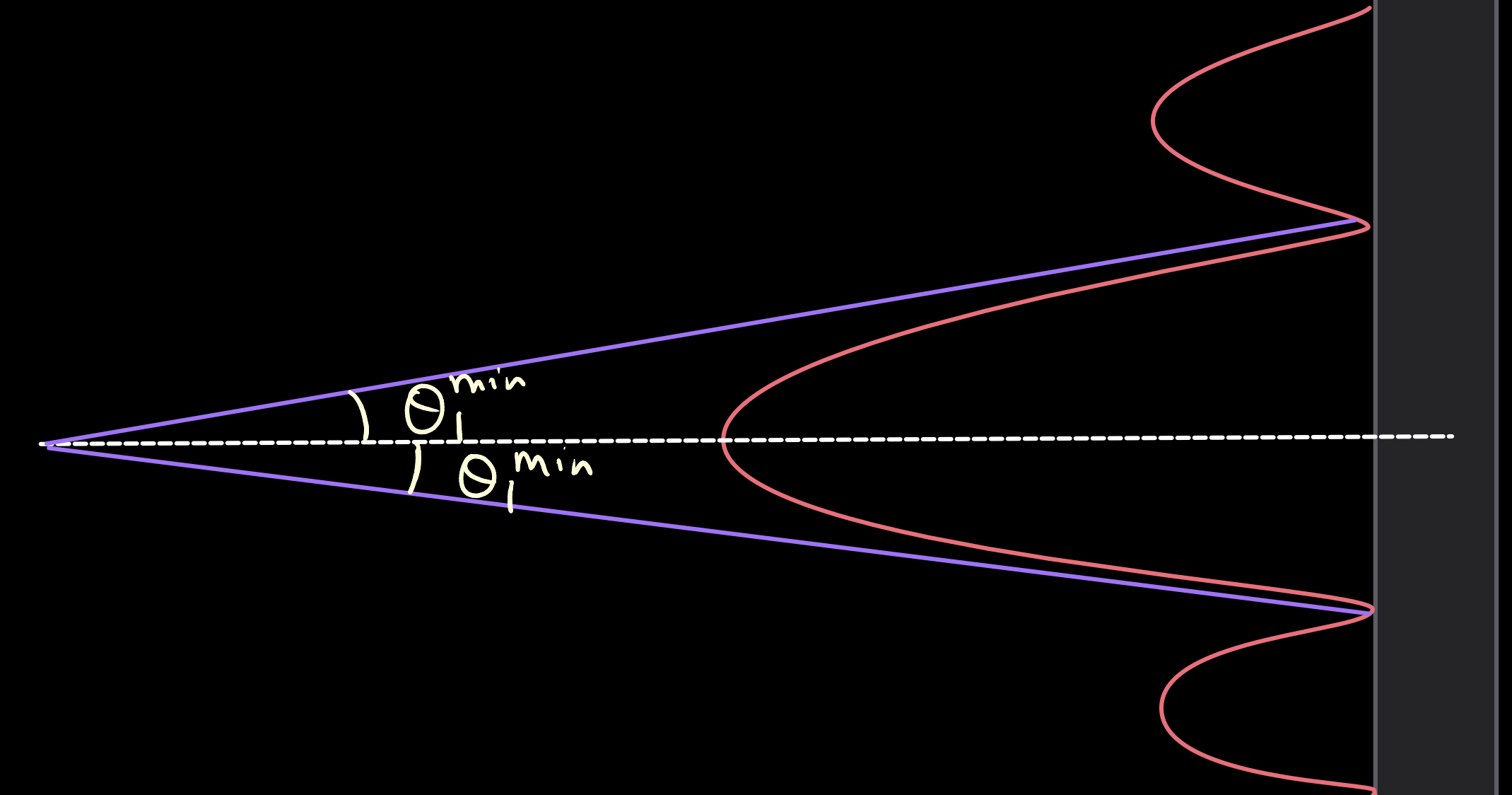
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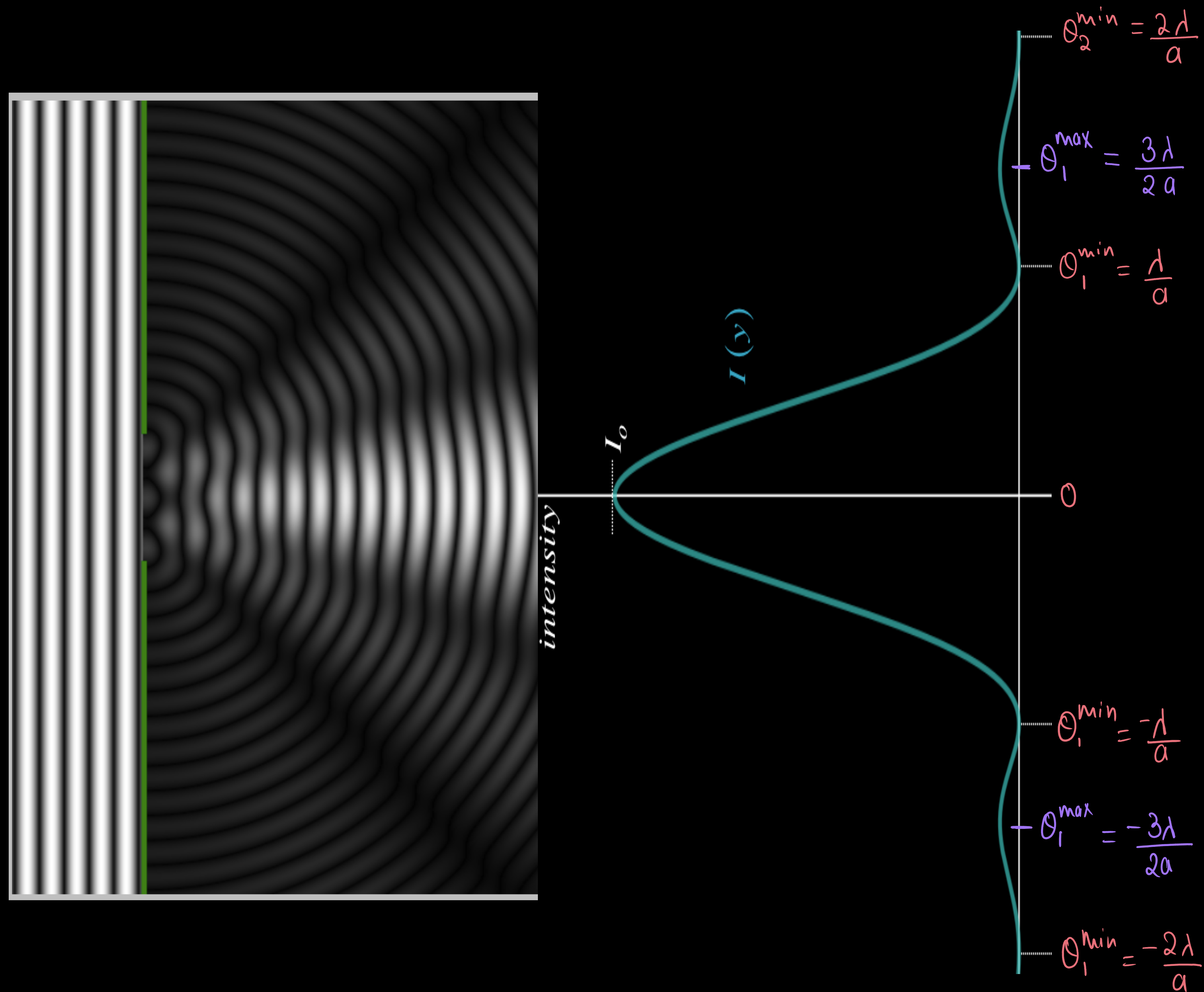


* the width of central max. is twice as that of secondary maxima or minima.

* The intensity of the secondary maxima goes on decreasing with the order (n) of Maxima.

* greater the wavelength (λ), greater the width of the maximum.

Intensity and photograph of fringes due to diffraction at single slit.



- * In the double-slit experiment, we must note that the pattern on the screen is actually a superposition of single-slit diffraction from each slit or hole, and the double-slit interference pattern.

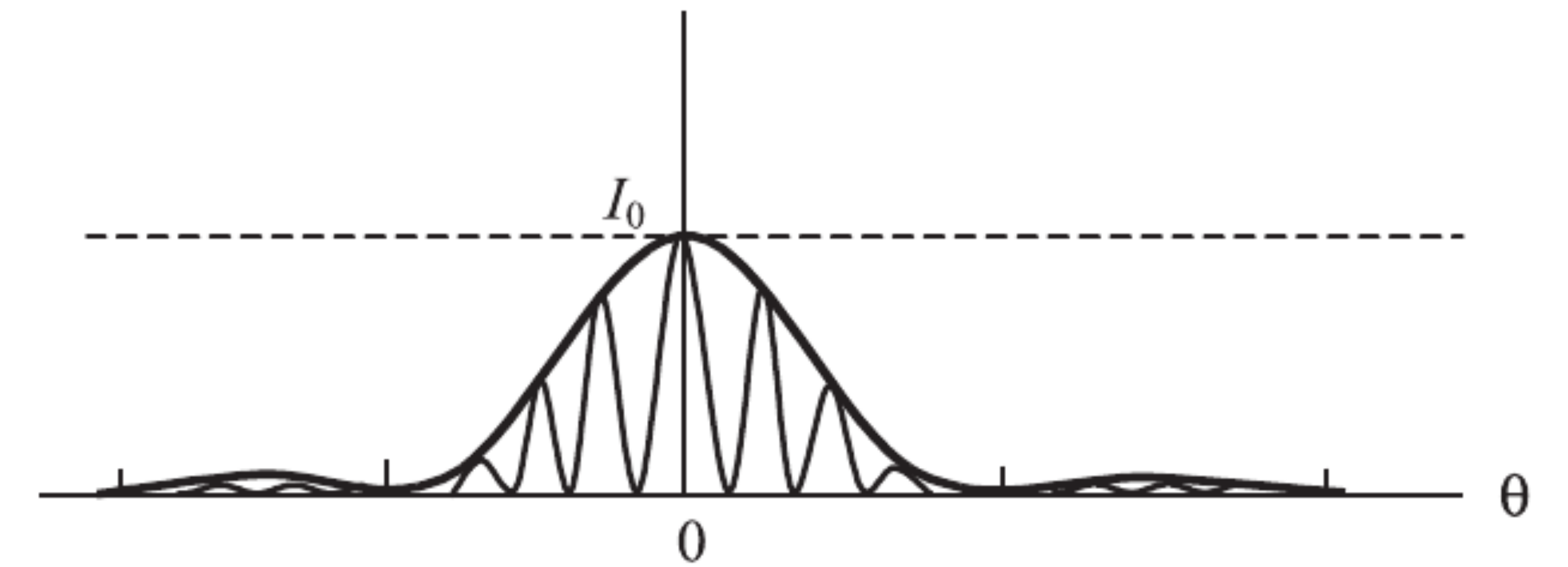


FIGURE 10.17 The actual double-slit interference pattern. The envelope shows the single slit diffraction.

- * In the double-slit experiment, we must note that the pattern on the screen is actually a superposition of single-slit diffraction from each slit or hole, and the double-slit interference pattern.
- * It shows a broader diffraction peak in which there appear several fringes of smaller width due to double-slit interference.

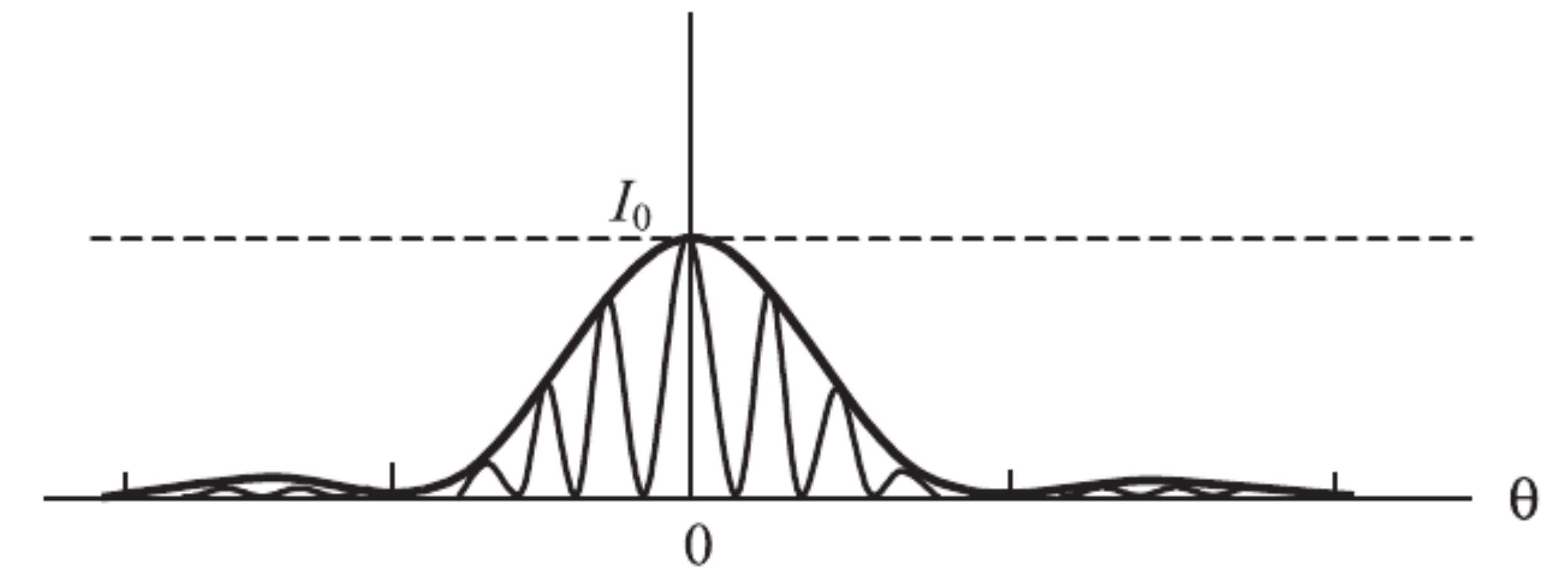
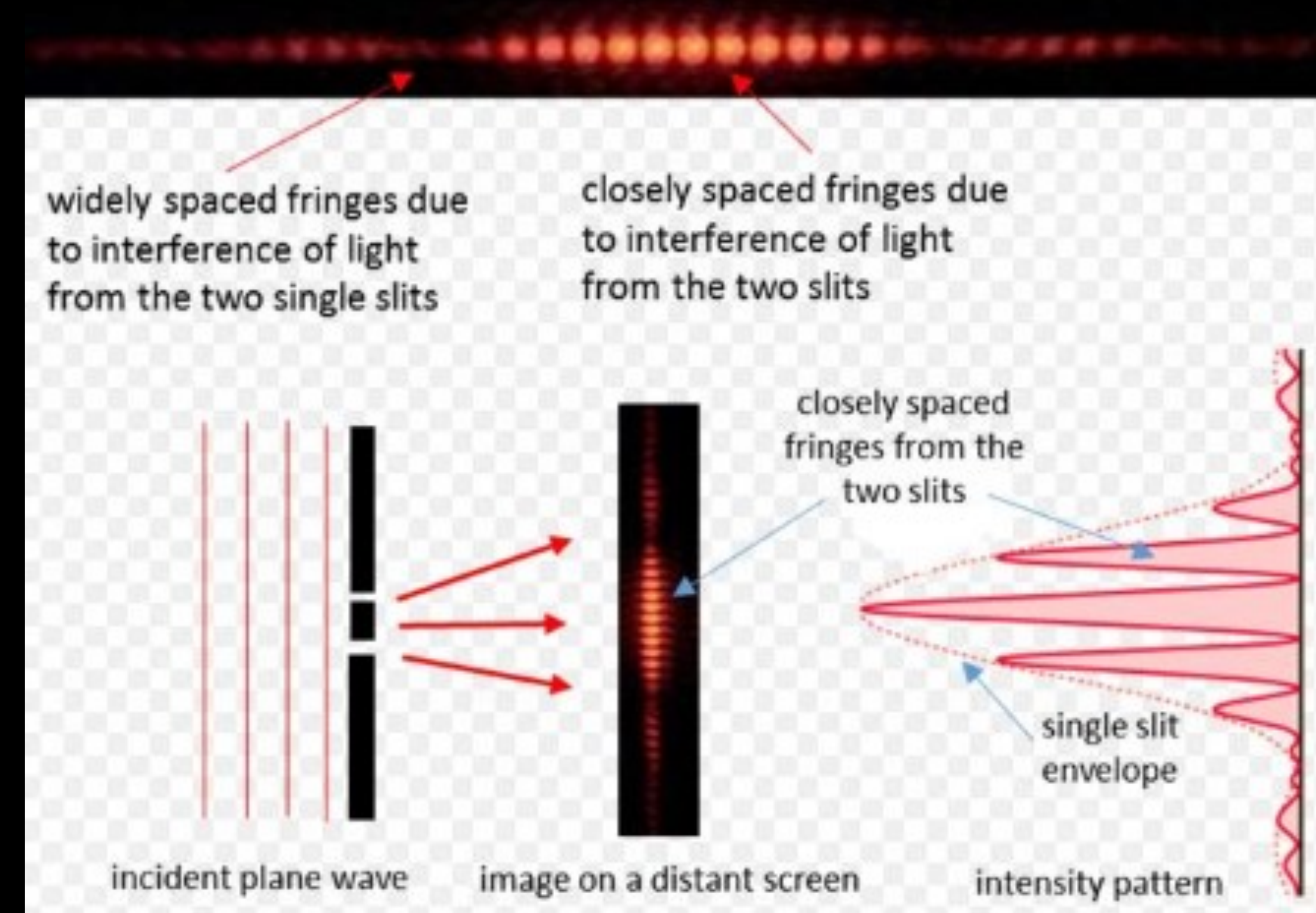


FIGURE 10.17 The actual double-slit interference pattern. The envelope shows the single slit diffraction.

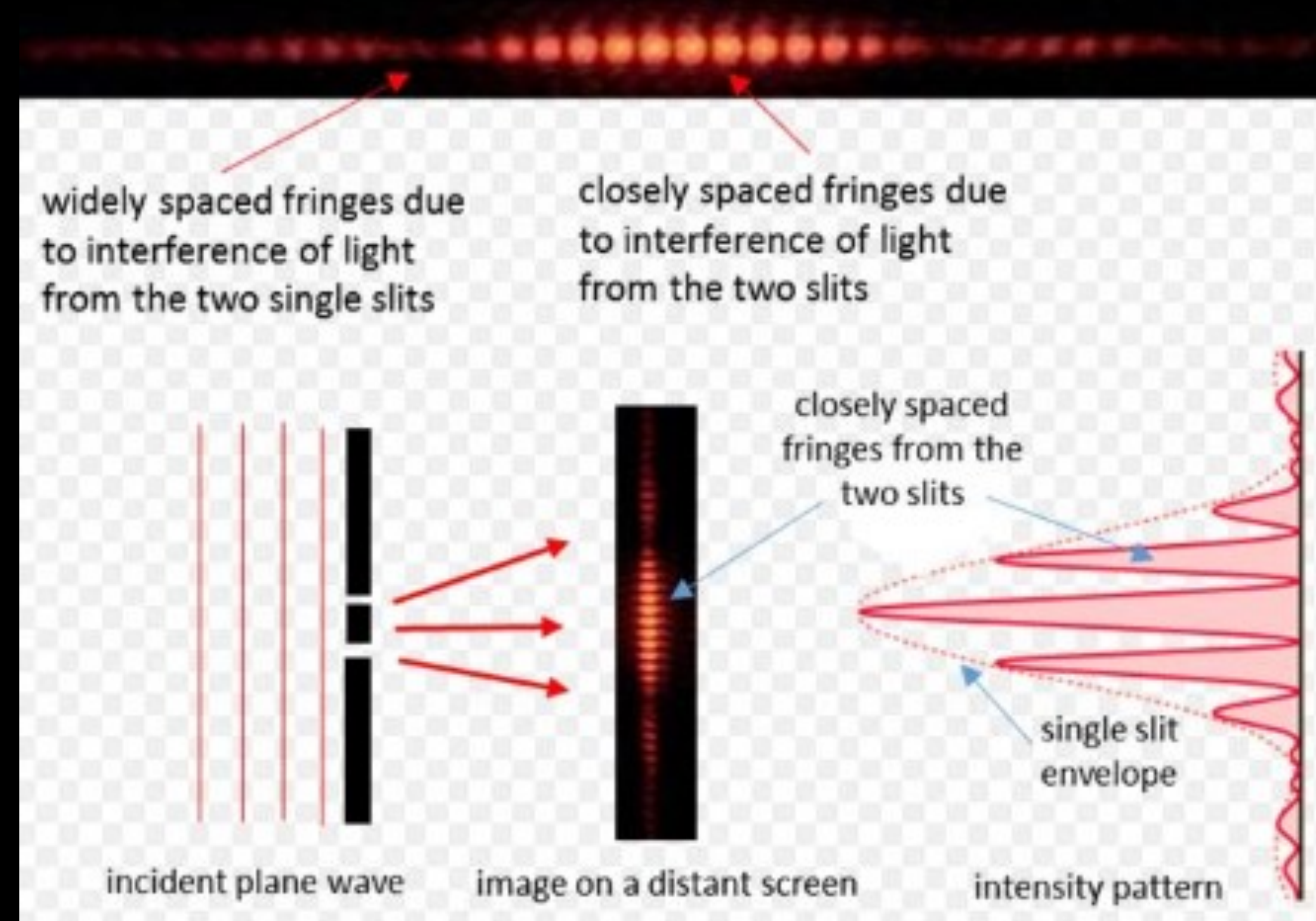
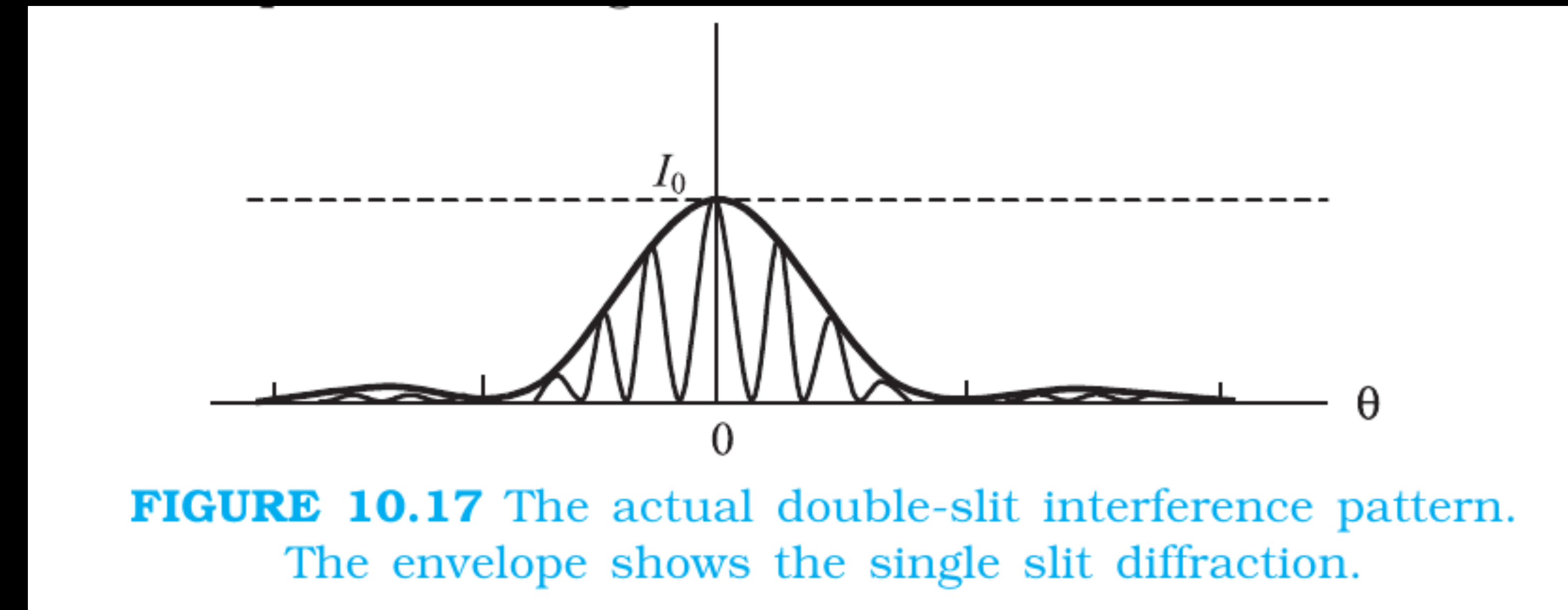


* In the double-slit experiment, we must note that the pattern on the screen is actually a superposition of single-slit diffraction from each slit or hole, and the double-slit interference pattern.

* It shows a broader diffraction peak in which there appear several fringes of smaller width due to double-slit interference.

* The number of interference fringes occurring in the broad diffraction peak depends on the ratio d/a , that is the ratio of the distance between the two slits to the width of a slit.

$$n = 2d/a$$



Difference between Interference Pattern and Single Slit Diffraction Pattern

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No one has ever been able to define the difference between interference and diffraction satisfactorily. It is just a question of usage, and there is no specific, important physical difference between them. The best we can do is, roughly speaking, is to say that when there are only a few sources, say two, interfering, then the result is usually called interference, but if there is a large number of them, it seems that the word diffraction is more often used.

—Richard Feynman, Feynman Lectures on Physics, Vol. 1

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We now compare and contrast the interference pattern with that seen for a coherently illuminated single slit (usually called the single slit diffraction pattern).

- (i) The interference pattern has a number of equally spaced bright and dark bands. The diffraction pattern has a central bright maximum which is twice as wide as the other maxima. The intensity falls as we go to successive maxima away from the centre, on either side.
- (ii) We calculate the interference pattern by superposing two waves originating from the two narrow slits. The diffraction pattern is a superposition of a continuous family of waves originating from each point on a single slit.
- (iii) For a single slit of width a , the first null of the interference pattern occurs at an angle of λ/a . At the same angle of λ/a , we get a maximum (not a null) for two narrow slits separated by a distance a .



* In interference and diffraction, light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe. There is no gain or loss of energy, which is consistent with the principle of conservation of energy.